COMPUTED TOMOGRAPHY IMAGING SPECTROSCOPY OF THE SOLAR TRANSITION REGION

by

Roy Thomas Smart

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ABSTRACT

The solar transition region, the thin layer between the cool lower atmosphere of the Sun and its million-degree corona, is threaded by jets and flows that change within seconds. Measuring how fast this gas moves requires its spectrum, the brightness of its light at each wavelength, at every point in an image. A conventional spectrograph records the spectrum along a narrow slit scanned across the target, so it cannot see a changing event everywhere at once. In computed tomography imaging spectroscopy, an instrument without a slit records several images, each mixing position and wavelength along a different direction, and a computer reconstructs the spectrum at every point, as a medical scanner reconstructs a body from many views. The reconstruction is only as accurate as its model of the instrument, including how the sensors blur each image and add noise to it.

This dissertation follows that chain from the sensor to the Sun in four studies. First, from the tracks of energetic particles crossing the sensors of the Interface Region Imaging Spectrograph at shallow angles, I measure in orbit how far charge spreads sideways inside them, which differs by a fifth among sensors of one design. Second, I show that ultraviolet images from two space telescopes are less noisy than standard theory predicts, because some electrons are lost before they can be counted, so these sensors perform better in the ultraviolet than was understood. Third, I describe the Extreme Ultraviolet Snapshot Imaging Spectrograph, which disperses the image of the Sun in several directions at once, as it was built and flown on a rocket on 30 September 2019. Fourth, I reconstruct the spectra of the largest event it observed, a jet in the quiet Sun, measure flows of up to **XX** kilometers per second, and compare them with the magnetic field beneath the jet and with its temperature of **XX** million kelvin, measured with several observatories.

Outline.

- The conclusions, in one or two sentences: what the jet shows about how transition region flows are driven, and what the dissertation shows about snapshot imaging spectroscopy.

1

INTRODUCTION

Outline.

- **The problem context.**
  - The solar transition region: between the chromosphere and the corona, about 10^4 to 10^6 K, thin and structured by the magnetic field.
  - Its dynamics: jets, explosive events, and flows, changing in seconds to minutes; magnetic reconnection as the proposed driver.
  - What diagnoses them: the profiles of emission lines formed there, such as O V, whose shifts, widths, and asymmetries give flows and turbulence; needed over a wide field, all at once.
  - The data cube $I(x, y, \lambda, t)$ against a two-dimensional detector, and the ways to flatten it: tunable filters (none in the extreme ultraviolet), multilayer imagers (no line profiles), rastered slits (IRIS, Hinode/EIS, SPICE; no simultaneity), many slits (MUSE), integral field units (SNIFS), and slitless spectrographs, whose overlappograms go back to Skylab S082A. A short version of the survey in Chapter 4.
  - Computed tomography imaging spectroscopy: each dispersed order as a projection of the cube, and the number of views limiting what can be recovered (Kak & Slaney 1988; Descour et al. 1997). Its origins outside solar physics (Okamoto & Yamaguchi 1991; Bulygin et al. 1991; Descour & Dereniak 1995), and the terms spectrometer and spectrograph.

  - The lineage in this group: MOSES (Fox et al. 2010; Fox 2011), what its flights found (Courrier & Kankelborg 2018; Rust & Kankelborg 2019; Parker & Kankelborg 2022) and where its design binds, then ESIS (the earlier design in Courrier 2020; the first flight in Parker et al. 2022).
  - Why the sensors matter: a reconstruction inverts a model of the instrument, so errors in the model become artifacts in the spectra.
    * The blur of the sensor, from charge diffusion, belongs in the point spread function of the model.
    * The noise of the sensor sets how closely a reconstruction should fit the images (for example, stopping at $\chi^2 \approx 1$) and the uncertainty of what it recovers.
    * Both are rarely measured on a flight sensor; ESIS flew back-illuminated CCDs whose quantum efficiency was not measured before flight (Chapter 4).
    * Why Chapters 2 and 3 use sensors on IRIS and the Hubble Space Telescope rather than those of ESIS, and how their results carry over.
      · Chapter 2: the noise model of Chapter 3 is tested against IRIS, and needs an accurate model of how charge diffuses in the IRIS sensors.
      · Chapter 3: no ultraviolet photon transfer curve was available for the ESIS sensors, so the noise model is tested against the measurements from IRIS and the Hubble Space Telescope.

- **The purpose and objectives.**
  - The purpose: tomographic imaging spectroscopy of the transition region, carried from the sensor to a measurement of the Sun.
  - To measure how back-illuminated sensors blur an image, in orbit.
  - To explain the noise of back-illuminated sensors in the ultraviolet.

  - To describe ESIS as it was built and flown, and how its design answers the limitations of MOSES.
  - To measure the flows, magnetic topology, and temperature of a transition region jet with ESIS.

- **How the manuscripts address them.**
  - Chapters 2 to 5: for each manuscript, the question it answers, what it found, and what the next chapter takes from it. The chapters are in the order of publication, so none refers forward.
    * Chapter 2: the charge-diffusion kernel of the IRIS sensors, from glancing particle tracks, which Chapter 3 needs to model the noise of those sensors.
    * Chapter 3: partial charge collection, and the noise of ultraviolet images from IRIS and the Hubble Space Telescope.
    * Chapter 4: ESIS, its design, and its first flight.
    * Chapter 5: event E, a jet observed by ESIS.
  - Chapter 6: the conclusion.

4

MEASURING THE CHARGE-DIFFUSION KERNEL OF A BACK-ILLUMINATED CCD IN ORBIT FROM GLANCING PARTICLE TRACKS

Contribution of Authors and Co-Authors

Manuscript in Chapter 2

Author: Roy T. Smart
Contributions: **XX**

Co-Author: Charles C. Kankelborg
Contributions: **XX**

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Measuring the Charge-diffusion Kernel of a Back-illuminated CCD in Orbit from Glancing Particle Tracks

Roy T. Smart[a,*], **Charles C. Kankelborg**[a]
[a]Montana State University, Department of Physics, P.O. Box 173840, Bozeman, MT 59717, USA

Abstract.
The lateral diffusion of photogenerated charge in the undepleted layer of a back-illuminated silicon sensor sets its point spread function and the statistics of its noise in the ultraviolet, but it is rarely measured on a flight sensor. We show that glancing particle tracks, which cross the full thickness of the sensor over a length of tens of pixels, sample the diffusion width at every depth in a single exposure, and we use them to measure the depth-dependent charge-diffusion kernel of the four CCDs of the Interface Region Imaging Spectrograph in orbit, from 10044 tracks found in level-1 images taken mostly inside the South Atlantic Anomaly. The width of the charge cloud at the back surface, the depth at which the field-free layer ends, the spread acquired drifting across the depletion region, and the probability that two electrons deposited at the same depth are collected in the same pixel are measured directly on each sensor. The field-free layer ends at 0.40 of the thickness on the Slit-Jaw Imager, where the cloud at the back surface is 5.5 μm wide and the same-pixel probability is 0.339 ± 0.004, and the tracks require a further spread of 0.80 μm inside the depletion region. The four sensors, of one design, differ by a fifth in the width of the cloud at the back surface. The method needs no laboratory access and applies to any back-illuminated sensor in orbit.

Keywords: charge-coupled devices, charge diffusion, point spread function, ultraviolet detectors, particle tracks, IRIS.
*Roy T. Smart, roytsmart@gmail.com

1 Introduction

A back-illuminated silicon charge-coupled device (CCD) absorbs ultraviolet (UV) light within a fraction of a micron of its back surface, in a layer where there is no electric field to sweep the photogenerated charge toward the gates. The charge diffuses until it reaches the edge of the depletion region, and the resulting charge cloud, which can be several microns wide, is shared among neighboring pixels. This sharing broadens the point spread function (PSF) of the sensor[1] and, because the electrons liberated by one photon are shared together, it changes the statistics of the image noise, entering the variance-to-mean ratio (VMR) of a photon-transfer curve through the probability that two electrons from the same photon land in the same pixel.[2]

The width of the charge cloud is set by the thickness of the undepleted layer, which depends on the resistivity of the wafer and the bias applied to the device, both of which vary between nominally identical sensors and are rarely known for a sensor in orbit. Laboratory measurements use light of varying penetration depth[3,4] or the split events of soft X-rays,[5] and neither is available once the instrument has launched.

Energetic particles offer a probe that arrives for free. A particle crossing the sensor at glancing incidence deposits charge along a track tens of pixels long, and because it enters at the back surface and exits at the gates, the fractional position along the track is the fractional depth of the charge. Cosmic-ray muons have been used in this way to measure the growth of charge diffusion with drift distance in thick, fully depleted CCDs,[6] where the cloud widens with depth. In a thin sensor with an undepleted back surface the dependence is reversed, the cloud is widest at the back surface and collapses to a pixel-sharp line in the depletion region, and the same tracks measure the field-free kernel that governs UV imaging.

Here we apply this method to the four CCDs of the Interface Region Imaging Spectrograph (IRIS),[7] whose far ultraviolet (FUV) and near-UV photon-transfer curves[8] motivated the question. Section 2 describes the tracks, the model of their cross-section, and the fit; Section 3 presents the kernel of each sensor; and Section 4 discusses the systematics, including the charge density of the tracks.

2 Method

2.1 Data

IRIS[7] carries four CCDs of the same design: two behind the FUV channel of the spectrograph (FUV1 and FUV2, which share one level-1 image), one behind its near ultraviolet (NUV) channel, and one behind the Slit-Jaw Imager (SJI). They are back-illuminated, 14 μm thick, with 13 μm pixels, and their level-1 images are the raw frames with only the dark and the readout corrections applied, so a particle hit appears exactly as the camera recorded it. We searched 23792 such images, 8260 of them taken inside the South Atlantic Anomaly (SAA), where the flux of trapped protons at the altitude of IRIS is greatest,[9,10] drawn from the campaigns of Table 1. The campaigns added for this work were chosen among those exposing for 8 s, since a longer exposure crowds the frame inside SAA with hits that spoil the tracks around them (the campaigns of the original measurement that pass through SAA expose for 15 s, and yielded per frame half as many of the tracks kept below), and among those whose spectrograph readout window lies mostly beyond the limb, as the pointing recorded in each frame's header places it. Figure 1 shows a spectrograph image and a slit-jaw image taken seconds apart during one pass through SAA: the particle hits are dense, and a handful of them are the long glancing tracks we want. Tracks were sought only where the pointing recorded in each frame's header places the pixel at least 15 arcseconds above the photospheric limb, so that no track lies on the disk or in the chromosphere and spicules seen just above the limb.

2.2 Finding tracks

The background of each frame was estimated as the trimmed mean of the frames in the same observing sequence and subtracted, and the read noise was estimated from the median absolute deviation of the residual. Pixels more than five times the read noise above the background were labeled, and a straight line was fit by least squares to the charge-weighted centroid of each row of every connected group of labeled pixels. We kept the groups spanning at least twelve rows (or columns) with a slope below 0.35 pixels per row, so that each track is nearly aligned with the pixel grid and its cross-section is sampled once per row without resampling the pixels, and extracted a seven-pixel-wide cutout centered on the integer part of the fitted line in every row. This yielded 10044 tracks. The slope cut is not as costly as it sounds: Figure 2 shows that the trapped protons arrive strongly aligned with one axis of the detector or the other, which one depending on the roll of the spacecraft, so that more of them pass it than the 43% an isotropic population would, 57% at the roll of most of the campaigns.

2.3 Model

Figure 3 shows the principle. A particle that enters the back surface and exits the front surface spans the full thickness of the sensor, so the fractional depth of slice i of an N-slice track is

Table 1 The observing campaigns searched for tracks, with the cameras kept from each (F for the far-ultraviolet spectrograph, N for the near-ultraviolet one, S for the slit-jaw imager), the spacecraft roll, the exposure time, and the number of frames searched, with the number taken inside SAA in parentheses. The tracks of every campaign but the two marked †, whose frames were taken outside SAA, are left by trapped protons; those two hold cosmic rays. The core tracks are the flat tracks with $0.25 < t_c < 0.6$, over which the mean t_c with its standard error and the median σ_{max} with its interquartile range are taken.

date	cameras	roll	exp. (s)	frames	tracks	flat	core	t_c	σ_{max} (μm)
2014-03-12	FN	0°	15	888 (114)	608	171	140	0.418 ± 0.006	5.0 [4.5, 6.0]
2014-04-07†	FN	0°	4	1950 (0)	34	15	13	0.408 ± 0.015	5.5 [4.5, 6.0]
2018-03-16†	FN	0°	15	622 (34)	118	50	38	0.392 ± 0.009	5.0 [4.0, 5.0]
2018-05-04	FN	−90°	15	722 (273)	611	174	129	0.406 ± 0.006	5.5 [4.5, 6.0]
2018-05-04/06	S	−90°	15	1156 (469)	471	157	139	0.410 ± 0.006	5.5 [5.0, 6.0]
2014-08-20	FNS	−75°	8	782 (333)	636	328	292	0.405 ± 0.004	5.5 [5.0, 6.0]
2014-08-21	FNS	−75°	8	768 (318)	800	367	328	0.403 ± 0.004	5.2 [4.5, 6.0]
2014-08-23	FNS	90°	8	1214 (333)	800	471	417	0.409 ± 0.003	5.5 [5.0, 6.0]
2014-08-24	FNS	90°	8	806 (330)	942	473	436	0.408 ± 0.003	5.0 [4.5, 6.0]
2014-08-25	FNS	90°	8	778 (321)	1482	641	547	0.408 ± 0.003	5.5 [4.5, 6.0]
2014-11-02	FNS	0°	8	762 (300)	1153	463	388	0.399 ± 0.003	5.5 [5.0, 6.0]
2014-11-27	FNS	0°	8	970 (383)	144	9	9	0.406 ± 0.024	5.5 [4.5, 5.5]
2015-03-26	FNS	−90°	8	616 (210)	65	33	30	0.398 ± 0.011	5.5 [4.5, 6.0]
2015-09-20	FNS	0°	8	1039 (575)	505	245	221	0.406 ± 0.005	5.5 [5.0, 6.0]
2016-01-02	FNS	0°	8	3836 (1134)	542	218	183	0.406 ± 0.005	5.5 [4.5, 6.0]
2016-01-06	FN	0°	8	638 (184)	90	61	58	0.423 ± 0.009	5.0 [4.5, 5.5]
2016-05-16	FNS	0°	8	1174 (732)	511	243	208	0.397 ± 0.004	5.5 [5.0, 6.0]
2016-10-28	FN	0°	8	556 (234)	186	98	82	0.415 ± 0.007	5.0 [4.5, 5.5]
2016-12-13	FNS	0°	8	1737 (990)	123	67	65	0.389 ± 0.007	5.0 [4.5, 5.5]
2016-12-17	FNS	0°	8	1247 (192)	3	2	2	0.350 ± 0.035	5.2 [4.9, 5.6]
2016-12-25	FNS	0°	8	1531 (801)	220	101	95	0.404 ± 0.007	5.0 [4.5, 6.0]

$t = (i + 1/2)/N$, up to the orientation of the track, and the charge deposited in each slice diffuses from a known depth. We model the charge in every slice as spread uniformly along the fitted centerline and diffused as a Gaussian whose variance has two parts,

$$\sigma^2(t) = \sigma^2_{\text{max}} \left(1 - \frac{t}{t_c}\right) + \sigma_d^2\, g(t). \tag{1}$$

The first part is the field-free layer. It applies only for $t < t_c$, where $t_c = z_f/D$ is the fractional thickness of the layer and σ_{max} is the width of the charge cloud at the back surface, and it is the depth dependence of a random walk to an absorbing plane at z_f. The second part is the depletion region. Charge that reaches the edge of the depletion region, whether it was deposited there or diffused there from the field-free layer, still has to drift to the gates and spreads a little as it does so. σ_d is the spread acquired crossing the full depleted thickness, and $g(t) = \min[(1-t)/(1-t_c), 1]$ scales it with the distance drifted, so that it is one for all charge deposited in the field-free layer and falls linearly to zero at the gates. We work in fractional depth because we observe the length of a track in slices rather than the thickness of the sensor in microns, and the uniform spread along the tilted centerline within a slice is approximated by adding $m^2/12$ to the variance, where m is the slope in pixels per row.

The three parameters are not fit alike. t_c and σ_{max} describe the field-free layer, whose thickness can vary across a wafer, and are fit to each track together with its orientation, by exhaustive search

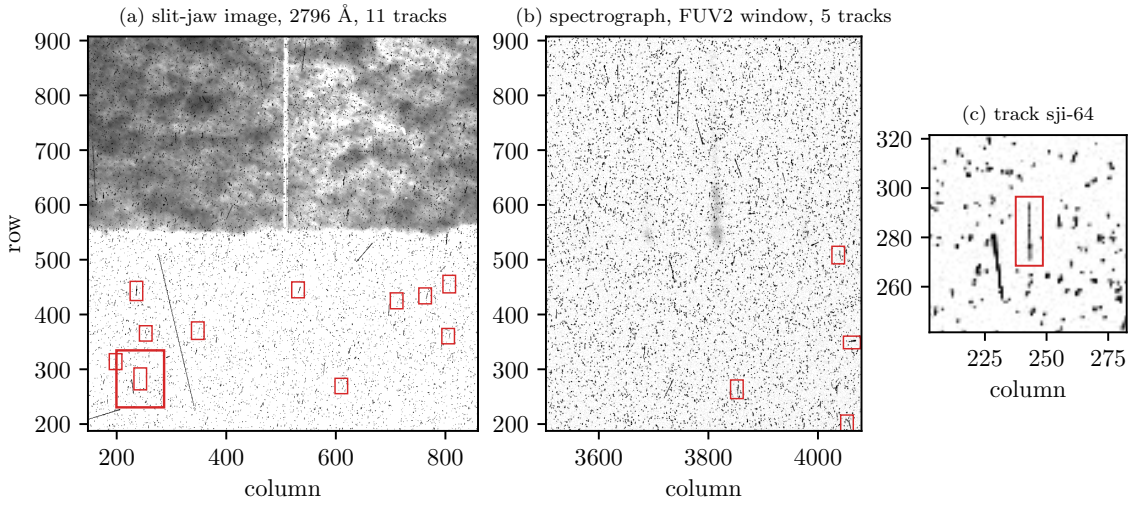


Fig 1 Level-1 images taken seconds apart during a pass through SAA on 2018 May 4, with the spacecraft rolled by $-90°$ so that the limb runs horizontally across the field. (a) The slit-jaw image at 2796 Å, whose lower half is off the limb. (b) The widest of the readout windows of the spectrograph image, on the FUV2 CCD. (c) One of the tracks in (a) at full resolution. The images are shown with an inverse hyperbolic sine stretch above the background, saturating at 400 data numbers, and the red boxes outline the tracks the finder kept from each frame. Most of the particle hits are the short streaks and dots that pepper both images; the tracks used here are the long, thin, nearly vertical ones.

on a grid of 21 values in each of $t_c \in [0, 1]$ and $\sigma_{\text{max}} \in [0, 10]$ μm. At each grid point the offset and tilt of the centerline are adjusted, by up to 0.6 pixels and 0.03 pixels per row, to minimize a robust misfit,

$$M = \sum_i \sum_j \ln\left(1 + \frac{r_{ij}^2}{2}\right), \qquad r_{ij} = \frac{f_{ij} - q_{ij}}{\epsilon_i}, \tag{2}$$

where f_{ij} is the fraction of the charge of slice i measured in pixel j of the cutout, q_{ij} the fraction the model puts there, and ϵ_i the read noise of a pixel divided by the charge of the slice, the uncertainty of a fraction from read noise alone, so that the brighter slices count for more. This is the negative log-likelihood of a Cauchy distribution rather than a Gaussian, so that a single stray pixel, from a delta ray or a second hit inside the cutout, cannot dominate the fit, and differences in M are what the thresholds below are stated in. σ_d describes the drift field, which is a property of the CCD and its bias rather than of any one track, and a single track constrains it only weakly, so it is fit to each CCD as a whole. Every track is fit as above at each of 16 values of σ_d from 0 to 1.5 μm, the misfits M of the flat tracks, defined next, are summed at each value, and the value that minimizes the sum is adopted for every track on that CCD. Which tracks are flat depends on their fits, so the selection and σ_d are iterated to consistency, which takes one or two rounds.

2.4 Selection

The useful tracks are those which cross the full thickness of the sensor at nearly constant energy loss. We kept only the tracks for which the fit constrains t_c to within 0.15 and improves on a model with no diffusion by at least ten in M, for which the median charge per slice in the last third of the track is within 50% of that in the first third, since a rise in the deposited charge along the track is

the Bragg peak of a particle that stopped inside the sensor and did not cross its full thickness, and for which the fit finds a depleted end, $t_c \leq 0.7$, since a particle that crossed the sensor is pixel-sharp where it left through the gates, whereas a feature that is wide from end to end is, on the slit-jaw imager, usually a spicule or other structure at the limb that the mask let through. We call these the flat tracks; 4387 tracks pass these cuts, 1016 of them on the SJI CCD, and Figure 9 shows the longest of them alongside a stopping track.

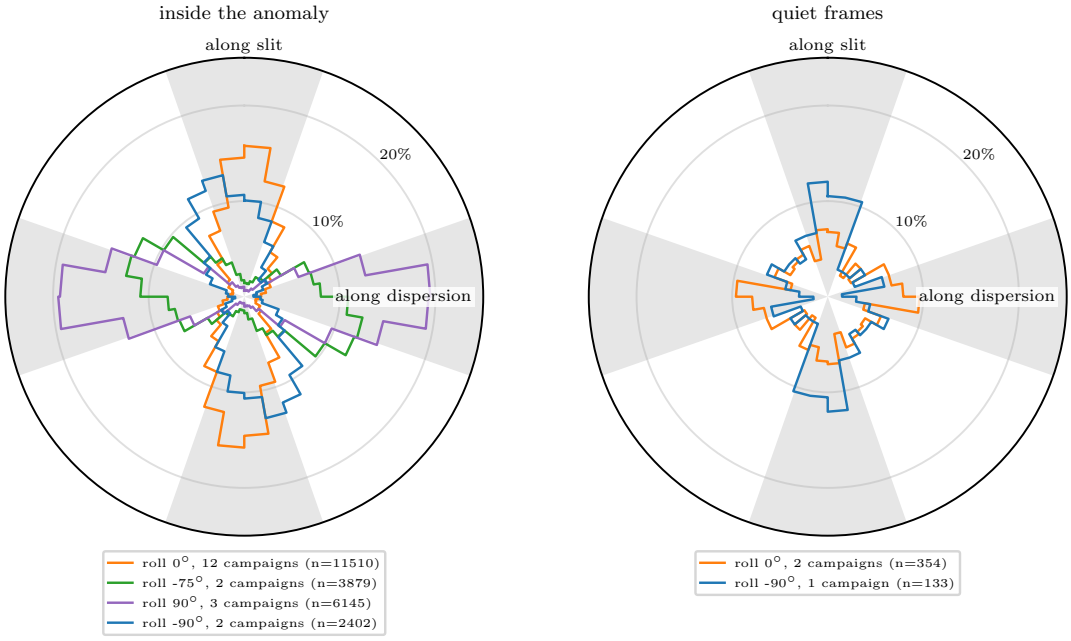


Fig 2 The azimuth in the detector frame of every elongated connected component above 5σ in the spectrograph frames of every campaign, with a charge-weighted width under 3 pixels and an aspect ratio over 4, before any of the finder's cuts, so that the whole circle is populated, grouped by the roll of the spacecraft. Tracks are undirected, so each is drawn at θ and $\theta + 180°$, and the grey wedges are the windows about the row and column axes that the finder accepts. Left: the frames taken inside SAA. Right: quiet frames. The trapped protons arrive strongly aligned, along the slit at rolls of $0°$ and $-90°$, along the dispersion at $90°$, and between the two at $-75°$, so the preferred direction is a property of the arrival direction as seen by the spacecraft rather than of the detector. The cosmic rays of the quiet frames at roll $0°$ are close to isotropic, while the quiet frames of the one campaign at $-90°$, taken at the edge of the anomaly, keep some of its alignment.

3 Results

3.1 The width against depth

Figure 4 shows the flat tracks stacked on their fitted centerlines and the diffusion width they imply at each depth, measured two ways. The first assumes no depth dependence at all: for each depth bin, the misfits of every slice in the bin are summed over a grid of trial widths and the width which minimizes the sum is taken, so the wedge shape emerges from the data rather than being imposed. The second is the average of the per-track fits of Equation 1. The two agree to within half

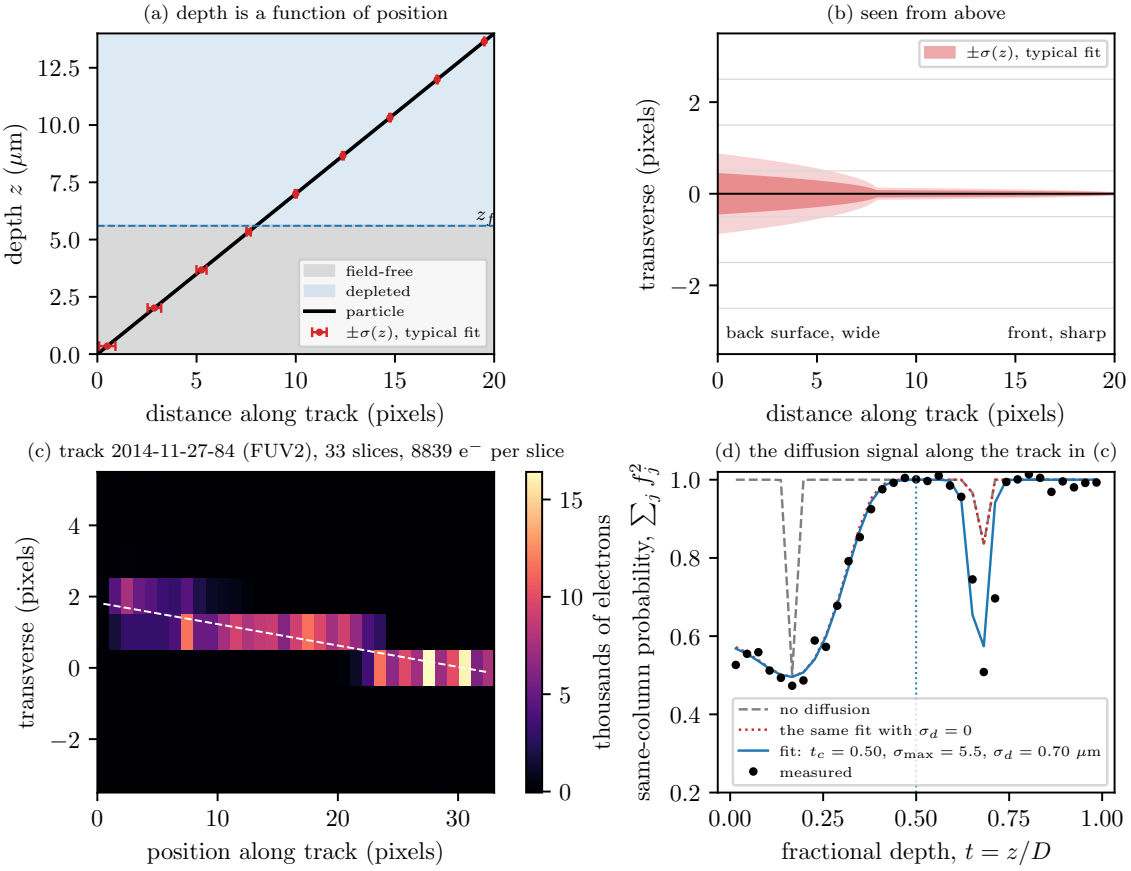


Fig 3 How a glancing track measures the diffusion width against depth. (a) A particle entering at the illuminated back surface ($z = 0$) and leaving at the gates ($z = D$) crosses the field-free layer first, where the charge it liberates diffuses until it reaches the depletion edge, so the lateral spread (red bars, to scale in pixels, at the median fit of the FUV2 tracks and its σ_d) is largest at the back surface and falls steeply to $z = z_f$; beyond it only the drift across the depletion region spreads the charge, by under a micron, narrowing to nothing at the gates. (b) Seen from above, the track is a wedge: about 0.4 pixels wide at one end and a few hundredths of a pixel beyond t_c. (c) A proton track on the FUV2 CCD, in the coordinates of its parent image, with the fitted centerline dashed. (d) The probability that two electrons deposited in the same slice of that track are collected in the same column, $\sum_j f_j^2$ corrected for read noise, for every slice, against the best fit, the same fit with $\sigma_d = 0$, and a model with no diffusion, all evaluated at the fitted centerline. The dips where the no-diffusion curve falls below one are where the centerline crosses a column boundary; the diffusion signal is the gap between the measured points and that curve. Beyond t_c the cloud is under a micron wide, a small fraction of a pixel, and changes $\sum_j f_j^2$ only where the centerline runs within a tenth of a pixel of a boundary, as it does near $t = 0.65$: there the measured dip follows the fit with σ_d and is deeper and wider than the fit without it.

a micron on all four CCDs at every depth. Beyond t_c the model-free profile does not go to zero but keeps a floor across the depletion region, about three quarters of a micron just past t_c and a quarter micron near the gates, and the fits follow it: this floor is the σ_d term of Equation 1. The third panel shows the kernel itself on FUV2 at four depths, integrated over a pixel: the fits follow its core as it narrows from the back surface to the gates, while the stray charge discussed with Figure 7 lowers

the measured peak by a few hundredths and raises the wings.

3.2 The spread inside the depletion region

Figure 5 shows how σ_d is determined. The misfit M of Equation 2 summed over the flat tracks on each CCD has a sharp minimum, at $\sigma_d = 0.80$ μm on SJI and at 0.70 μm on FUV1, FUV2, and NUV, with the neighboring grid points a tenth of a micron away already disfavored by one to four hundred units of misfit. The improvement over $\sigma_d = 0$ is one to three thousand units pooled but under two per track, and the value each track prefers on its own is spread over the whole grid, from 0 to 1.5 μm, so the effect is modest for any one track and consistent across them, which is why σ_d is fit to each CCD as a whole. The resolution of the centerline matters here: a centerline located only to 0.1 pixel is uncertain by 1.3 μm, the size of the effect being sought, and with such a grid the minimum washes out entirely. σ_d trades against the field-free wedge: as it rises from zero to its adopted value, the mean t_c of the flat tracks falls by 0.03 to 0.06, most on FUV1, since without it the wedge stretches to absorb the floor, while the mean $\sigma_{\max}$ rises by only a few percent. Leaving σ_d out of the model would therefore bias the critical depth, which is why it is part of the model rather than a correction applied afterwards.

3.3 The critical depth and back-surface width

Figure 6 shows the fitted t_c and $\sigma_{\max}$ of every flat track, and Table 1 their means in each campaign. The critical depth clusters between 0.30 and 0.50 on all four CCDs. Its mean over the core tracks is the same to within a hundredth for the two particle populations, the four spacecraft rolls, and the 8 and 15 s exposures, and every campaign with more than a handful of core tracks lies between 0.39 and 0.42; the four CCDs differ from one another by up to 0.03, FUV1 lowest and FUV2 highest. The back-surface width differs more between chips: FUV2 and SJI cluster at 5.5 to 6 μm, NUV at 4.5 to 5.5 μm, and FUV1 at 3.5 to 5 μm, so the pooled scatter is mostly this chip-to-chip difference, which is expected from the resistivity of the wafer and the applied bias and is one reason a sensor's kernel is worth measuring rather than assuming.

3.4 The same-pixel probability

For the noise statistics of a UV image the quantity that matters is not the width but the probability that two electrons deposited at the same depth are collected in the same pixel. For a slice with pixel fractions f_j that is $p = \sum_j f_j^2$, corrected for the read-noise contribution $\sum_j \epsilon_j^2$, and it can be read directly off each slice with no model of the shape of the kernel. Since the kernel is separable, the probability that two electrons are collected in the same pixel is $\mathcal{P} = p^2$. Figure 7 shows p against depth on each CCD, and Table 2 lists it for the slices within $D/10$ of the back surface. On the SJI CCD the back surface gives $\mathcal{P} = 0.339 \pm 0.004$, FUV2 the same to within the errors, while NUV and, further still, FUV1 spread their charge over fewer pixels, in the same order as their back-surface widths.

The figure gives p two ways in each depth bin, as the plain mean over the slices and as the median of how far each slice falls short of its own sharp-cloud value. The two agree at the back surface and part beyond t_c, where the mean settles near 0.88 on all four CCDs while the median stays within 0.02 of the 0.95 of a perfectly sharp cloud, with the per-track fits between them. The difference is charge that does not belong to the track. A frame taken inside the anomaly is crowded with hits, and one that touches a track is joined to it by the finder and cut out with it. Such charge

lands only off the peak of a slice, so it can only lower p, and it pulls the mean down in proportion to its charge while the median is unmoved until it reaches half the slices. Two checks say this is what the mean is seeing: it falls twice as far below the sharp-cloud value in campaigns exposed for 15 s as in those exposed for 8 s, at the same charge per slice, and it does not fall at all for the few tracks of cosmic rays recorded outside the anomaly, where the frames are empty. At the back surface the cloud already spans several columns and the same stray charge moves p by under 0.02, so Table 2 keeps the mean, which is the quantity that enters the variance of an image. Beyond t_c the median is the measurement, and it leaves the depletion region a few hundredths short of sharp, in the direction and about the size of the σ_d the fits carry.

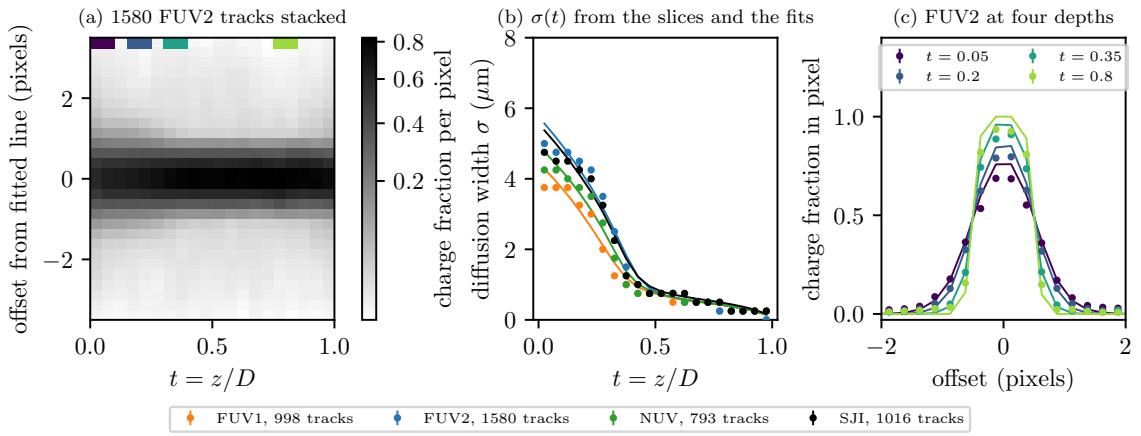


Fig 4 (a) Every slice of every flat track on the FUV2 CCD placed at its fractional depth and aligned on its fitted centerline, with each pixel's charge spread uniformly over its own width. The wedge is subtle: the wings at $t < 0.4$ carry a few percent of the charge. (b) Points: the diffusion width in each depth bin on each CCD, found by summing the misfit of Equation 2 over every slice in the bin on a grid of trial widths and minimizing, with no parametric model of the depth dependence; the error bars span the widths within two of the minimum M. Lines: the average of the per-track fits of Equation 1 on the same CCD, which reproduces the model-free profile at every depth, including the floor of a few tenths of a micron beyond t_c that the σ_d term supplies. (c) The kernel itself, integrated over a pixel: the mean fraction of the charge of the FUV2 slices in four depth bins, each a tenth of the thickness wide and marked along the top of (a), collected in a pixel whose center lies at the given offset from the fitted centerline, with its standard error. Lines: the same mean of the fractions the per-track fits predict for the same pixels. The measured peak falls a few hundredths below the fits at every depth, including the depletion region where the fits are pixel-sharp, and the measured wings stand above them: this is the stray charge that separates the mean from the median in Figure 7, which the robust misfit ignores and the mean includes.

4 Discussion

At the back surface, where UV photons are absorbed, the tracks give the kernel directly: on SJI the field-free layer is 0.40 of the thickness, the cloud is 5.5 μm wide, and the same-pixel probability is 0.339 ± 0.004, the quantity that enters the variance of a UV image. Deeper in the sensor the tracks see a spread of under a micron, σ_d, that a purely field-free kernel would neglect. It is about what the drift-time diffusion across the depleted thickness plus the micron-scale ionization column of the track itself should give, and it is small compared with the back-surface spread, so it has little

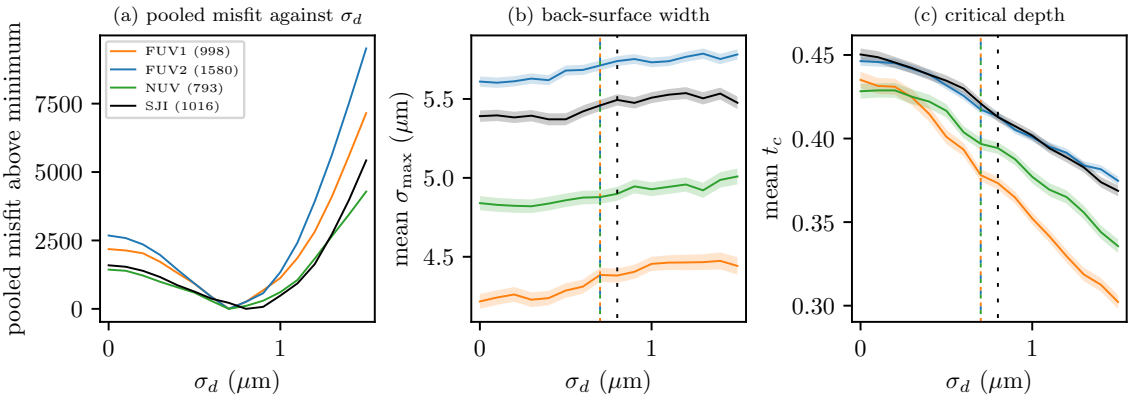


Fig 5 The spread inside the depletion region, σ_d in Equation 1, which is shared by every track on a CCD. (a) The misfit of Equation 2 summed over the flat tracks on each CCD at each σ_d, with each track refit in t_c, $\sigma_{\max}$, orientation, and centerline, shown relative to its minimum, which lies at $\sigma_d = 0.70$ μm on FUV1, 0.70 μm on FUV2, 0.70 μm on NUV, and 0.80 μm on SJI. (b) and (c) The mean $\sigma_{\max}$ and t_c of the same tracks at each σ_d, with the standard error of the mean shaded and the minimum of (a) on each CCD dashed: the spread inside the depletion region trades against the field-free wedge, so t_c falls steadily as σ_d rises, fastest on FUV1, while $\sigma_{\max}$ rises by a few percent.

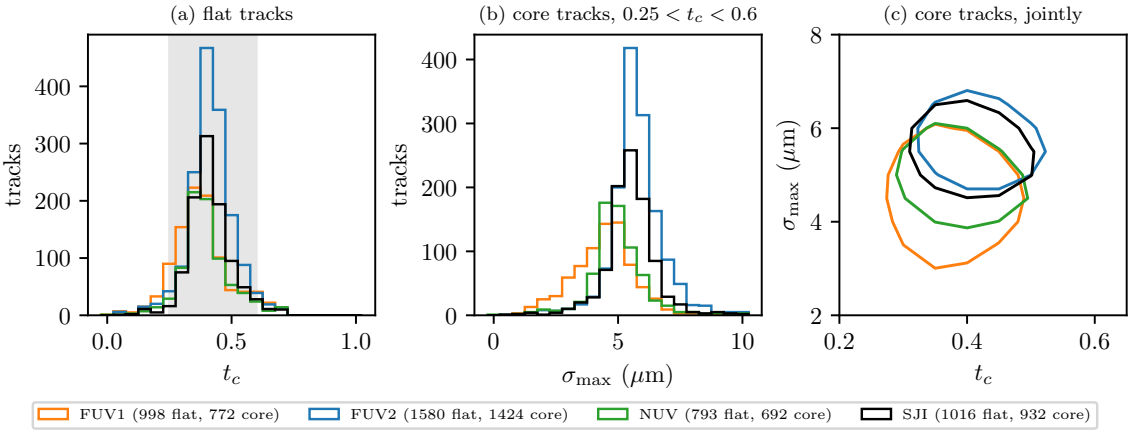


Fig 6 The fitted parameters. (a) The critical depth of every flat track on each CCD. The tails below 0.25 and above 0.6 are tracks Equation 1 describes poorly, sharp or diffuse along most of their length, and they are excluded from the shaded core used elsewhere. (b) The back-surface width of the core tracks. (c) The joint distribution of the core tracks on each CCD, as the contour enclosing half of them, on the grid the fits are searched over (0.05 in t_c and 0.5 μm in $\sigma_{\max}$) smoothed by one cell. The contours run wider in $\sigma_{\max}$ than in t_c, since a single track constrains its back-surface width less well than the depth at which it sharpens.

effect on UV imaging. It cannot be left out of the fit, though: without it the wedge stretches to absorb the floor and t_c comes out too large, which is why Equation 1 carries it from the start.

The FUV1 CCD is narrower than the other three, at the back surface and in the fitted $\sigma_{\max}$ alike, and NUV sits between it and the FUV2 and SJI pair. We have no explanation beyond chip-to-chip variation of the field-free thickness, which depends on the resistivity of the wafer and the applied bias; the same measurement on the Atmospheric Imaging Assembly (AIA) CCDs, which are of

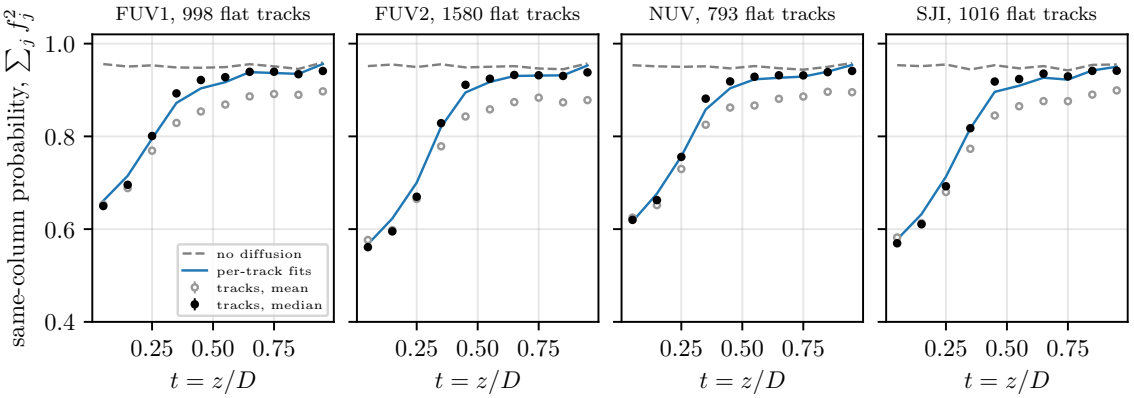


Fig 7 The probability that two electrons deposited in the same slice are collected in the same column, over the flat tracks on each CCD in bins of fractional depth. The dashed line is Equation 1 with no charge diffusion, averaged over the slices of each bin, which is below one only where a centerline runs near a pixel boundary, and the blue line is the same at each track's own fit. The black points are the dashed line minus the median, over the slices of the bin, of how far each slice falls short of its own no-diffusion value, with the standard error of the median; the open points are the plain mean of the slices. Charge from other hits in the frame that touched a track and was cut out with it can only lower the sum, so the mean sits below the median wherever such charge is common, beyond t_c on all four CCDs, while at the back surface the two agree. The back surface is lowest on the FUV2 and SJI CCDs and highest on FUV1.

Table 2 The charge diffusion measured from glancing particle tracks on the IRIS CCDs. The number of frames is the number of level-1 images searched, with the number taken inside SAA in parentheses. The flat tracks are those which constrain t_c to within 0.15 and show no Bragg rise along their length. σ_d is the spread inside the depletion region fit to each CCD as a whole, and t_c and $\sigma_{\max}$ are the medians of the per-track fits at that σ_d, with the interquartile range in brackets. p is the probability that two electrons deposited within $D/10$ of the back surface are collected in the same column, and $\mathcal{P} = p^2$ is the corresponding probability that they are collected in the same pixel.

CCD	frames	tracks	flat	σ_d (μm)	t_c	$\sigma_{\max}$ (μm)	p	$\mathcal{P}$
FUV1	8471 (2755)	2369	998	0.70	0.35 [0.30, 0.45]	4.5 [3.5, 5.0]	0.653 ± 0.004	0.426 ± 0.006
FUV2	8471 (2755)	3360	1580	0.70	0.40 [0.35, 0.45]	5.5 [5.0, 6.1]	0.576 ± 0.003	0.332 ± 0.003
NUV	8474 (2755)	2036	793	0.70	0.40 [0.35, 0.45]	5.0 [4.5, 5.5]	0.625 ± 0.004	0.390 ± 0.005
SJI	6847 (2750)	2279	1016	0.80	0.40 [0.35, 0.45]	5.5 [5.0, 6.0]	0.582 ± 0.003	0.339 ± 0.004

the same design, would show whether such variation is common.

Figure 8 turns the kernel into the quantity an instrument designer needs, the width of the charge cloud of a photon, by averaging it over the depth at which a photon of each wavelength is absorbed in silicon. Across the whole ultraviolet the absorption length is under 10 nm, so every photon sees the back-surface width of Table 2, a charge cloud 4 to 5 μm wide depending on the CCD. Only in the visible and the soft X-rays, where photons reach the depletion region, does the cloud narrow, and near the band gap the sensor is nearly transparent. The spread between the four CCDs is the spread a designer should expect between sensors of one design.

Three systematics deserve comment. First, the tracks are left by protons that deposit thousands of electrons per row, whereas a UV photon liberates one to three. In thick, fully depleted sensors, slow protons leave tracks visibly widened by the mutual repulsion of that charge,[11] and if the same occurred in the field-free layer here the tracks would overestimate the single-photon kernel. The tracks themselves can test this, by comparing the kernels of faint and bright tracks, and that

comparison is the first thing to add. Second, the depth assignment assumes every flat track crosses the full thickness of the sensor; a track that entered or left through the edge of the field-free layer, or that was clipped by the Bragg cut at one end, would have its depths compressed. The agreement between the model-free and parametric profiles, and between campaigns, suggests such tracks are rare, but a validation on synthetic tracks injected into real frames is the way to bound it. Third, the stray charge of Section 3, from hits that touch a track and are cut out with it, is kept out of the same-pixel profile by the median but not out of the widths of Figure 4 nor the σ_d scan, both of which sum the misfit of every slice. It is one-sided and it grows with the exposure of the frame, so the floor of a few tenths of a micron beyond t_c and the σ_d of Equation 1 are upper bounds on the true spread, and the campaigns exposed for 15 s, a tenth of the flat tracks, carry twice their share of it.

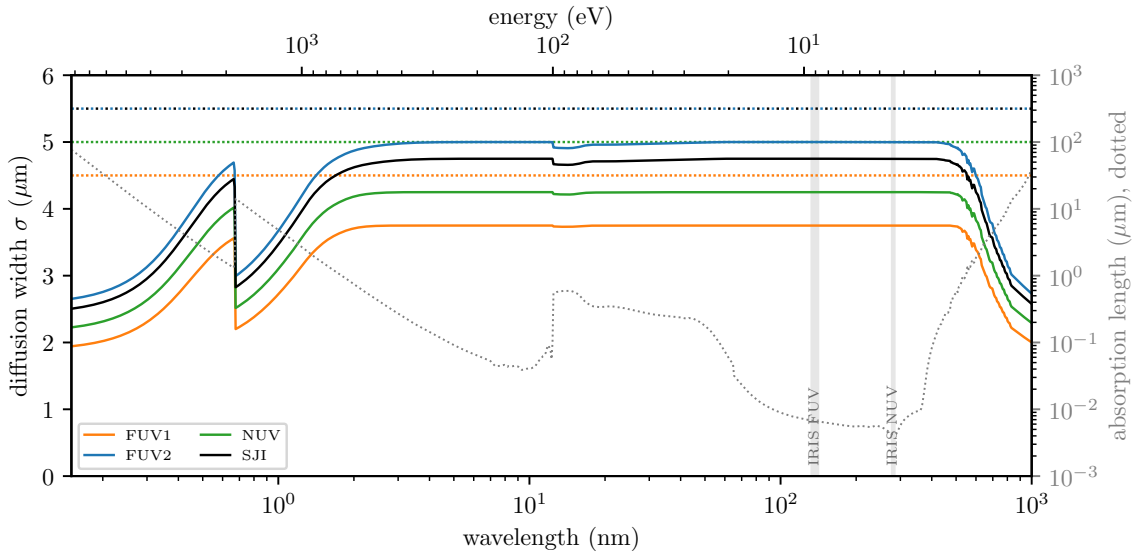


Fig 8 The width of the charge cloud a photon leaves in each CCD, against the wavelength of the photon. The width is the standard deviation of the cloud. A photon is absorbed at a random depth, exponentially distributed with the absorption length of silicon (gray dotted, from the tabulated optical constants[12,13]), and the charge it liberates spreads by the width measured at that depth in Figure 4b, held at the value of its first bin up to the back surface. The curve is the standard deviation of the cloud that results, which is the square root of the squared width averaged over depth, weighted by the fraction of photons absorbed at each depth; photons that pass through the 14 μm of silicon are not counted. Between 30 and 350 nm every photon is absorbed within 10 nm of the back surface, and the cloud has the width of the first depth bin of Figure 4b. This lies below the median $\sigma_{\max}$ of the per-track fits in Table 2 (dotted in the color of each CCD), as the points of Figure 4b lie below the fits near the back surface. In the visible and in the soft X-rays the photons penetrate to the depletion region and the cloud narrows. Near the band gap silicon is nearly transparent, and the few photons absorbed are spread through its thickness.

5 Conclusion

Glancing particle tracks measure the depth-dependent charge-diffusion kernel of a back-illuminated CCD in orbit, with no laboratory access. On the IRIS CCDs they give the thickness of the field-free layer, the width of the charge cloud at the back surface, which differs by a fifth between four

sensors of one design, and a spread of under a micron acquired inside the depletion region that a field-free kernel alone would miss.

Appendix A: The tracks as the finder cut them out

The cutouts distributed with this article are seven pixels wide and follow the fitted centerline, so they show the tracks after the finder has done its work. Figure 9 shows the longest of the flat tracks and one track that the Bragg cut rejects. Sub-pixel structure is invisible at this scale, which is why the wedge signature has to be extracted by fitting.

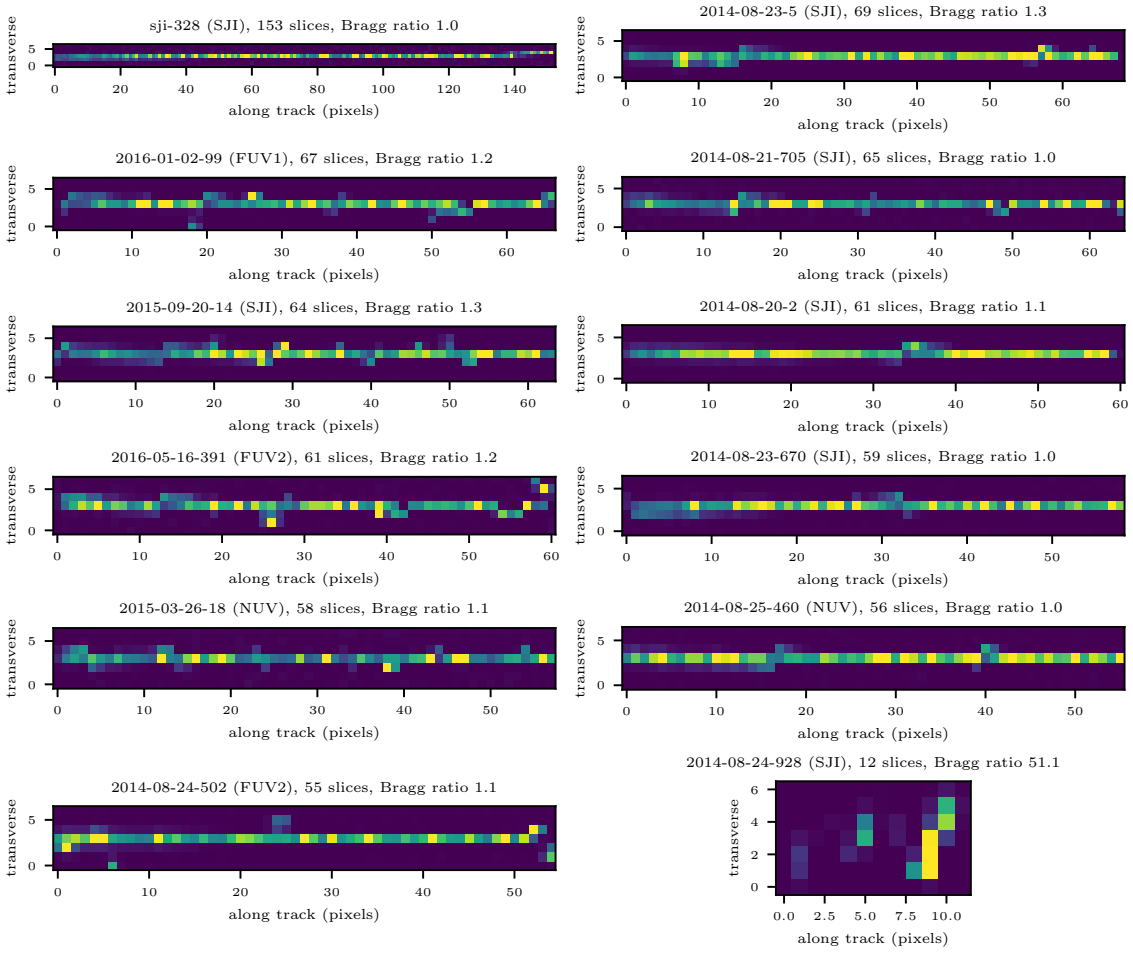


Fig 9 The eleven longest flat tracks and, in the last panel, the track with the strongest Bragg rise, as the finder cut them out: seven pixels wide, re-centered on the integer part of the fitted centerline in every slice, and oriented with the back surface at the left. Each panel's color scale is clipped at its 98th percentile so that a single bright pixel does not hide the track. The stopping track's charge per slice climbs by a factor of twenty toward one end, and the selection of Section 2 rejects it.

Disclosures

The authors declare that there are no financial interests, commercial affiliations, or other potential conflicts of interest that could have influenced the objectivity of this research or the writing of this paper. The analysis code and the text of this article were drafted with the assistance of Claude, a large language model, under the direction and review of the authors.

Code, Data, and Materials Availability

This article is executable:[14] every figure, table, and number quoted in the text is computed from the tracks when the article is built. The code, the track cutouts, and the fits are available at https://github.com/roytsmart/ccd-diffusion-paper, and the level-1 IRIS images they were extracted from are available from the IRIS data archive.

Acknowledgments

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THE NOISE ON ULTRAVIOLET ASTRONOMICAL IMAGES CAPTURED BY BACK-ILLUMINATED SILICON SENSORS

Contribution of Authors and Co-Authors

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Author: Roy T. Smart
Contributions: **XX**

Co-Author: Charles C. Kankelborg
Contributions: **XX**

Co-Author: Jacob D. Parker
Contributions: **XX**

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The Noise on Ultraviolet Astronomical Images Captured by Back-illuminated Silicon Sensors

Roy T. Smart [1], Charles C. Kankelborg [1], and Jacob D. Parker [1]

[1] *Montana State University, Department of Physics, P.O. Box 173840, Bozeman, MT 59717, USA*

ABSTRACT

Ultraviolet (UV) astronomy currently relies on back-illuminated silicon imaging sensors. While the noise in these sensors is typically assumed to be dominated by photon shot noise, measurements from HST's Wide Field Camera 3 (WFC3) and the Interface Region Imaging Spectrograph (IRIS) reveal a significant discrepancy: the noise measured in the ultraviolet (UV) is systematically lower than theoretical predictions. We propose that this discrepancy is caused by partial charge collection (PCC), whereby a fraction of photogenerated electron-hole pairs recombine before they can be measured. We present a simple theoretical model, valid for wavelengths from 1 Å to 10 000 Å, that incorporates the effects of PCC and shows better agreement with the noise measurements from both WFC3 and IRIS, resolving the previously unexplained discrepancy. This finding implies that the signal-to-noise ratio achievable with these sensors is higher in the UV than previously understood, potentially impacting both future instrument proposals and the uncertainties of our current UV imagery. At about 2000 Å to 3500 Å, PCC is the dominant noise source in the high-signal limit.

Keywords: Astronomical instrumentation (799); Astronomical detectors (84); Ultraviolet astronomy (1736)

1. INTRODUCTION

Back-illuminated, silicon-based image sensors such as charge-coupled devices (CCDs) and complementary metal–oxide–semiconductor (CMOS) sensors are ubiquitous in ultraviolet (UV) astronomy, and are currently used in many UV instruments such as the Atmospheric Imaging Assembly (AIA) (J. R. Lemen et al. 2012), the Interface Region Imaging Spectrograph (IRIS) (B. De Pontieu et al. 2014), and HST's Wide Field Camera 3 (WFC3) (R. A. Kimble et al. 2008). Predicting the noise measured by these sensors is important both for quantifying the uncertainty on current astronomical observations as well as for evaluating the performance of proposed future instruments.

In the high-signal limit (that is, neglecting readout noise), the largest source of noise measured by these sensors is thought to be shot noise (R. A. Stern et al. 1986). This noise is due to the quantized nature of light (W. Schottky 1918) and is not a property of the sensor *per se*, but its magnitude depends on the number of measured photons, which is determined by the sensor's geometry and optical constants. Another well-known source of noise inherent to silicon sensors is Fano noise (U. Fano 1947), the random variation of the number of electrons produced per interacting photon. This noise source is small (D. Rodrigues et al. 2023), and often ignored entirely, but it can be combined with the shot noise to form the traditional estimate of the total noise (R. A. Stern et al. 1986; J. R. Janesick 2001).

This traditional model of shot noise and Fano noise is very accurate in the visible (M. Marinelli & J. Green 2024) and X-ray (S. Ishikawa et al. 2011; P. S. Athiray et al. 2020; B. D. Schwab et al. 2020) wavelength regimes. However, this model appears to be less accurate in the UV. For example, J. P. Wülser et al. (2018) showed that IRIS measured *less* noise in the far ultraviolet (FUV) channel than the traditional model would suggest and similar results were found in the near ultraviolet (NUV) by WFC3. T. Borders et al. (2010) measured the quantum yield of the WFC3 CCDs from narrowband flat fields between 208 nm and 400 nm and found it to be about 30% smaller than the theoretical prediction, which falls from 1.7 electrons per interacting photon at 200 nm to unity at 340 nm. The WFC3 instrument handbook, which summarizes this measurement, still describes its cause as unclear (M. Marinelli & J. Green 2024). This discrepancy indicates that the traditional model is incomplete, and that a more detailed model is needed to predict the noise measured by these sensors in the UV.

Corresponding author: Roy T. Smart
Email: roytsmart@gmail.com

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In this work, we aim to resolve the UV noise discrepancy by developing a noise model which adds two additional effects to the traditional model. The first effect is partial charge collection (PCC), which occurs when electron-hole pairs recombine before they can be measured by the sensor (R. A. Stern et al. 1994; J. R. Janesick 2001). Noise due to PCC is a well-known effect, for example D. Rodrigues et al. (2023) accounted for PCC in their measurement of the Fano noise in soft X-ray (SXR), but its influence on the noise measured in UV has been rarely modeled in detail. The second effect is charge diffusion (or charge spreading), the tendency for photoelectrons to migrate into neighboring pixels (J. R. Janesick 2001), a mechanism suggested by both J. P. Wülser et al. (2018) and T. Borders et al. (2010) to explain the discrepancies they measured. Flat fields are already known to depart from Poisson statistics in the visible, where the variance is suppressed and reappears as a covariance between neighboring pixels, an effect attributed to the electrostatic repulsion of charge already collected and known as the brighter-fatter effect (A. Guyonnet et al. 2015; P. Astier et al. 2019). That mechanism is distinct from the one considered here. It distorts the pixel boundaries by an amount which grows with the accumulated signal, whereas charge diffusion displaces each electron independently of every other, and independently displacing the points of a Poisson process yields another Poisson process. Charge diffusion by itself therefore cannot alter the variance of a flat field at all. It becomes a noise term only when a single photon liberates several electrons which would otherwise be collected together, so it switches on with the quantum yield in the UV rather than with the signal level. The closest treatment of this combination that we are aware of is J. J. Givans et al. (2022), who extended a flat-field correlation formalism to include both quantum yield and charge diffusion for the infrared detectors of the Nancy Grace Roman Space Telescope. We will model the contribution of each of these effects to the total noise and compare our predicted noise to the noise measured by IRIS and WFC3. We will also provide noise predictions for AIA and the Multi-slit Solar Explorer (MUSE) (B. De Pontieu et al. 2020) which can be used for estimating the uncertainty of observations captured by these instruments. Finally, we have made available a reference implementation of our noise model in Python which is available to be installed from the Python Package Index (PyPI).

2. SENSOR MODEL

In this work, we will use the back-illuminated CCD model described in R. A. Stern et al. (1994) as a basis for our sensor model to make it directly comparable with previous works in the literature. Figure 1 is a schematic of our sensor model and it shows the light-sensitive region of the sensor to be an epitaxial silicon layer with a thickness D. A fraction of the incident photons are either absorbed in the oxide layer (thickness δ) or reflected at the first or second interface. The illuminated side of the epitaxial layer is considered to have a PCC region of thickness W, where electron-hole pairs can recombine before being measured, and there is a depletion region of thickness z_d on the non-illuminated side.

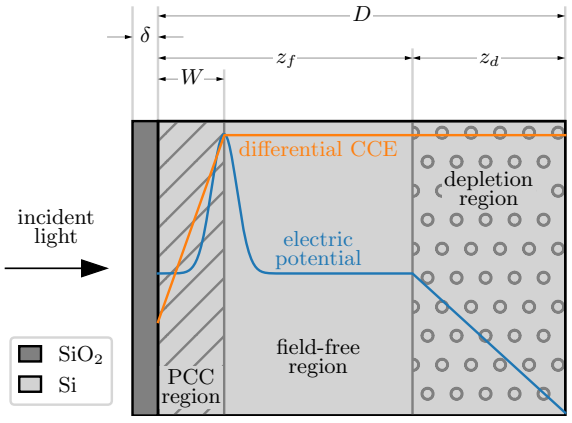


Figure 1. A schematic (not to scale) of the back-illuminated sensor model used in this work, labeled with the thicknesses of each layer. Overplotted is a *qualitative* example of the electric potential within the sensor that motivates the differential CCE.

In Section 2.1 we will calculate the signal predicted by this sensor model using the method described in R. A. Stern et al. (1994), adapted to more recent measurements. In Section 2.2 we will calculate the noise predicted by this sensor model in a self-consistent manner, and in Section 2.3 we will model the charge diffusion of this sensor which is needed for a complete description of the noise. Throughout this section we will adopt the convention that unprimed variables represent quantities incident on the sensor, primed variables represent quantities internal to the sensor, and the double-primed variables represent quantities measured by the sensor.

2.1. *Signal*

The average number of photoelectrons measured per incident photon is known as the quantum efficiency (QE). This is a common performance metric for measuring sensor sensitivity and it is given in J. R. Janesick (2001) as

$$\mathrm{QE}(\lambda, T) = \left\langle \frac{N_e''}{N_\gamma} \right\rangle = A(\lambda)\,\overline{n}(\lambda, T)\,\overline{\eta}(\lambda), \qquad (1)$$

Table 1. The sensor model parameters which best fit the measurements in P. Boerner et al. (2012) and J. Heymes et al. (2020).

Source	implementation	η_0	W (nm)	δ (nm)	D (μm)	substrate roughness (nm)
P. Boerner et al. (2012)	`e2v_ccd203`	0.452	335	1.4	16	1.1
J. Heymes et al. (2020)	`e2v_ccd97`	0.334	153	4.6	14	4.8

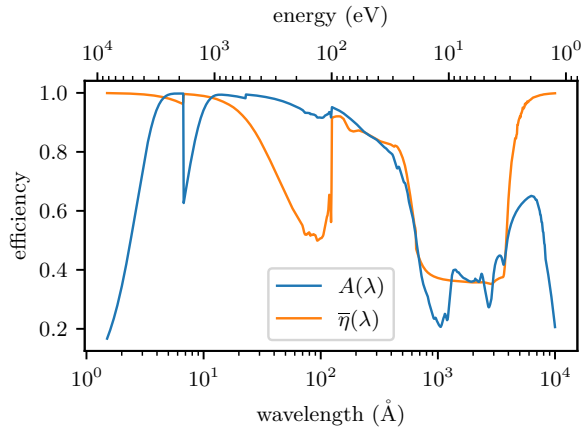


Figure 2. The fraction of incident light absorbed by the light-sensitive silicon layer and the CCE as a function of wavelength for the parameters inferred from the J. Heymes et al. (2020) measurements.

where λ is the wavelength of the incident photons in vacuum, T is the temperature of the sensor, $\langle X \rangle = \overline{X}$ denotes the expected value of a random variable X, N_e'' is the number of electrons measured by the sensor, N_γ is the total number of photons incident on the sensor, $A(\lambda)$ is the absorbance of the light-sensitive layer, $\overline{n}(\lambda, T)$ is the average quantum yield, the number of photoelectrons generated per absorbed photon, and $\overline{\eta}(\lambda)$ is the average charge-collection efficiency (CCE), the fraction of generated photoelectrons measured by the sensor.

The fraction of incident energy that is absorbed by the light sensitive layer is the absorbance, $A(\lambda)$, which is reduced by reflections at each interface, absorption in the oxide layer (UV), or by penetration through the device (SXR and infrared (IR)). $A(\lambda)$ can be determined from the optical constants of Si and SiO_2 using, for example, the popular IMD code (D. L. Windt 1998). For this work, we used our Python library, `optika` (R. T. Smart et al. 2026), which uses the transfer matrix method described in P. Yeh (1988) with the optical constants from E. D. Palik (1997), B. Henke et al. (1993), and L. V. R. de Marcos et al. (2016) to compute the electric field for every interface in the sensor. In Figure 2 we have plotted $A(\lambda)$ for the J. Heymes et al. (2020) parameters in Table 1.

In R. A. Stern et al. (1994), the authors assume no reflections from the unilluminated side of the sensor for simplicity. In this work, we compute the total change in Poynting flux into and out of the light-sensitive region of the sensor to determine $A(\lambda)$. This treatment introduces interference effects for infrared wavelengths, which can be seen on the right side of Figure 2.

Determining $\overline{n}(\lambda, T)$ over the entire wavelength range considered in this study is an area of ongoing research (G. W. Fraser et al. 1994; J. Geist 1996; F. Scholze et al. 1998; J. Fang et al. 2019, just to name a few). This work will use the quantum yield model developed by K. Ramanathan & N. Kurinsky (2020) which uses a phenomenological model of impact ionization to bridge the "UV gap", an area where direct measurements of the quantum yield are not yet available. They give the average quantum yield as

$$\overline{n}(E,T) = \begin{cases} \sum_{n=0}^{\infty} n p_n(E,T), & 0 < E \leq 50\,\mathrm{eV} \\ E/\epsilon_{eh}(T), & 50\,\mathrm{eV} < E < \infty, \end{cases} \tag{2}$$

where E is the energy of the incident photon, n is the actual quantum yield, $p_n(E,T)$ is a table of pair-creation probabilities which is provided in their supplementary material,

$$\epsilon_{eh}(T) = 1.7 E_g(T) + (0.084\,\mathrm{eV}^{-1})A + 1.3\,\mathrm{eV}, \tag{3}$$

is the asymptotic mean energy per electron-hole pair, $A = 5.2\,\mathrm{eV}^2$, and

$$E_g(T) = 1.1692\,\mathrm{eV} - \frac{(4.9 \times 10^{-4}\,\mathrm{eV\,K}^{-1})\,T^2}{T + 655\,\mathrm{K}}, \tag{4}$$

is the bandgap energy of silicon.

In R. A. Stern et al. (1994), the average CCE is expressed in terms of a differential CCE, $\eta(z)$, which is the fraction of photoelectrons collected for a photon absorbed at a depth z into the epitaxial layer. The average CCE is then the first moment of the differential CCE weighted by the probability of absorbing a photon at a depth z,

$$\overline{\eta}(\lambda) = \frac{\int_0^\infty \eta(z)\, e^{-\alpha(\lambda) z}\, dz}{\int_0^\infty e^{-\alpha(\lambda) z}\, dz}, \tag{5}$$

where $\alpha(\lambda)$ is the absorption coefficient of silicon for the given wavelength.

In principle, $\eta(z)$ is a function of the exact implant profile which is usually impractical to measure, but see

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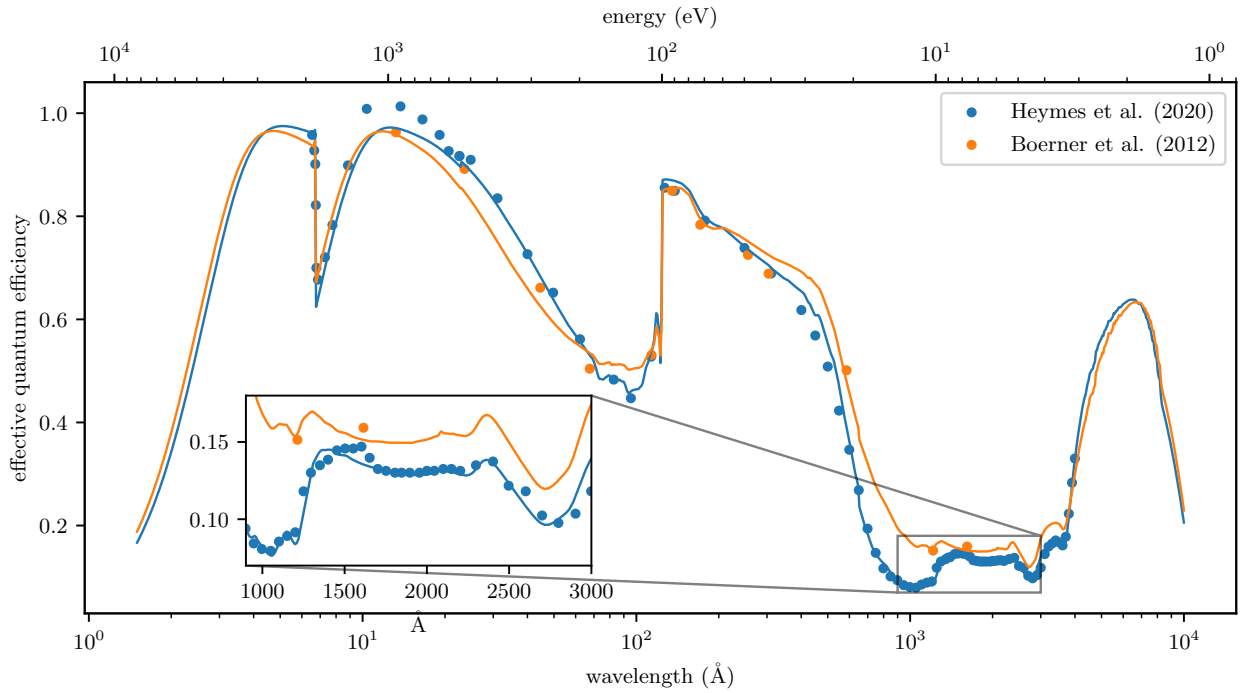


Figure 3. A comparison of EQE(λ) measured by P. Boerner et al. (2012) and J. Heymes et al. (2020), along with the best-fit models described in Table 1 plotted as lines with the same color as the data. The inset zooms into the ultraviolet to better visualize the difference between the measurement and model in this range.

R. A. Stern et al. (2004); P. Boerner et al. (2012) for a case where the authors did have a measurement of the implant profile provided by the manufacturer. In R. A. Stern et al. (1994), the authors instead adopt a piecewise-linear approximation of the differential CCE,

$$\eta(z) = \begin{cases} \eta_0 + (1-\eta_0)z/W, & 0 < z < W \\ 1, & W < z < \infty \end{cases} \tag{6}$$

where η_0 is the differential CCE at the back surface of the sensor. Plugging Equation 6 into Equation 5 yields an expression for the CCE,

$$\overline{\eta}(\lambda) = \eta_0 + \left(\frac{1-\eta_0}{\alpha W}\right)(1 - e^{-\alpha W}), \tag{7}$$

which can be used in Equation 1 to determine the QE. In Figure 2 we have plotted the average CCE for the J. Heymes et al. (2020) parameters in Table 1.

In R. A. Stern et al. (1994), the authors define an effective QE as

$$\mathrm{EQE}(\lambda) = A(\lambda)\,\overline{\eta}(\lambda), \tag{8}$$

which is found by taking the ratio of the current measured by the sensor to the current measured by a calibrated photodiode and is the quantity that is typically measured when calibrating an image sensor (R. A. Stern et al. 1994, 2004; P. Boerner et al. 2012). In Figure 3, we have plotted the measured, effective QE for two sources: P. Boerner et al. (2012) which measured the AIA CCDs at a few discrete wavelengths, and J. Heymes et al. (2020) which measured a Teledyne e2v CCD97 sensor over a wide wavelength range with high resolution using a monochromator.

Using the Nelder-Mead minimization algorithm (F. Gao & L. Han 2010), we found the free parameters of our model which best fit the data in P. Boerner et al. (2012) and J. Heymes et al. (2020). These models are plotted as solid lines in Figure 3, and the corresponding values of the free parameters are shown in Table 1. Throughout the remainder of this work we will use the model which best fits the J. Heymes et al. (2020) data.

2.2. *Noise*

Our noise model will consider three sources: shot noise from the quantized nature of the photons striking the sensor, Fano noise due to inherent randomness in the process which converts photons to electrons, and noise due to electrons randomly recombining before they can be measured by the sensor. This section will describe the statistics of each noise source and demonstrate a

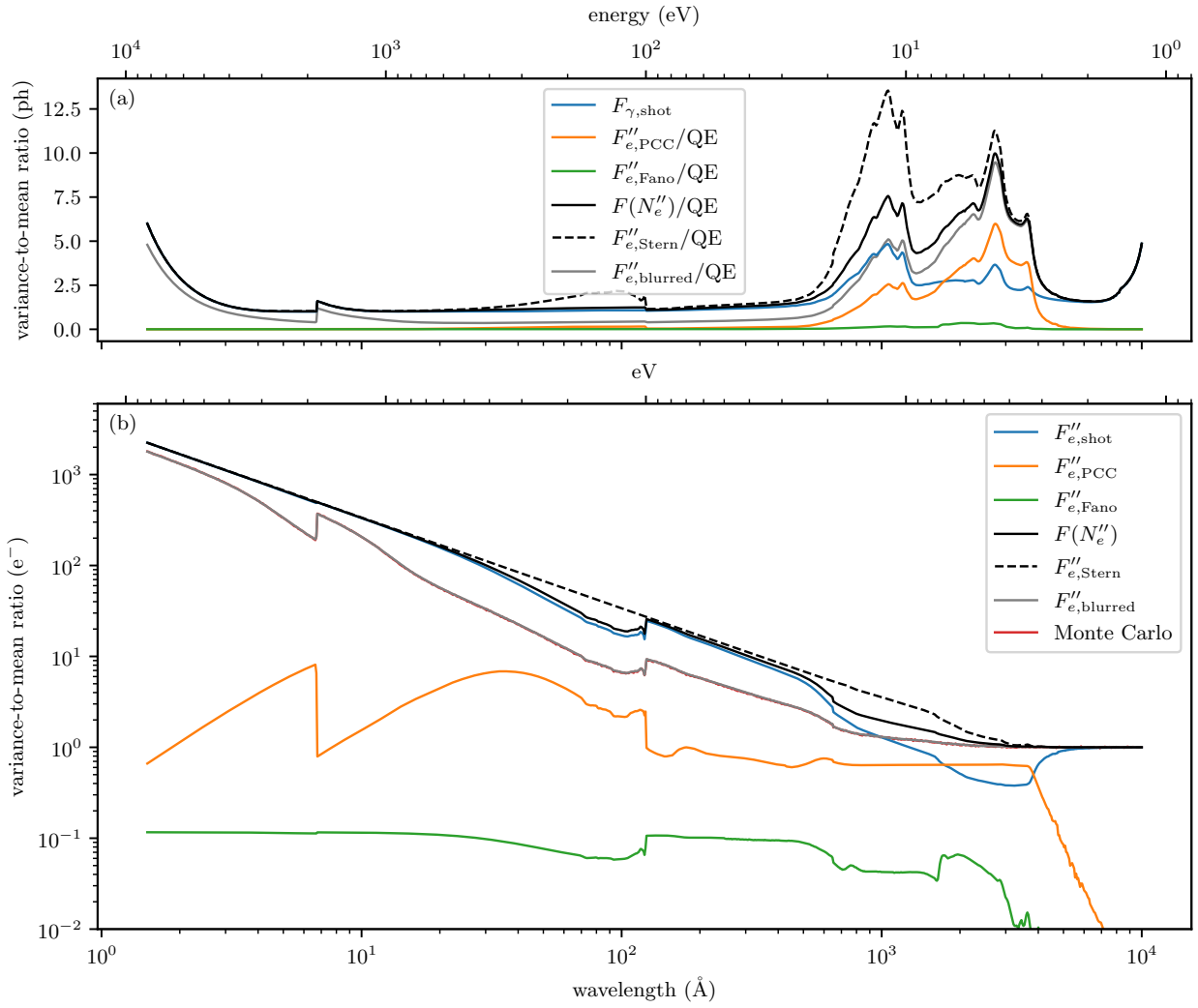


Figure 4. The total and component-wise VMR of our sensor model expressed in two different units: photons incident on the sensor (top) and electrons measured by the sensor (bottom). Plotted for comparison is the VMR of the R. A. Stern et al. (1986) noise model (dashed) and a Monte Carlo simulation of $F(N''_e)$ in red.

procedure which can simulate the noise measured by our model sensor for a given number of incident photons.

Throughout this work, we will measure noise in terms of a variance-to-mean ratio (VMR),

$$F(X) = \frac{\mathrm{Var}(X)}{\overline{X}}, \tag{9}$$

where $\mathrm{Var}(X)$ is the variance of X. Using the VMR instead of the signal-to-noise ratio (SNR) to express the noise is convenient since it is constant as a function of signal for the distributions studied here. For example, the VMR of a Poisson random variable is always unity since its variance and expectation value are equal. The VMR is not dimensionless (it has the same units as X), so we must take care to interpret the VMR in terms of the correct units. For a constant multiple of a random variable, aX,

$$F(aX) = aF(X). \tag{10}$$

We will use this relationship throughout this work to convert from incident photons to measured electrons and vice versa.

In Figure 4 we have plotted the VMR for the noise sources considered in this study in two different units: number of incident photons and number of measured electrons. Figure 4a is useful when *designing* an instrument since the number of photons incident on the sensor can be determined by the radiance of the source and the effective area of the instrument. Figure 4b is useful

when *calibrating* an instrument since we directly measure these electrons. The details of the calculations for Figure 4 are presented in the subsections below.

2.2.1. *Shot Noise*

Photon shot noise is often assumed to be the leading noise contributor in UV astronomy (R. A. Stern et al. 1986; J. R. Lemen et al. 2012; B. De Pontieu et al. 2014). If the expected number of photons that interact with the light-sensitive layer is

$$\overline{N}'_\gamma = A\overline{N}_\gamma. \tag{11}$$

then the actual number of interacting photons, N'_γ, is sampled from the Poisson distribution,

$$N'_\gamma \leftarrow \text{Poisson}\big(\overline{N}'_\gamma\big). \tag{12}$$

As mentioned in the previous section, the VMR of this process is

$$F(N'_\gamma) = 1, \tag{13}$$

but this in terms of the number of absorbed photons, which is internal to the sensor. In terms of the expected number of incident photons given the number of absorbed photons the VMR is

$$F_{\gamma,\text{shot}} \equiv F\left(\frac{N'_\gamma}{A}\right) = \frac{1}{A} \tag{14}$$

since A is the conversion factor between incident photons and absorbed photons. $F_{\gamma,\text{shot}}$ is plotted in Figure 4a in blue and demonstrates that this expression is nearly unity across almost the entire wavelength range of the sensor except in the UV, where the absorbance is poor (Figure 2). We can also express the VMR of the shot noise in terms of the number of measured electrons by using the QE as the conversion factor between the unprimed and double-primed variables,

$$F''_{e,\text{shot}} = \overline{n}\,\overline{\eta}. \tag{15}$$

This equation is plotted in Figure 4b in blue, where we can see that it increases linearly with decreasing wavelength since the average quantum yield is increasing.

2.2.2. *Fano Noise*

The energy resolution of silicon detectors is ultimately limited due to Fano noise (U. Fano 1947), the random variation of the quantum yield n. Fano noise is usually expressed in terms of the Fano factor, $\mathcal{F} = F(n)$. Silicon is commonly accepted to have $\mathcal{F} \approx 0.1$ (J. R. Janesick 2001). In part due to variations of the Fano noise as a function of wavelength and temperature (G. W. Fraser et al. 1994), there is some disagreement in the literature around a more precise value for $\mathcal{F}$ (G. W. Fraser et al. 1994; B. Lowe 1997; M. N. Mazziotta 2008; I. V. Kotov et al. 2018; D. Rodrigues et al. 2021, 2023, & references therein). However, because the Fano noise is so small compared to the other noise sources considered in this study, the precise form of this noise model does not appreciably influence our conclusions.

As we mentioned in Section 2.1, this work uses the quantum yield model of K. Ramanathan & N. Kurinsky (2020). They provide the probability mass function (PMF) of the quantum yield distribution, $p_n(E,T)$, which we can use directly to generate random samples of the quantum yield for $E \leq 50\,\text{eV}$. For $E > 50\,\text{eV}$, we will assume that the quantum yield is a Gaussian distribution centered around $\overline{n}(\lambda,T)$, with standard deviation $\sqrt{\overline{n}(\lambda,T)\,\mathcal{F}(T)}$, where $\mathcal{F}(T)$ is the asymptotic Fano factor given by K. Ramanathan & N. Kurinsky (2020) as

$$\mathcal{F}(T) = (-0.028\,\text{eV}^{-1})\,E_g(T) + (0.0015\,\text{eV}^{-2})\,A + 0.14. \tag{16}$$

At 300 K, this corresponds to a Fano factor of approximately 0.114, which is generally consistent with the best-available measurement of the Fano factor by D. Rodrigues et al. (2021) at 6 keV.

$\mathcal{F}$ is a quantity internal to the sensor. The measured Fano factor is instead

$$F''_{e,\text{Fano}} = \overline{\eta}\mathcal{F} \tag{17}$$

since the conversion factor between the generated electrons and the measured electrons is the CCE. In Figure 4b we have plotted $F''_{e,\text{Fano}}$ in green. This shows that the Fano noise is the smallest contributor to the total noise across the entire wavelength range and is often at least an order of magnitude smaller than the other noise sources.

2.2.3. *Partial Charge Collection Noise*

In Figure 5, we have plotted the penetration depth in silicon vs. the thickness of the PCC region. We can see that there are two regions in the UV where the penetration depth is less than the thickness of the PCC region. In these regions, there is a significant chance that some of the electron-hole pairs will randomly recombine before being measured by the sensor. The stochastic nature of this process leads us to consider this as a new noise source, PCC noise.

In the R. A. Stern et al. (1994) charge-collection model, each photoelectron generated in the PCC region has a probability $\eta(z)$ of *not* recombining and subsequently being measured by the sensor. If n'_i is the quantum yield of the ith photon generated using the K. Ramanathan & N. Kurinsky (2020) model discussed in the

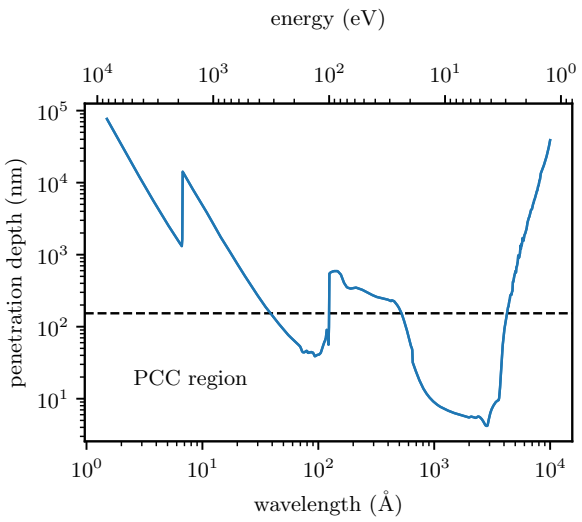


Figure 5. The penetration depth in silicon as a function of wavelength plotted against the depth of the PCC region.

previous section, we can express the *measured* quantum yield as

$$n_i'' \leftarrow \text{Binomial}\big(n_i', \eta(z_i)\big), \tag{18}$$

where z_i is the penetration depth of the ith photon and $\text{Binomial}(n, p)$ is a sample from the binomial distribution with n trials and probability of success p. We can use 18 to compute the total number of measured electrons given N_γ' absorbed photons as

$$N_e'' = \sum_{i=0}^{N_\gamma'} n_i''. \tag{19}$$

This expression is intended to be used in an instrument forward model to sample the distribution of measured electrons given an expected number of incident photons.

At first glance the VMR of this process looks intractable, since Equation 19 is a sum of binomial draws in which both the number of trials and the probability of success vary from one term to the next. It is made tractable by the observation made at the start of this section, that the VMR is constant as a function of signal, which leaves us free to evaluate it at whichever illumination is most convenient. The most convenient is the faintest. Consider an exposure so brief that the entire pixel array receives a single photon with probability ε and no photon at all otherwise. Let X be the number of electrons collected by a given pixel, and X_1 the number that same pixel collects on those occasions when the photon is present. The empty exposures contribute nothing to either average, so

$$\overline{X} = \varepsilon\,\overline{X_1}, \qquad \overline{X^2} = \varepsilon\,\overline{X_1^2}, \tag{20}$$

and the VMR of the exposure is

$$F(X) = \frac{\overline{X^2} - \overline{X}^2}{\overline{X}} = \frac{\overline{X_1^2}}{\overline{X_1}} - \varepsilon\,\overline{X_1}. \tag{21}$$

As the exposure is made fainter the second term vanishes, and the VMR is fixed entirely by the first two moments of what a single photon delivers to a single pixel. Since the VMR does not depend on the illumination, this limit is the answer at any illumination.

In this subsection charge diffusion is neglected, so the electrons liberated by that photon are all collected by the pixel it struck, and X_1 is simply the measured quantum yield n''. Both of its moments follow from Equation 18. For a photon which liberates n' electrons at a depth z, the measured yield n'' is binomial, with mean $n'\eta(z)$ and variance $n'\eta(z)\big(1-\eta(z)\big)$, so its second moment is

$$\overline{(n'')^2}\big|_{n',z} = n'\eta(1-\eta) + \big(n'\eta\big)^2. \tag{22}$$

Averaging over the absorption depth and the quantum yield, which are independent at a given wavelength, and using $\text{Var}(n') = \mathcal{F}\overline{n}$ to evaluate $\overline{(n')^2}$, gives

$$\overline{(n'')^2} = \overline{n}\,\overline{\eta} + \overline{n}\big(\overline{n} + \mathcal{F} - 1\big)\overline{\eta^2}. \tag{23}$$

The mean is simply $\overline{n''} = \overline{n}\,\overline{\eta}$, so dividing Equation 23 by it gives the VMR of the electrons measured by the sensor,

$$F_e'' = 1 + \big(\overline{n} + \mathcal{F} - 1\big)\frac{\overline{\eta^2}}{\overline{\eta}}. \tag{24}$$

The two terms of Equation 23 are worth distinguishing, since they respond differently to charge diffusion in Section 2.3. Each factor of η is the probability that one electron survives recombination, so the number of factors counts the electrons a term describes. The first term carries a single factor and is the contribution of the electrons taken one at a time; it becomes the leading unity of Equation 24, which is why the VMR measured in electrons can never fall below one. The second term carries two factors, through $\overline{\eta^2}$, and describes the electrons two at a time. It is nonzero only because a single photon can liberate more than one electron, and it is the entire excess above the Poisson floor.

It is conventional to express this result in terms of the VMR of the CCE rather than its second moment. Taking the second moment of Equation 6 in a similar fashion to Equation 5 gives

$$F(\eta) = \frac{2e^{-\alpha W}}{\overline{\eta}}\left(\frac{1-\eta_0}{\alpha W}\right)^2\big(\sinh(\alpha W) - \alpha W\big), \tag{25}$$

and rearranging the definition of the VMR shows that the ratio appearing in Equation 24 is simply $\overline{\eta} + F(\eta)$.

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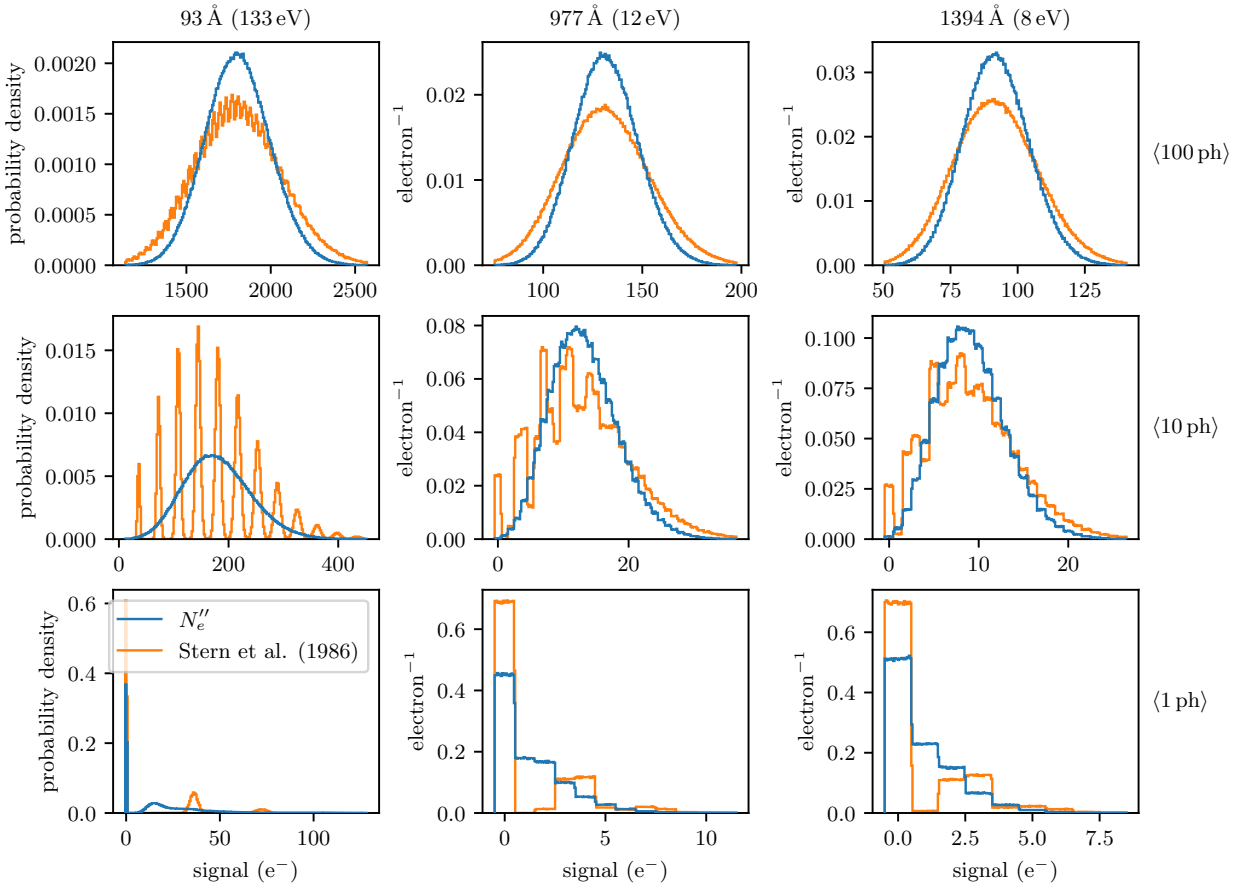


Figure 6. The probability distribution of the number of measured electrons for a given wavelength and expected number of *absorbed* photons calculated using 1000000 samples of Equation 19 and the R. A. Stern et al. (1986) noise model.

Substituting this and setting aside the shot noise of Equation 15 leaves the combined Fano and PCC contribution,

$$F''_{e,\text{sensor}} = F''_e - F''_{e,\text{shot}} = 1 + (\mathcal{F}-1)\overline{\eta} + \left(\overline{n} + \mathcal{F} - 1\right) F(\eta). \tag{26}$$

The contribution from only PCC noise is then

$$F''_{e,\text{PCC}} = F''_{e,\text{sensor}} - F''_{e,\text{Fano}}, \tag{27}$$

which is plotted in Figure 4 in orange. We can see that the PCC noise is usually very small compared to the shot noise, but in the UV from about 2000 Å to 3500 Å, it actually exceeds the shot noise and is the dominant contributor to the noise.

In Figure 6, we have plotted the PMFs of Equation 19 and the R. A. Stern et al. (1986) noise model to visualize the differences between these two distributions. Each model has been evaluated on a grid where each column of the grid represents a different wavelength (chosen to demonstrate the worst-case differences between the two models) and each row represents a different expected number of absorbed photons. When the number of absorbed photons is low, like in the bottom two rows of Figure 6, we can see that the R. A. Stern et al. (1986) model (orange) has a comb-like appearance caused by the Fano noise slightly blurring the PMF of the photon shot noise. In contrast, our model shows much less of this comb pattern since the PCC noise tends to blur the distribution further into a single peak. As the number of photons increases, both distributions tend towards a Gaussian, but our model remains narrower.

To ease adoption of this model, we've provided a reference implementation of Equation 19 in Python, `optika.sensors.signal()`, which is designed to be simple to use for existing and future instrument pipelines.

2.3. *Charge Diffusion*

In most back-illuminated imaging sensors used for UV astronomy, the depletion region (the region with signifi-

cant electric field) does not extend all the way from the gate structure through the device. As a result, there is a so-called field-free region near the back of the sensor where photoelectrons must undergo a random walk to find their way to the depletion region where they can then be conducted to the terminals and measured (J. R. Janesick 2001). This random walk generally leads to a loss of spatial resolution measured by the sensor since electrons can diffuse to adjacent pixels. It can also lead to a reduction in the variance of a flat field measured by the sensor, but only where a single photon liberates more than one electron. The random walk displaces each electron independently of every other, so for a quantum yield of unity it carries one Poisson distribution of electrons into another and leaves the variance of a flat field unchanged. Where the quantum yield exceeds unity, the electrons liberated by a single photon would otherwise be collected together, and spreading them across neighboring pixels both reduces the variance and induces a covariance between those pixels.

Using Monte Carlo modeling, J. R. Janesick (2001) found the following expression for the standard deviation of the charge diffusion kernel:

$$\sigma(z) = \begin{cases} z_f\sqrt{1 - z/z_f}, & 0 < z < z_f \\ 0, & z_f < z < D, \end{cases} \tag{28}$$

where z is the distance from the back surface at which the photon is absorbed, $z_f = D - z_d$ is the thickness of the field-free region of the sensor, and z_d is the thickness of the depletion region. Using Equation 28, we can find the average variance of the charge diffusion kernel by taking a mean across the entire thickness of the sensor weighted by the probability of a photon being absorbed at that depth,

$$\overline{\sigma^2} = \frac{z_f\left(\alpha z_f + e^{-\alpha z_f} - 1\right)}{\alpha\left(1 - e^{-\alpha D}\right)}. \tag{29}$$

The thickness of the depletion region or the field-free region is difficult to measure, and depends on the voltage applied to the sensor and the charge collected at the terminals (R. A. Stern et al. 2004).

However, R. A. Stern et al. (2004) did estimate the size of the charge diffusion kernel, for two discrete wavelengths, of a 15 μm-thick (100 Ω-cm resistivity) CCD for the GOES Soft X-ray Imager. We can use these measurements to estimate the size of the depletion region and model the size of the charge diffusion kernel as a function of wavelength. R. A. Stern et al. (2004) did not directly measure the size of the charge diffusion kernel, instead they measured a quantity they named the mean charge capture (MCC), the fraction of charge captured by the central pixel. Naively, the MCC would be the integral of the charge diffusion kernel over the extent of a pixel. However, since a photon can strike anywhere within the central pixel, we need to convolve with a rectangle function the width of a pixel before integrating. If we assume that the charge diffusion kernel is a Gaussian with variance $\overline{\sigma^2}$, then the MCC is

$$m = \left[\frac{1}{\sqrt{\pi a}}\left(e^{-a} - 1\right) + \operatorname{erf}\left(\sqrt{a}\right)\right]^2, \tag{30}$$

where $a = d^2/2\overline{\sigma^2}$, d is the width of a pixel, and $\operatorname{erf}(x)$ is the error function.

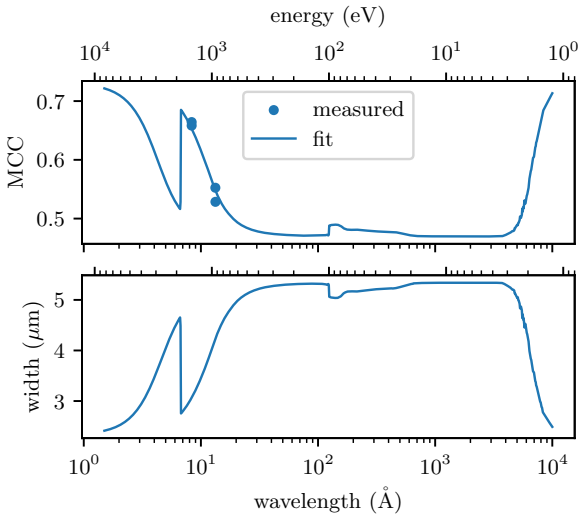


Figure 7. The top panel plots the MCC measured by R. A. Stern et al. (2004) and the fit of our model. The bottom panel shows the corresponding standard deviation of the charge diffusion kernel for the J. Heymes et al. (2020) sensor.

In the top panel of Figure 7, we have plotted a fit of Equation 30 to the measurements in R. A. Stern et al. (2004) which found $z_d = 8.7$ μm best matched the data. In the lower panel of Figure 7, we have plotted the corresponding standard deviation of the charge diffusion kernel as a function of wavelength which predicts that the charge diffusion is reasonably constant over much of the SXR and ultraviolet wavelengths since the penetration depth is low in this regime.

To visualize the extent of this charge diffusion, we've included Figure 8, a 3×3 kernel where the value in each pixel has been computed in a manner similar to Equation 30, just with different limits of integration. We used the IRIS pixel size and a representative wavelength to demonstrate the worst-case scenario for IRIS. Convolving this kernel with an image approximates the spatial

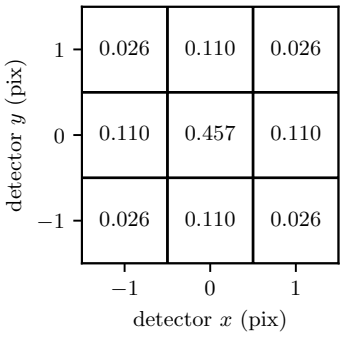


Figure 8. The charge diffusion kernel at 1400 Å convolved with a 13 μm IRIS pixel and integrated over the extent of each pixel.

extent of the charge diffusion, but it does not describe the process faithfully. A convolution assigns a fractional number of electrons to each pixel, whereas every electron is in fact collected by exactly one pixel, and it treats the displacement of each electron as independent, whereas the electrons liberated by a single photon share both an absorption depth and a sub-pixel origin. We therefore diffuse each electron individually, displacing it by $\mathcal{N}\big(0, \sigma^2(z_i)\big)$ in each direction and rounding to the nearest pixel, which conserves charge exactly and preserves the correlation between electrons from the same photon. This is implemented by `optika.sensors.signal()`, and the kernel in Figure 8 was computed with `optika.sensors.kernel_diffusion()`, which can be used to visualize the charge diffusion for other wavelengths and pixel sizes.

Charge diffusion does not add a noise source of its own. It moves electrons between pixels without creating or destroying any, so it leaves the mean of a flat-field image unchanged, and it reduces the variance only by weakening the correlation between electrons that came from the same photon. To find its effect on the noise we return to the faint exposure of Section 2.2, in which the array receives at most one photon. The image is then nothing but the charge liberated by that photon, and the only thing diffusion changes is how it is divided between the pixels, so we now ask how many of the photon's electrons each *pixel* collects rather than how many survive in total. Let m be the number of electrons which survive recombination, as in Section 2.2, and let q_j be the probability that any one of *those* electrons is collected by pixel j. Recombination has therefore already been accounted for in m, and q_j describes only where the surviving charge is delivered, so that $\sum_j q_j = 1$; the CCE enters this section through m and not through q_j. The electrons are displaced independently of one another, so the photon contributes $X_j \leftarrow \mathrm{Binomial}(m, q_j)$ to pixel j, and the second moment summed over the pixels is

$$\sum_j \overline{X_j^2} = m\sum_j q_j - m\sum_j q_j^2 + m^2\sum_j q_j^2 = m + m(m-1)\sum_j q_j^2. \tag{31}$$

Two quantities have appeared which were not present in Section 2.2. The sum $\sum_j q_j^2$ is the probability that two electrons displaced independently are collected by the same pixel, and the factor $m(m-1)$ multiplying it is the number of ordered pairs of distinct electrons the photon delivered. Neither was introduced by hand. Both are produced by the algebra, and they arrive together, because the only way that spreading the charge can alter the second moment is by separating electrons which would otherwise have been counted in the same pixel. Where no charge is shared, $\sum_j q_j^2 = 1$, Equation 31 collapses to m^2, and the result of Section 2.2 is recovered; this is why the pairs are invisible there and unavoidable here. The opposite extreme is equally simple. Where the charge is spread so widely that no two electrons share a pixel, $\sum_j q_j^2$ falls to zero and the sum of squares collapses to m, which is to say the image becomes a set of electrons bearing no relation to one another and the VMR falls to unity. Between these limits the VMR measured in electrons is nothing more than the ratio of the sum of the squares of a single-photon image to the sum of that same image, and charge diffusion moves it from one end of that range to the other.

It remains to evaluate $\sum_j q_j^2$ for the charge diffusion described by Equation 28. Two electrons liberated by the same photon begin their random walks from a common absorption depth and a common position within a pixel, so how often they are collected together depends on how far the charge spreads at the depth where the photon was absorbed, expressed in units of the pixel width,

$$\begin{aligned} \mathcal{P}(z) &= p\big(\sigma(z)/d\big)^2, \\ p(s) &= \operatorname{erf}\left(\frac{1}{2s}\right) - \frac{2s}{\sqrt{\pi}}\left(1 - e^{-1/4s^2}\right), \end{aligned} \tag{32}$$

where $\mathcal{P}(z)$ is $\sum_j q_j^2$ averaged over the position of the photon within its pixel, $p(s)$ is the corresponding probability for a single axis, d is the width of a pixel, and the square accounts for the two directions across the face of the sensor, which spread independently of one another. In the limit of a narrow charge cloud this probability approaches unity and the electrons remain together; in the opposite limit it falls to zero and they are scattered independently.

Averaging Equation 31 over the quantum yield and the absorption depth, exactly as in Section 2.2, requires

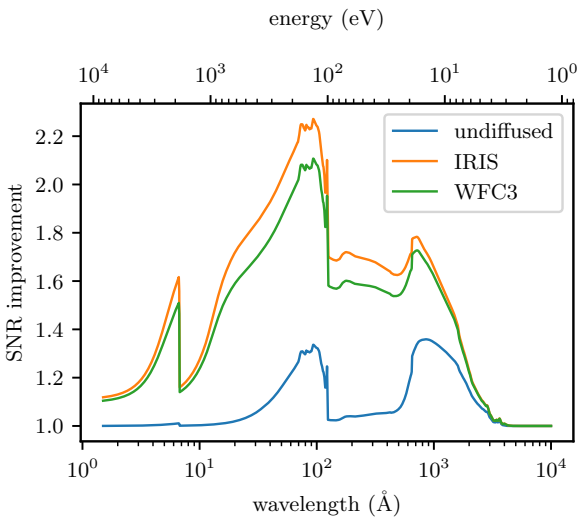


Figure 9. The SNR improvement predicted by our noise model compared to the R. A. Stern et al. (1986) model. In blue we have plotted the SNR improvement ignoring charge diffusion. In orange and green we have plotted the SNR improvement, including charge diffusion, for a flat-field image observed by IRIS and WFC3.

$\overline{m} = \overline{n}\,\overline{\eta}$ as before, and $\overline{m(m-1)} = \overline{n}\big(\overline{n}+\mathcal{F}-1\big)\overline{\eta^2}$, since a pair survives recombination only if both of its electrons do. Dividing by the mean gives the VMR measured by a sensor with charge diffusion,

$$F''_{e,\mathrm{diff}} = 1 + \big(\overline{n} + \mathcal{F} - 1\big)\frac{\overline{\mathcal{P}\eta^2}}{\overline{\eta}}. \tag{33}$$

Comparing with Equation 24, the effect of charge diffusion is to discount the two-electron term by the probability that the two electrons are in fact measured together, and to leave the one-electron term untouched. The discount is applied inside the average over absorption depth rather than outside it, since the distance the charge spreads and the fraction of it that survives recombination both depend on where in the sensor the photon was absorbed, and the two are correlated: photons absorbed near the back surface are the ones which both recombine most readily and diffuse the furthest. Equation 33 is the form implemented by `optika.sensors.vmr_signal()` and plotted for IRIS in Figure 4. It assumes the illumination is uniform, so that the charge leaving each pixel is balanced on average by the charge arriving from its neighbors; near the edge of the sensor, where diffused charge is lost, it is an approximation.

3. RESULTS AND DISCUSSION

Since EQE(λ) can be interpreted as an efficiency, it is tempting to think that the expected number of measured photons and their variance is EQE$\times\overline{N}_\gamma$. However, using the effective QE in this way is equivalent to an all-or-nothing charge collection model where either all the electrons associated with an absorbed photon are measured or none of them are. This simple noise model is formalized in Equations 9 and 10 of R. A. Stern et al. (1986), which gives the VMR of the number of measured electrons as

$$F''_{e,\mathrm{Stern}} = \overline{n} + \mathcal{F} \tag{34}$$

This is the noise model used by T. Borders et al. (2010) and J. P. Wülser et al. (2018). For comparison, the VMR predicted by our undiffused noise model is the sum of $F''_{e,\mathrm{shot}}$ and $F''_{e,\mathrm{sensor}}$,

$$F(N''_e) = 1 + \big(\overline{n} + \mathcal{F} - 1\big)\big(\overline{\eta} + F(\eta)\big). \tag{35}$$

In Figure 4, we have compared $F''_{e,\mathrm{Stern}}$ (dashed) to our undiffused model, $F(N''_e)$ (black). Over most of the SXR and visible wavelengths the R. A. Stern et al. (1986) model is a good approximation of the undiffused model developed in this work since the penetration depth is deeper than the PCC region. However, in the UV, where the penetration depth is very shallow, R. A. Stern et al. (1986) overestimates the noise compared to our undiffused model.

It may seem paradoxical to add a noise source and then measure less total noise, but this is because more photons are measured in our model than the all-or-nothing model. Even though PCC introduces noise, it is less noise than there would be if the photon was undetected. This is good news for designers of UV astronomical instruments since there is less noise than expected from the R. A. Stern et al. (1986) model in this wavelength range.

In Figure 9, we have plotted the ratio of the SNR predicted by our model to the SNR predicted by the traditional model. The blue line is the improvement ratio for the undiffused model and it shows that there are two regions, one from 30-100 Å and another from 500-2000 Å, where the noise statistics deviate from the traditional model, in some cases by up to ∼30%. This figure can be used to quickly estimate the importance of PCC noise in a given wavelength range.

In Table 2, we have calculated the VMR in terms of incident photons and measured electrons for the target wavelengths of a few popular and upcoming solar instruments: AIA (J. R. Lemen et al. 2012), IRIS (B. De Pontieu et al. 2014), and MUSE (B. De Pontieu et al. 2020). The results show that for the extreme ultraviolet (EUV) channels, AIA and MUSE are nearly shot-noise-limited since the VMR in units of incident photons is near unity. Tabulated in the second-to-last column of

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Table 2. The VMR predicted by our model (including charge diffusion) for prominent wavelengths in selected UV observatories, in both incident photon and measured electron units. Also included is the SNR improvement ratio between our model and the traditional noise model as well as the standard deviation of the charge diffusion kernel.

Instrument	wavelength (Å)	VMR (ph)	VMR (e^-)	SNR improvement	diffusion width(μm)
AIA	94	0.43	6.96	2.27	5.32
	131	0.40	8.98	1.69	5.05
	171	0.42	6.72	1.72	5.15
	193	0.44	5.96	1.71	5.17
	211	0.45	5.53	1.70	5.17
	304	0.51	3.96	1.68	5.21
	335	0.53	3.64	1.67	5.21
	1600	3.83	1.15	1.42	5.34
	1700	4.51	1.12	1.35	5.34
IRIS	1330	3.22	1.20	1.52	5.34
	1400	3.23	1.19	1.49	5.34
	2796	9.21	1.02	1.08	5.34
	2832	8.93	1.02	1.09	5.34
MUSE	108	0.43	6.75	2.15	5.32
	171	0.42	6.72	1.72	5.15
	284	0.49	4.21	1.69	5.20
WFC3	2080	6.08	1.07	1.19	5.34
	2240	6.57	1.05	1.15	5.34
	2400	6.40	1.04	1.13	5.34
	2560	7.90	1.03	1.10	5.34
	2720	9.51	1.03	1.09	5.34
	2880	8.51	1.03	1.08	5.34
	3040	6.56	1.02	1.05	5.34
	4000	3.01	1.01	1.01	5.30

Table 2 is the improvement factor between the VMR of the R. A. Stern et al. (1986) noise model and our noise model. These ratios show that the AIA 94 Å and 1600 Å, IRIS 1330 Å and 1400 Å, and MUSE 108 Å channels are predicted to have at least 20% more SNR than the traditional noise model would suggest. These results include the influence of charge diffusion, and we've also tabulated the size of the charge diffusion kernel in the last column, since it sets the scale over which the charge from a single photon is shared between neighboring pixels.

In Table 3, we've attempted to reproduce the measurements of J. P. Wülser et al. (2018) and T. Borders et al. (2010) by taking the ratio of the VMR of a simulated UV flat-field image to the VMR of a simulated visible flat-field image. The flat-field images were created by drawing samples from Equation 19 and then diffusing each measured electron individually, using the pixel size of the corresponding instrument. This table shows that the discrepancy discussed in the introduction is substantially reduced, though our model does not reproduce the measurements exactly. For WFC3, T. Borders et al. (2010) measured 1.09 at 2080 Å where the traditional model expects about 1.7, and our model predicts 1.06, close to the measured value. For IRIS, J. P. Wülser et al. (2018) measured 1.5 at 1350 Å where the traditional model expects about 2, and our model predicts 1.19. This is much nearer the measurement than the traditional model, but it underestimates it, where the traditional model overestimates it. Our model therefore accounts for most of the reported discrepancy in both cases, but it overcorrects for IRIS. The remaining disagreement appears to be confined to our treatment of charge diffusion. Evaluating the same model without charge diffusion raises the predicted IRIS ratio to approximately the measured value, which suggests that the quantum yield and the PCC are not responsible.

Table 3. The ratio of the VMR of a UV flat-field image to the VMR of a visible-light flat-field image. The second column from the right is the value of this ratio measured by T. Borders et al. (2010) and J. P. Wülser et al. (2018). For IRIS this is the ratio of the inverse camera gains measured from photon-transfer curves, 12 photons per data number using a deuterium lamp at 1350 Å and 18 photons per data number using the onboard blue light-emitting diode (J. P. Wülser et al. 2018), which we take to be the 430 nm device flown on the GOES Soft X-ray Imager (R. A. Stern et al. 2004). The rightmost column is the value of this quantity predicted by our noise model (including charge diffusion).

Instrument	λ_{UV} (Å)	λ_{vis} (Å)	measurement	model
IRIS	1350	4300	1.50	1.19
WFC3	2080	4000	1.09	1.06
	2240	4000	1.09	1.05
	2400	4000	1.07	1.04
	2560	4000	1.04	1.03
	2720	4000	1.04	1.02
	2880	4000	1.03	1.02
	3040	4000	1.01	1.01

Our diffusion model has three limitations. The first two would each raise our predicted ratios and could therefore account for the disagreement. First, we have adopted the thickness of the depletion region fit to the measurements of R. A. Stern et al. (2004), which were made on a different sensor operated at a different voltage. The depletion thickness depends on both the resistivity of the silicon and the applied bias, so it need not carry over between sensors, and since the charge diffusion depends only on the difference between the substrate and depletion thicknesses, an error in one cannot be distinguished from an error in the other. A thicker depletion region implies a thinner field-free region and therefore less charge diffusion, which would raise our predicted ratios and improve the agreement for IRIS. Second, we model the charge cloud as a Gaussian, whereas G. G. Pavlov & J. A. Nousek (1999) show that the true radial distribution is peaked more strongly and has heavier tails, which would place more of the charge liberated by a single photon in a single pixel than we have assumed. The third limitation works in the opposite direction: we neglect the further diffusion that the charge undergoes while drifting across the depletion region. Including it would spread the charge liberated by a single photon over more pixels and lower our predicted ratios further, so it cannot explain the underestimate, and it implies that the first two effects would have to be larger still to account for the disagreement on their own.

4. CONCLUSIONS AND FUTURE WORK

This work aims to resolve the discrepancy identified by T. Borders et al. (2010) and J. P. Wülser et al. (2018) in the predicted vs. observed noise measured by back-illuminated silicon imaging sensors. To accomplish this, we reconsidered the effects of PCC and charge diffusion on the noise by developing a simple model of a back-illuminated sensor based on the CCE introduced by R. A. Stern et al. (1994) and a charge diffusion model found by J. R. Janesick (2001). By analyzing these effects in a self-consistent manner, we were able to mostly resolve the discrepancy, and we propose that PCC is the most important effect to consider in the UV. The width of the charge diffusion kernel is nearly independent of wavelength, but its effect on the VMR is not, since diffusion suppresses the photon-correlated term, which grows with the quantum yield.

Our sensor model predicts that there are two UV bands, 30-100 Å and 500-2000 Å, where PCC effects should be considered for a correct noise estimate. We evaluated our sensor model for some popular and upcoming UV astronomical instruments and found that there are a few channels in each instrument where our model implies that the SNR is better than the traditional model would suggest. We also have provided a reference implementation of our sensor model in Python, `optika.sensors`, to make this noise model simple to integrate with existing instrument data processing pipelines.

This work does not consider read noise or other noise sources since those will be specific to the particular camera electronics of each instrument, and also cannot be characterized using a VMR. Instead, this work focuses only on the noise intrinsic to the charge-generation process, since this is comparatively consistent across different instruments.

We plan to use this work to model the noise for the EUV Snapshot Imaging Spectrograph (ESIS) (J. D. Parker et al. 2022), as well as the Full-sun Ultraviolet Rocket Spectrometer (FURST), sounding-rocket-based spectrographs developed by our research group for observing the solar atmosphere.

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2020), all of the simulations, figures, and tables are generated dynamically when this article is created. This helps prevent mistakes and allows this research to be as repeatable as possible. The code to create this document is available at https://github.com/roytsmart/ccd-noise-paper.

Software: optika (R. T. Smart et al. 2026)

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THE EUV SNAPSHOT IMAGING SPECTROGRAPH

Contribution of Authors and Co-Authors

Manuscript in Chapter 4

Author: Roy T. Smart
Contributions: **XX**

Co-Author: Hans T. Courrier
Contributions: **XX**

Co-Author: Jacob D. Parker
Contributions: **XX**

Co-Author: Charles C. Kankelborg
Contributions: **XX**

Co-Author: Amy R. Winebarger
Contributions: **XX**

Co-Author: Ken Kobayashi
Contributions: **XX**

Co-Author: Brent Beabout
Contributions: **XX**

Co-Author: Dyana Beabout
Contributions: **XX**

Co-Author: Ben Carrol
Contributions: **XX**

Co-Author: Jonathan Cirtain
Contributions: **XX**

Co-Author: James A. Duffy
Contributions: **XX**

Co-Author: Eric Gullikson
Contributions: **XX**

Co-Author: Micah Johnson
Contributions: **XX**

Co-Author: Laurel Rachmeler

Contributions: **XX**

Co-Author: Larry Springer
Contributions: **XX**

Co-Author: David L. Windt
Contributions: **XX**

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The EUV Snapshot Imaging Spectrograph

Roy T. Smart [1], Hans T. Courrier,[1] Jacob D. Parker [2], Charles C. Kankelborg [1], Amy R. Winebarger [3], Ken Kobayashi [3], Brent Beabout,[3] Dyana Beabout,[3] Ben Carrol,[1] Jonathan Cirtain,[3] James A. Duffy,[3] Eric Gullikson [4], Micah Johnson,[1] Laurel Rachmeler [3], Larry Springer,[1] and David L. Windt [5]

[1] *Montana State University, Department of Physics, P.O. Box 173840, Bozeman, MT 59717, USA*
[2] *NASA Goddard Space Flight Center, Greenbelt, MD 20771, USA*
[3] *NASA Marshall Space Flight Center, Huntsville, AL 35812, USA*
[4] *Lawrence Berkeley National Laboratory, 1 Cyclotron Road, Berkeley, CA 94720, USA*
[5] *Reflective X-ray Optics LLC, 425 Riverside Dr., #16G, New York, NY 10025, USA*

1. INTRODUCTION

The light emitted by the solar transition region (TR) and corona varies significantly as a function of position, wavelength, and time. When viewed from Earth, the spectral radiance from the Sun can be written as: $I(x, y, \lambda, t)$, where x and y are the helioprojective Cartesian coordinates (W. T. Thompson 2006), λ is wavelength, and t is time. The ideal solar imaging spectrograph would capture $I(x, y, \lambda, t)$ with high resolution in x, y, λ, and t *and* over a wide field of view (FOV), wavelength range, and time period. Of course, the temporal dimension is privileged, so we often reduce the problem to capturing a 3D spatial/spectral cube at a particular time t_0: $I(x, y, \lambda, t_0)$. Since we use 2D detectors, this means that we must find a way to flatten the 3D cube into two dimensions without losing information.

One obvious way to accomplish this is to multiplex one of the three remaining dimensions in time. Narrowband, tunable filters, such as the GREGOR Fabry–Pérot Interferometer (K. G. Puschmann et al. 2012) or CRISP at the Swedish Solar Telescope (G. B. Scharmer et al. 2008), multiplex the wavelength dimension in time, and can change the selected wavelength in 50 ms or less (G. B. Scharmer et al. 2008), but the technology does not exist to use this technique for wavelengths shorter than ∼150 nm (J.-P. Wuelser et al. 2000). The nearest extreme ultraviolet (EUV) equivalent is a multilayer-coated imager such as the Transition Region and Coronal Explorer (TRACE) (B. N. Handy et al. 1999) or the Atmospheric Imaging Assembly (AIA) (J. R. Lemen et al. 2012), which abandons the wavelength dimension entirely and integrates over a passband holding many emission lines. Two such passbands can be compared to infer a Doppler shift (T. Sakao et al. 1999), but weaker lines within them bound the velocity resolution near $\sim 1000\,\mathrm{km\,s^{-1}}$ (K. Kobayashi et al. 2000), coarser than the flows that characterize TR dynamics.

A spatial dimension can be multiplexed instead. An entrance slit admits a single column of the scene, so the detector records λ against y without ambiguity, and x is recovered by stepping the slit across the target. The Interface Region Imaging Spectrograph (IRIS) (B. De Pontieu et al. 2014) and the Spectral Imaging of the Coronal Environment (SPICE) aboard Solar Orbiter (SPICE Consortium et al. 2020) work this way, and the spectra they return are the most faithful of any technique discussed here. What is sacrificed is simultaneity: neighboring columns of the reconstructed cube are separated in time by the raster, and structure that evolves faster than the raster completes is smeared along x. The Multi-slit Solar Explorer (MUSE) (B. De Pontieu et al. 2020, 2022; M. C. M. Cheung et al. 2022) narrows this gap considerably by ruling 37 slits across the field and separating their overlapping spectra afterward, which shortens the raster of an active region to as little as 12 s.

A third strategy divides the detector rather than the exposure. An integral field spectrograph slices the field of view and reformats those slices so that each is dispersed onto its own region of the detector, so the entire cube arrives in one exposure, at the cost of trading field of view against wavelength range. The microlensed hyperspectral imager of M. van Noort et al. (2022) demonstrates the technique at visible wavelengths, and the Solar eruptioN Integral Field Spectrograph (SNIFS) (V. L. Herde et al. 2024), flown in 2025, demonstrates it in the far ultraviolet (FUV). Extending it to the EUV remains out of reach of available optics.

The last strategy is to allow the flattening to be ambiguous and to undo it afterward. A spectrograph with no entrance slit disperses the whole field at once,

Corresponding author: Roy T. Smart
Email: roytsmart@gmail.com

so every emission line forms an image of the Sun and those images superimpose on the detector. The resulting frame, an *overlappogram*, encodes x and λ along a single axis, and the two cannot be separated within one exposure. Overlappograms are as old as spaceborne solar spectroscopy, having been recorded by the Naval Research Laboratory's (NRL's) S082A spectroheliograph (R. Tousey et al. 1973, 1977), which yielded both a census of EUV transitions (U. Feldman et al. 1985) and, much later, flare line ratios (F. P. Keenan et al. 2006); the technique is in use today at soft X-ray wavelengths by the Marshall Grazing Incidence X-ray Spectrometer (MaGIXS) (S. L. Savage et al. 2023). What has changed since S082A is that the ambiguity has become tractable. Reconstructing the cube from superimposed images is a tomographic problem (A. C. Kak & M. Slaney 1988), in which each diffraction order views the cube from a different angle and the number of independent views limits how much of the line profile can be recovered (M. R. Descour et al. 1997). Computed tomography imaging spectrographs (CTISs) (T. Okamoto & I. Yamaguchi 1991; F. V. Bulygin et al. 1991; M. Descour & E. Dereniak 1995) carry this to its extreme, projecting as many as 25 orders onto a single detector, with computational cost rising accordingly (N. Hagen & E. L. Dereniak 2008; N. Hagen & M. W. Kudenov 2013). Fewer views can suffice when the scene is sparse: C. DeForest et al. (2004) synthesized a magnetogram from a single exposure using only two dispersed orders. Inversion methods for overlapping spectral images have advanced considerably since (A. R. Winebarger et al. 2019; J. M. Davila et al. 2019; U. Kamaci & F. Kamalabadi 2026).

The Multi-Order Solar EUV Spectrograph (MOSES) (J. L. Fox et al. 2010; J. L. Fox 2011) brought this strategy to the solar EUV. A single concave grating forms three images at once, the undispersed $m = 0$ order and the $m = \pm 1$ orders, on three detectors, while a multilayer coating narrows the passband to a few emission lines so that the cube to be recovered is sparse enough to invert. Two flights showed that the concept works. J. L. Fox et al. (2010) measured the Doppler shift and width of explosive event line profiles in the TR; T. Rust & C. C. Kankelborg (2019) resolved bimodal profiles in quiet Sun events; H. T. Courrier & C. C. Kankelborg (2018) recovered Doppler maps using local correlation tracking; and J. D. Parker & C. C. Kankelborg (2022) separated the contributions of individual spectral lines. The same flights exposed where the design binds. Only three views are available, one of which is undispersed and therefore contributes no spectral information, which caps the number of degrees of freedom that an inversion can recover; and because the dispersed orders lie in a plane, they fill the cylindrical volume of a sounding rocket payload poorly.

The EUV Snapshot Imaging Spectrograph (ESIS) was designed to relax both constraints. An array of gratings arranged about the optical axis re-images the prime focus of a single primary mirror, each dispersing at a different angle, so that every detector records a dispersed view and the views are distributed around the payload rather than confined to a plane. ESIS flew for the first time on 2019 September 30, and first results from that flight are reported by J. D. Parker et al. (2022). An earlier form of the design was described by H. T. Courrier (2020); the instrument that flew differs from it, most consequentially in the removal of the primary mask, which admits vignetting into the system. This paper describes the instrument as it was built and flown. In Section 2 we explain how experience with MOSES shaped the design of ESIS. Section 3 states the scientific objectives and the requirements they impose. Section 4 describes the instrument itself, and Section **??** the mission profile.

2. THE ESIS CONCEPT

A primary goal of the ESIS instrument is to improve upon the imaging spectroscopy demonstrated by MOSES. Therefore, the design of the new instrument draws heavily from experiences and lessons learned through two flights of MOSES. ESIS and MOSES are both CTIS instruments. As such, both produce overlappograms of a narrow portion of the solar spectrum, with the goal of enabling the reconstruction of a spectral line profile at every point in the field of view. The similarities end there, however, as the optical layout of ESIS differs significantly from that of MOSES. In this section, we detail some difficulties and limitations encountered with MOSES, then describe how the new design of ESIS addresses these issues.

2.1. *Limitations of the MOSES Design*

The MOSES design features a single concave diffraction grating forming images on three charge-coupled device (CCD) detectors (J. L. Fox et al. 2010) (Figure 1). The optical path is folded in half by a single flat secondary mirror (omitted in Figure 1). Provided that the three cameras are positioned correctly, this arrangement allows the entire telescope to be brought into focus using only the central (undispersed) order and a visible light source. Unfortunately this design uses volume inefficiently for two reasons. First, the lack of magnification by the secondary mirror limits the folded length of the entire telescope to be no less than half of the 5 m focal length of the grating (J. L. Fox et al. 2010;

J. L. Fox 2011). Second, the resolving power of the instrument is limited by the ratio of camera separation to pixel size. To achieve the maximum dispersion of 29 km pix^{-1} s^{-1} (J. L. Fox et al. 2010), the outboard orders are imaged as far apart as possible in the ~ 0.56 m diameter envelope of the rocket payload. This planar arrangement leaves much unused space in the cylindrical volume of the payload.

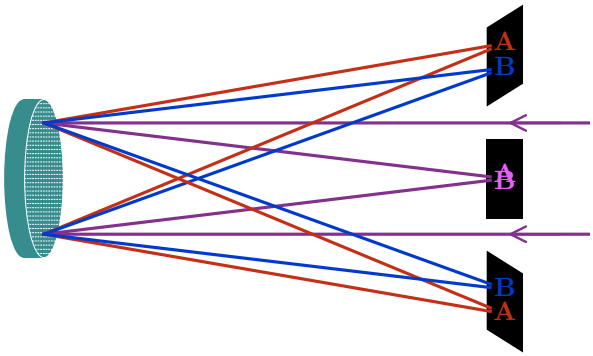


Figure 1. Schematic diagram of the MOSES instrument. Incident light from the right forms an undispersed image on the central $m = 0$ CCD. Dispersed images are formed on the outboard $m = \pm 1$ CCDs. The imaging of longer and shorter wavelengths is indicated by the use of red and blue letters respectively.

Furthermore, the monolithic secondary, though it confers the focus advantage noted above, does not allow efficient placement of the $m = \pm 1$ CCDs. For all practical purposes, the payload can only accommodate three diffraction orders ($m = -1, 0, +1$). Therefore, *MOSES can only collect, at most, three pieces of information at each point in the field of view.* From this, it is not reasonable to expect the reconstruction of more than three degrees of freedom for each spectral line, except in the case of very compact, isolated features such as those described by J. L. Fox et al. (2010) and T. Rust & C. C. Kankelborg (2019). Consequently, it is a reasonable approximation to say that MOSES is sensitive primarily to spectral line intensities, shifts, and widths (C. C. Kankelborg & R. J. Thomas 2001). With any tomographic apparatus, the degree of detail that can be resolved in the object depends critically on the number of viewing angles (A. C. Kak & M. Slaney 1988; M. R. Descour et al. 1997; N. Hagen & E. L. Dereniak 2008). So it is with the spectrum we observe with MOSES: more dispersed images are required to confer sensitivity to finer spectral details such as higher moments of the spectral line shape.

A related issue stems from the use of a single dispersion plane. Since the solar corona and transition region are structured by magnetic fields, the scene tends to be dominated by field-aligned structures such as loops (R. Rosner et al. 1978; R. M. Bonnet et al. 1980). When the MOSES dispersion direction happens to be aligned nearly perpendicular to the magnetic field, filamentary structures on the transition region serve almost as spectrograph slits unto themselves. The estimation of Doppler shifts then becomes a simple act of triangulation, and broadenings are also readily diagnosed (J. L. Fox et al. 2010; H. T. Courrier & C. C. Kankelborg 2018). A double-peaked profile can also be observed with sufficiently isolated features (T. Rust & C. C. Kankelborg 2019). Unfortunately, solar magnetic fields in the transition region are quite complex and do not have a global preferred direction. In cases where the field is nearly parallel to the instrument dispersion, spectral shifts and broadenings are not readily apparent.

The single diffraction grating also leads to a compromise in the optical performance of the instrument. Since the MOSES grating forms images in three orders simultaneously, there are not enough degrees of freedom in the optical system to control aberrations in all three spectral orders. During the first mission, MOSES was flown with a small amount of defocus (T. Rust & C. C. Kankelborg 2019), which exacerbated the inter-order point-spread function (PSF) variation and caused the individual PSFs to span several pixels (T. Rust & C. C. Kankelborg 2019; S. Atwood & C. Kankelborg 2018). The outboard PSFs were elongated, with their major axes at different angles. This resulted in spurious spectral features that require additional consideration (S. Atwood & C. Kankelborg 2018) and further increase the complexity of the inversion process (T. Rust & C. C. Kankelborg 2019; H. T. Courrier & C. C. Kankelborg 2018).

As MOSES lacks a field stop, it is possible for the grating to image off-band radiation on the detector from solar features that lie outside the intended FOV. Although the intensity of this off-band radiation is relatively low, it is not negligible. J. D. Parker & C. C. Kankelborg (2022) compared synthetic MOSES images to the real data and found that approximately ten percent of the intensity in the zeroth order image originated from more than ten dim lines in the MOSES passband, most of them too faint to be visible in the dispersed images. This study revealed that an undispersed ($m = 0$) channel, although attractive due to its lack of spatial-spectral ambiguity, is of limited utility due to spectral contamination. Moreover, the FOV should be clearly defined, and the same, for each image, so that the spectral and spatial content of each image is unambiguous.

Finally, the exposure cadence of MOSES is hindered by a $\sim$6 s readout time for the CCDs (J. L. Fox 2011).

The observing interval for a solar sounding rocket flight is typically about five minutes. Consequently, every second of observing time is precious, both to achieve adequate exposure time and to catch the full development of dynamical phenomena. The MOSES observing duty cycle is ∼50 % since it is limited by the readout time of its CCDs. Thus, valuable observing time is lost. The readout data gap compelled us to develop a sequence with exposures ranging from 0.25-24 s, a careful tradeoff between deep and fast exposures.

In summary, our experience leads us to conclude that the MOSES design has the following primary limitations:

1. inefficient use of volume
2. dispersion constrained by payload dimensions
3. too few dispersed images (orders)
4. single dispersion plane
5. lack of aberration control
6. poorly-defined and wavelength dependent FOV
7. spectral contamination
8. low duty cycle

In designing ESIS, we have sought to improve upon each of these points.

2.2. *ESIS Features*

The layout of ESIS (Figure 2) is a modified form of Gregorian telescope. Incoming light is brought to focus at an octagonal field stop by a parabolic primary mirror. In the ESIS layout, the secondary mirror of a typical Gregorian telescope is replaced by a segmented, octagonal array of diffraction gratings. From the field stop, the gratings re-image to CCD detectors arranged radially around the primary mirror. The gratings are blazed for first order, so that each CCD is fed by a single corresponding grating, and all the gratings are identical in design. The features of this new layout address all of the limitations described in Section 2.1, and are summarized here.

Replacing the secondary mirror with an array of concave diffraction gratings confers several advantages to ESIS over MOSES. First, the concavity of the gratings creates magnification in the ESIS optical system, which results in a shorter axial length than MOSES, without sacrificing spatial or spectral resolution. Second, the magnification and tilt of an individual grating controls the position of the dispersed image with respect to the optical axis, so that the spectral resolution is not constrained by the payload dimensions. Third, the radial symmetry of the design places the cameras closer together, resulting in a more compact instrument. Furthermore, by arranging the detectors around the optical axis, more dispersed grating orders can be populated; up to eight gratings can be arrayed around the ESIS primary mirror (up to six with the current optical table). This contrasts with the three image orders available in the planar symmetry of MOSES. Taken together, these three design features make ESIS more compact than MOSES (Section 2.1 item 1), improve spectral resolution (item 2) and allow the collection of more projections to better constrain the interpretation of the data (item 3).

The ESIS gratings are arranged in a segmented array, clocked in 45° increments, so that there are four distinct dispersion planes. This greatly aids in reconstructing spectral line profiles since the dispersion space of ESIS occupies a 3D volume rather than a 2D plane as with MOSES. For ESIS, there is always a dispersion plane within 22.5° of the normal to any loop-like feature in the solar atmosphere. As discussed in Section 2.1, a nearly perpendicular dispersion plane allows a filamentary structure to serve like a spectrographic slit, resulting in a clear presentation of the spectrum. This feature addresses Section 2.1 item 4.

Rather than forming images at three spectral orders from a single grating, each ESIS imaging channel has a dedicated grating. Aberrations are controlled by optimizing the grating design to form images in first order, over a narrow range of ray deviation angles. This design controls aberration well enough to allow pixel-limited imaging, avoiding the PSF mismatch problems inherent to the MOSES design (Section 2.1 item 5). In its flight configuration with gratings optimized around a 629.732 Å wavelength, the instrument cannot be aligned and focused in visible light like MOSES. Visible gratings and a special alignment transfer procedure (Section 4.6) are used to align and focus ESIS.

The ESIS design also includes an octagonal field stop placed at prime focus. This confers two advantages. First, the field stop fully defines the instrument FOV, so that ESIS is not susceptible to the spectral confusion observed in MOSES data (Section 2.1 item 6). Second, each spectral line image observed by ESIS is bordered by the outline of the field stop (e.g., Section 4.1). This aids the inversion process since outside of this sharp edge the intensity is zero for any look angle through an ESIS data cube. The size and octagonal shape of the field stop are defined by the requirement that all CCDs must see the entire FOV from edge to edge, while leaving a small margin for alignment.

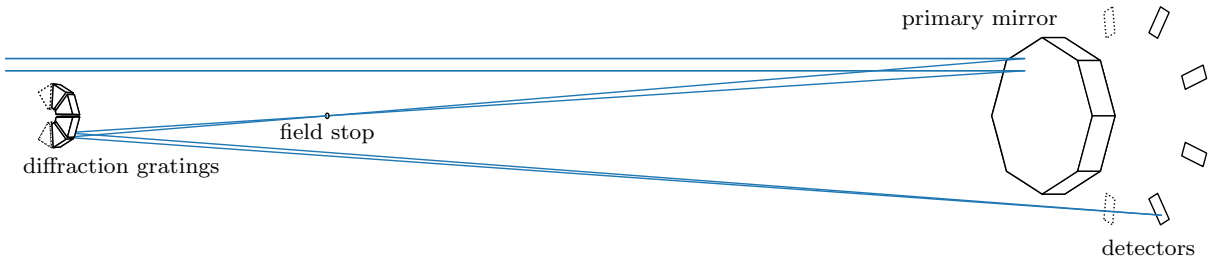


Figure 2. The ESIS optical layout. Dotted lines indicate the positions of unpopulated channels. The blue lines represent the path of O V through the system. The ESIS instrument is a pseudo-Gregorian design. The secondary mirror is replaced by a segmented array of concave diffraction gratings. The field stop at prime focus defines instrument spatial/spectral FOV. CCDs are arrayed around the primary mirror, each associated with a particular grating. Eight grating positions are available in principle; only six fit within the volume of the rocket payload. Four channels are populated for the first flight.

Lastly, in contrast to MOSES, ESIS employs frame transfer CCDs to make optimum use of our five minutes of observing time. The ESIS design is shutterless, so that each detector is always integrating. The result is a 100 % duty cycle. The lack of downtime for readout also allows ESIS to operate at a fixed, rapid cadence of 10 s. Longer integration times can be achieved for faint features by exposure stacking (Section 2.1 item 8).

In summary, the ESIS concept addresses all the limitations of the MOSES design enumerated in Section 2.1. The volume of the ESIS optical layout is smaller than MOSES by almost a factor of two, yet with a smaller PSF, improved spectral resolution, and faster exposure cadence. ESIS offers several features to improve the recovery of spectral information, including more channels, crossed dispersion planes, and a field stop.

3. SCIENCE OBJECTIVES

Early flights of MOSES demonstrated a working concept of simultaneous EUV imaging and spectroscopy. This concept adds a unique capability to the science that we can obtain from the EUV solar atmosphere. ESIS as designed improves upon the MOSES concept, as discussed in the previous section, and therefore improves our ability to accomplish our scientific objectives. In this section, we set forth the specific scientific objectives of the ESIS mission. It is from these objectives that we derived the quantitative science requirements (Section 3.3) that drove the ESIS design.

The ESIS mission was designed to achieve the following two overarching science goals: (1) observe magnetic reconnection in the TR, and (2) map the transfer of energy through the TR with emphasis on magnetohydrodynamic (MHD) waves. These objectives have significant overlap with the missions of IRIS (B. De Pontieu et al. 2014), the EUV Imaging Spectrograph (EIS) (J. L. Culhane et al. 2007) aboard Hinode, the EUV Normal-incidence Spectrometer (EUNIS) (J. W. Brosius et al. 2007, 2014), and a long history of FUV and EUV slit spectrographs. The ESIS instrument, however, can obtain both spatial and spectral information co-temporally. This allows us to resolve complicated morphologies of compact TR reconnection events (as was done with MOSES (J. L. Fox 2011; T. L. Rust 2017; H. T. Courrier & C. C. Kankelborg 2018)) and observe signatures of MHD waves over a large portion of the solar disk. Therefore, in support of goal 1, we will use ESIS to map flows as a function of time and space in multiple TR reconnection events. To achieve goal 2, we will cross-correlate the evolution at multiple temperatures in the TR to map the vertical transport of energy over a wide FOV.

3.1. *Magnetic Reconnection Events*

Magnetic reconnection describes the re-arrangement of the magnetic topology wherein magnetic energy is converted to kinetic energy resulting in the acceleration of plasma particles. Reconnection is implicated in many dynamic, high-energy solar events. Solar flares are a well-studied example (e.g., E. R. Priest & T. G. Forbes (2002) and the references therein); however, we have little hope of pointing in the right place at the right time to observe a significant flare event in a rocket flight lasting only five minutes. Instead, we will search for signatures of magnetic reconnection in TR spectral lines. A particular signature of reconnection in the TR is the explosive energy release by ubiquitous, small-scale events. These explosive events (EEs) are characterized as spatially compact (≈1.5 Mm length (K. P. Dere 1994)) line broadenings on the order of 100 $\mathrm{km\,s^{-1}}$ (K. P. Dere et al. 1991). They are observed across a range of TR emission lines that span temperatures of 20 000–250 000 K (C II–O V) (D. Moses et al. 1994). The typical lifetime of an EE is 60–90 s (D. Moses et al. 1994; K. P. Dere 1994; K. P. Dere et al. 1991). Due to their location near quiet Sun magnetic network elements, and the presence

of supersonic flows near the Alfvén speed, K. P. Dere et al. (1991) first suggested that EEs may result from the model of fast Petschek (H. E. Petschek 1964) reconnection.

The spectral line profile of EEs may indicate the type of reconnection that is occurring in the TR (e.g., T. L. Rust (2017)). For example, the Petschek model of reconnection predicts a 'bi-directional jet' line profile with highly Doppler-shifted wings, but little emission from the line core (D. E. Innes & G. Tóth 1999). D. E. Innes et al. (2015) developed a reconnection model resulting from a plasmoid instability (A. Bhattacharjee et al. 2009). In contrast to the bi-directional jet, this modeled line profile has bright core emission and broad wings. Both types of profile are seen in slit spectrograph data (e.g., D. E. Innes et al. (1997, 2015), and the references therein); however, MOSES observed EEs with more complicated morphologies than either of these two models suggest (J. L. Fox et al. 2010; T. L. Rust 2017). It is unclear whether the differing observations are a function of wavelength and temperature, a result of a limited number of observations, or a consequence of the morphology of the event being difficult to ascertain from slit spectrograph data.

ESIS will observe magnetic reconnection in the context of EEs, by extending the technique pioneered by MOSES to additional TR lines. Explosive events are well suited to sounding rocket observations; a significant portion of their temporal evolution can be captured in >150 s (e.g., the analysis by T. L. Rust (2017)) and they are sufficiently common to provide a statistically meaningful sample in a 5-minute rocket flight (e.g., K. P. Dere et al. (1989, 1991)). In similarity with MOSES, we sought a TR line for ESIS that is bright and well enough isolated from neighboring emission lines so as to be easily distinguished.

3.2. *Energy Transfer*

Tracking the mass and energy flow through the solar atmosphere is a long-standing goal in solar physics. Bulk mass flow is evidenced by Doppler shifts or skewness in spectral lines. However, the observed non-thermal broadening of TR spectral lines may result from a variety of physical processes, including MHD waves (B. De Pontieu et al. 2015, 2007), high-speed evaporative upflows (e.g., nanoflares, S. Patsourakos & J. A. Klimchuk (2006)), turbulence, and other sources (e.g., J. T. Mariska (1992)). This is a broad topic which ESIS can address in many ways. Here we will focus on a single application; ESIS will search for sources of Alfvén waves in the solar atmosphere by observing Doppler shifts and line broadening as the spectroscopic signature of these waves. The spectroscopic signature of Alfvén waves is most obvious near the limb of the solar disk, since the motion of the ions is transverse with respect to the propagation direction. Pointing near the limb may make it more difficult to observe distinct EEs, described in Section 3.1, since the average width of TR lines often increases near the limb of the Sun (T. R. Ayres 2021), thus decreasing the signal-to-noise ratio (SNR) of single events.

Alfvén waves in coronal holes are observed to carry an energy flux of $7 \times 10^5\,\mathrm{erg/cm^2/s}$, enough to energize the fast solar wind (M. Hahn et al. 2012; M. Hahn & D. W. Savin 2013). The source and frequency spectrum of these waves is unknown. Here, we hypothesize that MHD waves are similarly ubiquitous in quiet Sun and active regions, and play an important role in the energization of the quiescent corona.

The magnitude of non-thermal broadening of optically thin spectral lines is a direct measure of the wave amplitude (D. Banerjee et al. 2009; M. Hahn et al. 2012; M. Hahn & D. W. Savin 2013). We may estimate a lower limit on the non-thermal velocity to be observed as follows. We assume that the magnetic field is constant for small changes in scale height in the TR and that line-of-sight effects are negligible for observations sufficiently far from disk center. Since the solar wind is not accelerated to an appreciable fraction of the Alfvén wave velocity at altitudes $R \leq 1.15 R_\odot$ (S. R. Cranmer & A. A. van Ballegooijen 2005), the wave amplitude, v_{nt}, depends only weakly on electron density, n_e, so that $v_{nt} \propto n_e^{-1/4}$ (M. Hahn & D. W. Savin 2013; T. G. Moran 2001). Assuming pressure balance between the low corona and transition zone, we may infer non-thermal velocities in the TR by scaling according to the temperature drop, $v_{nt} \propto T^{1/4}$. The measured non-thermal velocity of $24\,\mathrm{km\,s^{-1}}$ for Si VIII (J. G. Doyle et al. 1998) (0.8 MK (T. G. Moran 2003)) near the limb should, neglecting damping, correspond to velocities of at least $18\,\mathrm{km\,s^{-1}}$ in the O V (0.25 MK) line. This is larger than the thermal width of $11\,\mathrm{km\,s^{-1}}$.

More recently, A. K. Srivastava et al. (2017) observed torsional Alfvén waves with amplitude $\sim$20 $\mathrm{km\,s^{-1}}$ and period $\sim$30 s in the chromosphere. Modeling shows that these torsional waves can transfer a significant amount of energy to the corona (T. Kudoh & K. Shibata 1999). The torsional motion is observed as Doppler shifts when viewed from the side. The oscillation period is long enough to be well resolved but short enough to see $\sim$10 cycles in a single rocket flight. An ESIS-like instrument is therefore well suited to observations of torsional Alfvén wave propagation over multiple heights in the TR.

By mapping Doppler velocities over a wide field of view in the TR, ESIS can address questions about both the origin of waves and whether they are able to propagate upward into the corona. Independent of the two propagation modes discussed above, there is a range of possible sources for Alfvén (and other MHD) waves in the solar atmosphere. Three potential scenarios are: (1) Waves originate in the chromosphere or below and propagate through the TR at a spatially uniform intensity; (2) Intense sources are localized in the TR, but fill only a fraction of the surface; and (3) Weak sources are localized in the TR, but cover the surface densely enough to appear like the first case. The resulting non-thermal widths for localized sources are significantly higher than the $\sim$18 km s^{-1} lower limit derived above. The concentration of non-thermal energy observed by ESIS will serve as an indicator of source density. Comparison of Doppler maps captured at different temperatures by ESIS will indicate whether a uniform source density originates in the chromosphere or below (scenario 1) or is associated with spatially distributed TR phenomena (scenario 3) such as explosive events, or macrospicules. Comparison with a wider selection of ground- and space-based imagery will allow us to determine whether intense, localized sources (scenario 2) are associated with converging or emerging magnetic bipoles, type II spicules, spicule bushes, or other sources beneath the TR. For these comparisons, we need only to localize, rather than resolve, wave sources. A spatial resolution of $\sim$2 Mm is sufficient to localize sources associated with magnetic flux tubes that are rooted in photospheric intergranular network lanes (e.g., T. E. Berger et al. (1995)).

3.3. *Science Requirements*

ESIS will investigate two science targets: reconnection in explosive events and the transport of mass and energy through the transition region. The latter may take many forms, from MHD waves of various modes to EUV jets or macro-spicules. To fulfill these goals, ESIS will obtain simultaneous intensity, Doppler shift and line width images of the O V 629.732 Å line in the solar transition region (0.25 MK) at rapid cadence. The bright, optically thin O V emission line is well isolated except for the two coronal Mg X lines. These coronal lines can be viewed as contamination or as a bonus; we expect that with the four ESIS projections it is possible to separate the O V emission from that of Mg X. From the important temporal, spatial, and velocity scales referenced in Sections 3.1 and 3.2 we define the instrument requirements in Table 1 that are needed to meet our science goals.

4. THE ESIS INSTRUMENT

ESIS is a multi-projection slitless spectrograph that obtains line intensities, Doppler shifts, and widths in a single snapshot over a 2D FOV. Starting from the notional instrument described in Section 2, ESIS has been designed to ensure all of the science requirements set forth in Table 1 are met. The final design parameters are summarized in Table 2.

A schematic diagram of a single ESIS channel is presented in Figure 4a, while the mechanical features of the primary mirror and gratings are detailed in Figures 4b and 4c, respectively.

4.1. *Optics*

Figure 4a shows the relative layout of the optics and detectors for a single ESIS channel. Here we give specific details of the primary mirror and gratings. The features of the field stop have been described previously in Section 2.2, while the CCD and cameras are covered in Section 4.8.

The primary mirror (Figure 4b) is octagonal in shape. The octagonal shape of the primary allows dynamic clearance for filter tubes that are arranged radially around the mirror (Section 4.4). The mirror is attached to a backing plate by three *bipods*: thin titanium structures that are flexible in the radial dimension, perpendicular to the mirror edge, but rigid in the other two dimensions. The bipods form a kinematic mount, isolating the primary mirror figure from mounting stress.

The mirror had to maintain its figure under direct solar illumination, so a Corning ultra-low expansion (ULE) substrate was used. The transparency of ULE, in conjunction with the transparency of the mirror coating in visible and near-IR wavelengths (e.g., Table 2 and Section 4.4), helps minimize the heating of the mirror. Surface figure specifications for the ESIS optics are described in Section **??**.

The spherical gratings (Figure 4c) re-image light from the field stop to form dispersed images at the CCDs. Each grating is individually mounted to a backing plate in a similar fashion as the primary mirror. For these much smaller optics, lightweight bipods were photochemically etched from thin titanium sheet. The bipods are bonded to both the grating and backing plate along the three long edges of each grating. The individual mounts allow each grating to be adjusted in tip and tilt to center the image on the CCD.

The gratings have a varied line space ruling pattern optimized to provide, in principle, pixel-limited imaging from the field stop to the CCDs. The pitch at the center of the grating is $d_0 = 0.38661$ μm (0.38677 μm as measured by the manufacturer) resulting in a dispersion of

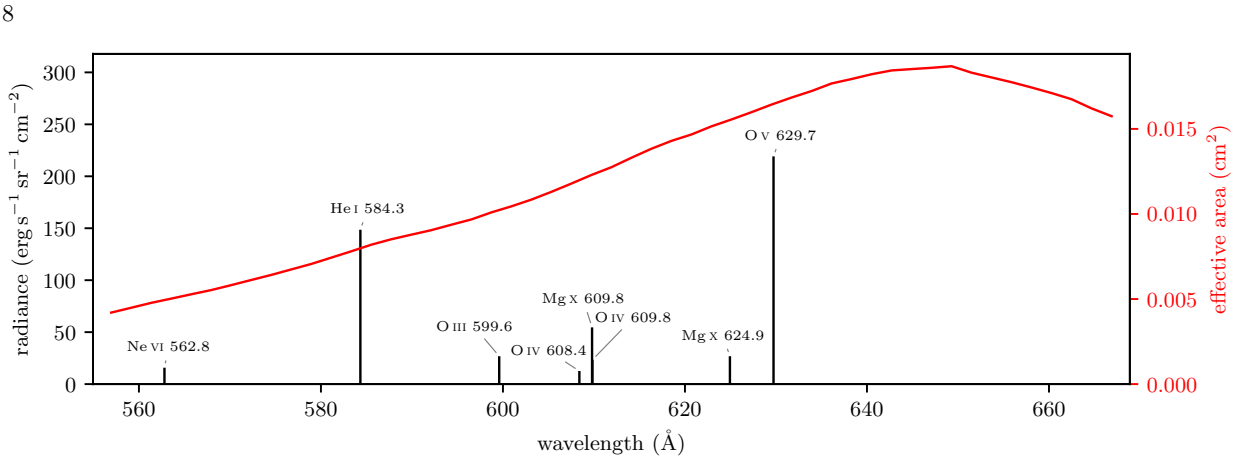


Figure 3. The eight brightest emission lines in the ESIS passband, and the effective area of a single channel beneath them. The lines are computed from version 11.0.2 of the CHIANTI atomic database, using the J. T. Schmelz et al. (2012) abundances of `sun_coronal_2012_schmelz_ext`, the quiet Sun differential emission measure (DEM) file `quiet_sun.dem`, and $n_e T = 1 \times 10^{15}$ K cm^{-3}.

Table 1. ESIS instrument requirements and capabilties. Note that MTF exceeds the Rayleigh criterion of 0.109.

Parameter	Requirement	Science Driver	Capabilities
Spectral line	O V 629.732 Å	EEs	O V, Mg X, He I, Figure 3
Spectral sampling	18 km s^{-1}	Broadening from MHD waves	17.5 km s^{-1} pix^{-1}, Table 2
Spatial resolution	2.1″(1.5 Mm)	EEs	**??**, Table **??**
SNR	17.3 (CH)	MHD waves in CH	17.7 (17 × 10 s exp.), Table 4
Cadence	15 s	Torsional waves	10 s eff., Section 4.5
Observing time	150 s	EEs	306.8 s, Section **??**
FOV diameter	10′	Span QS, AR, and limb	11.5′, Table 2

17.5 km s^{-1} pix^{-1} at the center of the O V 629.732 Å FOV. Each grating serves only a single CCD. The groove profile is laminar, with depth optimized for $m = 1$. Modeled groove efficiency in this order is 40 %.

Figure specification and groove profile are difficult to control near the edges of the gratings. Therefore, an uncoated margin was left around the edge of the grating, minimizing reflection in EUV (Figure 4c).

The ESIS passband is defined through a combination of the field stop, the grating dispersion, and the CCD size. The passband includes the He I 584.334 Å spectral line through Mg X (609.793 Å and 624.941 Å) to O V 629.732 Å. Figure 5 shows where images of each of the strong spectral lines fall on the CCD. The instrument dispersion satisfies the spectral resolution requirement in Table 1 and ensures that the spectral images are well-separated; Figure 5 shows that He I 584.334 Å is completely separated from the target O V 629.732 Å line.

4.2. *Vignetting*

The original design of ESIS had no vignetting thanks to a stop placed at the primary mirror that was designed to perfectly fill the grating with the same amount of light for each point in the FOV. This is the ESIS design that was used for the optimization procedure of the grating parameters described in Section **??**, for example. All other results described in the paper use the fully-open system. Before flight, we decided to remove the primary aperture stop to increase the sensitivity of the instrument at the expense of introducing vignetting to the ESIS FOV. This was acceptable since the vignetting was found to be a simple linear field as shown in Figure 6, and could be removed in the post-processing phase.

4.3. *Distortion*

Imaging from the prime focus to the detector introduces distortion. Since the four ESIS channels are arranged in four different orientations, the distortion complicates the comparison of the channels, and we must correct for it. The distortion is due to two factors: first,

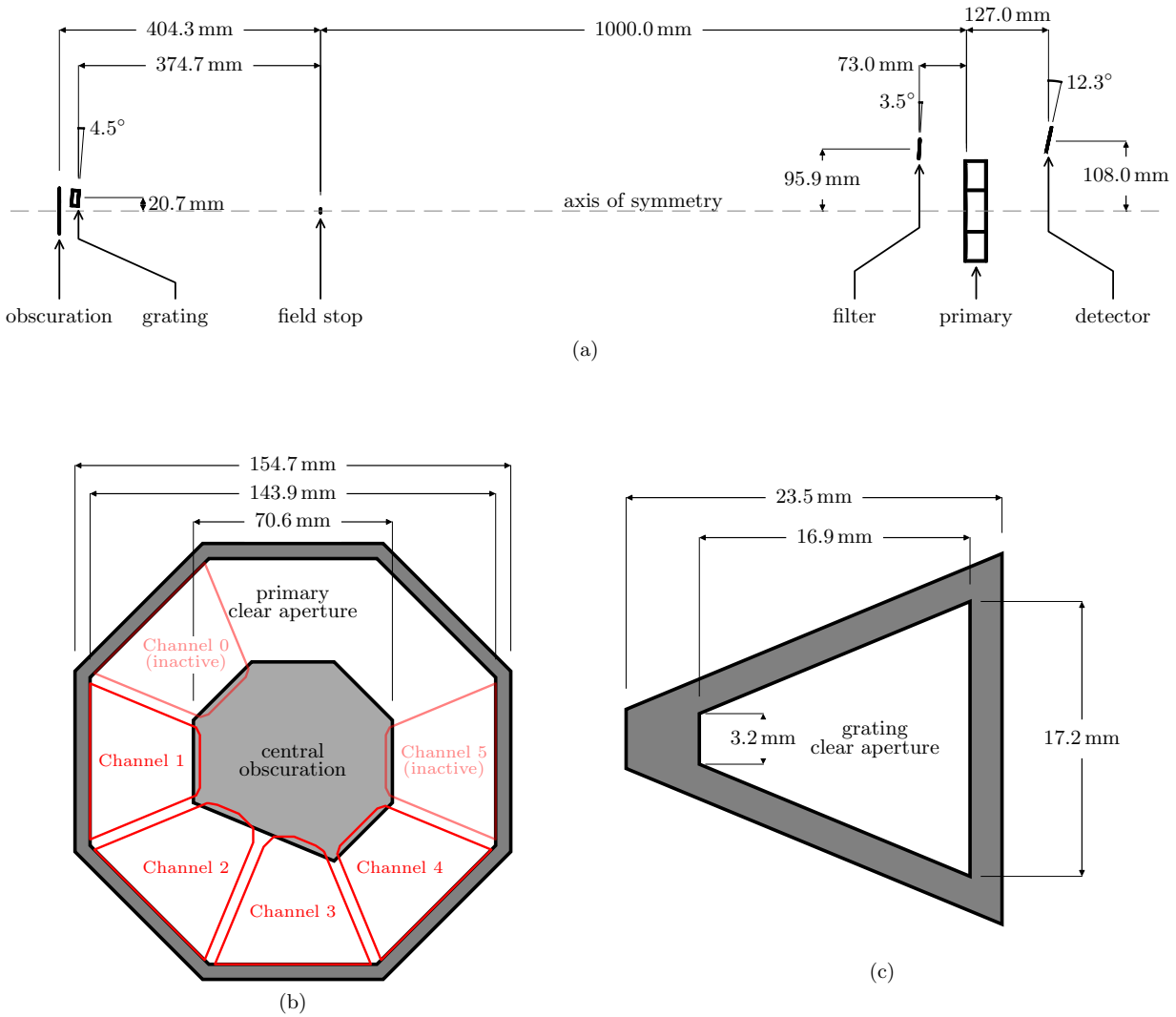


Figure 4. (a) Schematic diagram of a single channel of the ESIS optical system. (b) Clear aperture of the primary mirror, size of the central obscuration, and the footprint of the beam for each channel. (c) Clear aperture of one diffraction grating.

the tilt of the detector as needed to maintain good focus over the FOV (L. Poletto & R. J. Thomas 2004); second, the anamorphic magnification of the grating (see F. Schweizer (1979)).

$$\begin{aligned}(x', y') = \mathbf{C} + \mathbf{C}_x x + \mathbf{C}_y y + \mathbf{C}_\lambda \lambda \\ + \mathbf{C}_{xx} x^2 + \mathbf{C}_{xy} xy + \mathbf{C}_{x\lambda} x\lambda \\ + \mathbf{C}_{yy} y^2 + \mathbf{C}_{y\lambda} y\lambda + \mathbf{C}_{\lambda\lambda} \lambda^2\end{aligned} \quad (1)$$

4.4. *Coatings and Filters*

The diffraction gratings are coated with a multilayer optimized for a center wavelength of O V 629.732 Å, developed by a collaboration between Reflective X-ray Optics LLC and Lawrence Berkeley National Laboratory (LBNL). In Figure 9, characterization of a single, randomly selected multilayer coated grating at LBNL shows that the grating reflectivity is constant over the instrument FOV in the $m = 1$ order while the $m = 0$ order is almost completely suppressed. Figure 10 shows a schematic of the coating that achieves peak reflectiv-

Table 2. ESIS design parameters.

Component	Parameter	Calculated value (measured value)
Primary	Surface shape	Parabolic
	Focal length	1000.0 mm
	Aperture shape	Octagonal
	Aperture diameter	144 mm
	Coating	SiC single layer, optimized for O v 629.732 Å
	Efficiency (O v 629.732 Å)	25 % (19 %)
Field stop	Sky plane diameter	$11.5'$
	Aperture shape	Octagonal
	Aperture diameter	3.363 mm
Gratings (4)	Surface shape	Spherical
	Surface radius	597.830 mm (597.036 ± 0.156 mm)
	Aperture shape	Trapezoidal
	Aperture height	16.9 mm
	Aperture long base	17.2 mm
	Aperture short base	3.2 mm
	Ruling type	Varied line spacing
	Constant ruling spacing coefficient	0.38661 μm (0.38677 μm)
	Linear ruling spacing coefficient	-3.385×10^{-5} μm mm^{-1}
	Quadratic ruling spacing coefficient	-1.362×10^{-7} μm mm^{-2}
	Input angle	1.301°
	Output angle (O v 629.732 Å)	8.057°
	Manufacturing process	Individual master gratings
	Coating	Mg/SiC/Al multilayer, optimized for O v 629.732 Å
	Groove efficiency (O v 629.732 Å)	40 % (42 %)
	Efficiency (O v 629.732 Å)	14 % (13 %)
Filters (4)	Aperture shape	Circular
	Aperture diameter	30.0 mm
	Material	Al
	Thickness	100 nm
	Mesh transmission	82 %
	Mesh material	Ni
	Efficiency (O v 629.732 Å)	14 %
Detectors (4)	Manufacturer	Teledyne/e2v
	Model	CCD230-42
	Active area	2048 × 1024
	Pixel size	15 μm
	Quantum efficiency (O v 629.732 Å)	30 %
	Minimum cadence	2 s
System	Magnification	4.01
	f-number	$f/89$
	Plate scale along dispersion	$0.734''$ pix^{-1}
	Plate scale across dispersion	$0.770''$ pix^{-1}
	Dispersion	36.7 mÅ pix^{-1} (17.5 km s^{-1} pix^{-1})
	Passband	556.95 Å to 666.89 Å
	Efficiency (O v 629.732 Å)	0.10 %

ity and selectivity in the $m = 0$ order using four layer pairs of silicon carbide (SiC) and magnesium (Mg). The

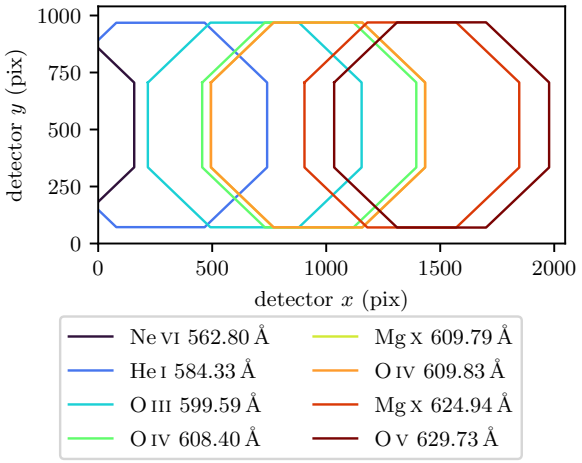


Figure 5. Areas occupied by strong spectral lines on the ESIS detectors. The plot axes are sized exactly to the CCD active area. The ESIS passband is defined by a combination of the field stop and grating dispersion.

Table 3. Coefficients of the quadratic distortion model, Equation 1, of a single ESIS channel, fit at the three target lines in the passband. The field angles x and y are measured from the center of the FOV, and the wavelength λ from 607.95 Å, the mean of the three lines. The detector coordinates x' and y' are in pixels from the lower left corner of the light-sensitive area of the detector.

Coefficient		x'	y'
$\mathbf{C}$	(pix)	9.13×10^{2}	5.20×10^{2}
$\mathbf{C}_{x}$	(pix arcsec^{-1})	1.36	8.95×10^{-7}
$\mathbf{C}_{y}$	(pix arcsec^{-1})	-1.25×10^{-7}	1.30
$\mathbf{C}_{\lambda}$	(pix Å^{-1})	2.72×10^{1}	-7.79×10^{-6}
$\mathbf{C}_{xx}$	(pix arcsec^{-2})	6.92×10^{-6}	3.70×10^{-9}
$\mathbf{C}_{xy}$	(pix arcsec^{-2})	1.09×10^{-9}	4.40×10^{-6}
$\mathbf{C}_{x\lambda}$	(pix Å^{-1} arcsec^{-1})	2.52×10^{-4}	-4.10×10^{-8}
$\mathbf{C}_{yy}$	(pix arcsec^{-2})	2.07×10^{-6}	1.21×10^{-9}
$\mathbf{C}_{y\lambda}$	(pix Å^{-1} arcsec^{-1})	2.50×10^{-8}	9.37×10^{-5}
$\mathbf{C}_{\lambda\lambda}$	(pix Å^{-2})	2.45×10^{-3}	3.35×10^{-7}

aluminum (Al) layers are deposited adjacent to each Mg layer to mitigate corrosion.

The maximum reflectance for the coating alone in the nominal instrument passband is ∼35 % in the upper panel of Figure 11, measured from witness samples coated at the same time as the diffraction gratings. Combined with the predicted groove efficiency from Section 4.1 and, given the relatively shallow groove profile and near normal incidence angle, the total reflectivity in first order is ∼14 % at O V 629.732 Å. This is confirmed by the first order efficiency measured from a single ESIS grating in the lower panel of Figure 11.

Unlike EUV imagers (e.g., TRACE (B. N. Handy et al. 1999), AIA (J. R. Lemen et al. 2012), and the High-Resolution Coronal Imager (Hi-C) (K. Kobayashi et al. 2014)) the ESIS passband is defined by a combination of the field stop and grating (Section 4.1, Figure 5) rather than multilayer coatings. The coating selectivity is therefore not critical in this respect, allowing the multilayer to be manipulated to suppress out-of-band bright, nearby emission lines. The lower panel of Figure 11 shows the peak reflectance of the grating multilayer is shifted slightly towards longer wavelengths to attenuate the He I 584.334 Å emission line, reducing the likelihood of detector saturation. A similar issue arises with the bright He II 303.785 Å line. Through careful design of the grating multilayer, the reflectivity at this wavelength is ∼2.5 % of that at O V 629.732 Å (upper panel of Figure 11). In combination with the primary mirror coating (described below) the rejection ratio at 303.785 Å is ∼37 dB. Thus, He II 303.785 Å emission is completely attenuated at the CCD.

The flight and spare primary mirrors were coated with the same Al/SiC/Mg multilayer. Corrosion of this multilayer rendered both mirrors unusable. The failed coating was stripped from primary mirror SN001. The mirror was then re-coated with a 5 nm thick layer of chromium (Cr) to improve adhesion followed by a 25 nm thick layer of SiC. The reflectance of this coating deposited on a silicon (Si) wafer witness sample appears in Figure 11. The spare primary mirror (SN002) retains the corroded Al/SiC/Mg multilayer.

The Si CCDs are sensitive to visible light as well as EUV. Visible solar radiation is much stronger than EUV, and visible stray light can survive multiple scatterings while retaining enough intensity to contaminate the EUV images. Luxél (F. R. Powell et al. 1990) Al filters 100 nm thick were used to shield each CCD from visible light. The Al film is supported by a 70 line per inch (lpi) nickel (Ni) mesh, with 82 % transmission. The theoretical filter transmission curve, modeled from CXRO data (B. L. Henke et al. 1993), is displayed in Figure 11. We conservatively estimate filter oxidation at the time of launch as a 4 nm thick layer of Al_2O_3.

An Al filter is positioned in front of the focal plane of each CCD by a filter tube, creating a light-tight box with a labyrinthine evacuation vent (e.g., Figure 15). The placement of the filter relative to the CCD is optimized so that the filter mesh shadow is not visible. By modeling the filter mesh shadow, we find that a position

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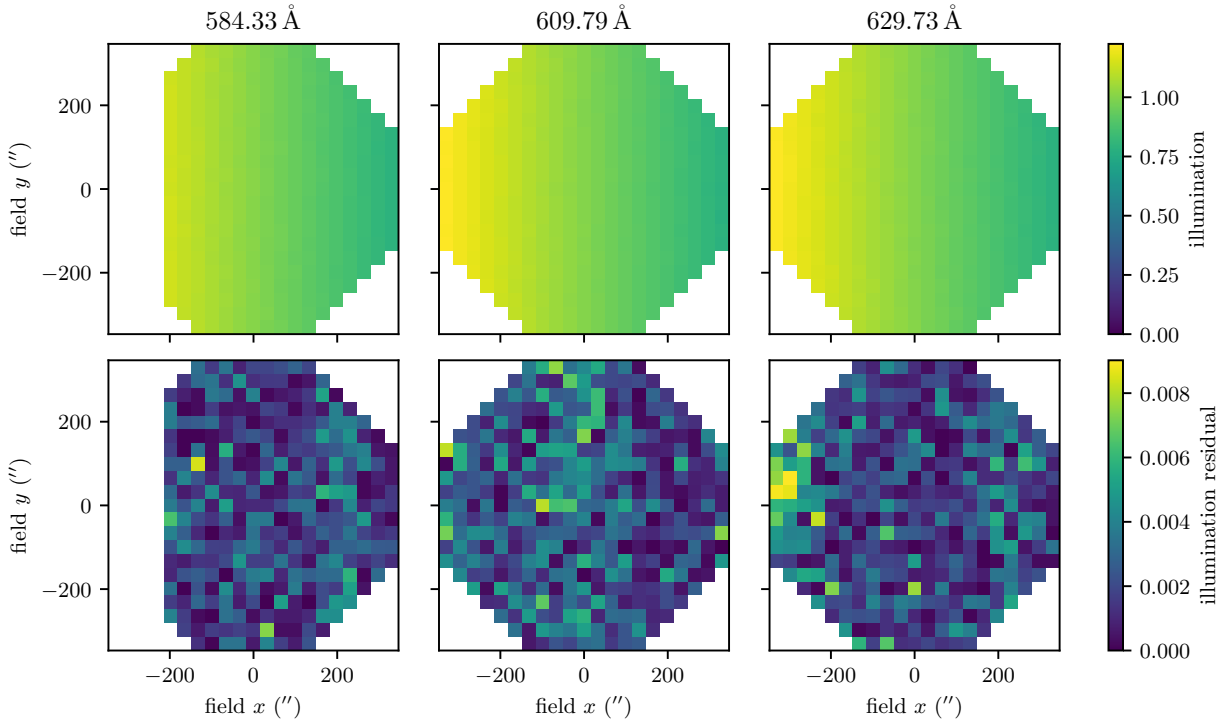


Figure 6. (Top) The relative illumination of a single ESIS channel as a function of position in the FOV, at each of the three target lines in the passband. The illumination is the fraction of the pupil which is unvignetted, normalized so that its average over the FOV is unity. (Bottom) The residual between that illumination and a linear model of it. Both rows are plotted only where the model was fit: the cells inside the FOV whose light lands on the detector. In the leftmost column, at He I, the FOV is cut off on the left by the edge of the detector.

far from the CCD (200 mm) and mesh grid clocking of 45° to the detector array reduces the shadow amplitude well below photon statistics. The MOSES instrument utilizes a similar design; no detectable signature of the filter mesh is found in data and inversion residuals from the 2006 MOSES flight.

To prevent oxidation, and to minimize the risk of tears, pinholes, and breakage from handling, the filters were stored in a nitrogen purged environment until after payload vibration testing.

4.5. *Sensitivity and Cadence*

Count rates for ESIS are estimated using the expected component throughput from Section 4.4 and the CCD quantum efficiency (QE) listed in Table 2. Line intensities are derived from J. E. Vernazza & E. M. Reeves (1978) (V&R) and the Solar and Heliospheric Observatory (SOHO)/Coronal Diagnostic Spectrometer (CDS) (R. A. Harrison et al. 1995) data, and are given in a variety of solar contexts: quiet sun (QS), coronal holes (CHs), and active regions (ARs). The 100 % duty cycle of ESIS (Section 4.8) gives us the flexibility to use the shortest exposures that are scientifically useful. So long as the shot noise dominates over read noise (which is true even for our coronal hole estimates at 10 s exposure length), we can stack exposures without a significant SNR penalty. Table 4 shows that ESIS is effectively shot noise limited with a 10 s exposure. The signal requirement in Table 1 is met by stacking exposures. Good quality images (∼300 counts) in active regions can be obtained by stacking 30 s worth of exposures. This cadence is sufficient to observe explosive events, but will not resolve torsional Alfvén waves described in Section 3. However, by stacking multiple 10 s exposures, sufficient SNR *and* temporal resolution of torsional Alfvén wave oscillations can be obtained.

4.6. *Alignment and Focus*

In the conceptual phase of ESIS, the decision was made to perform focus and alignment in visible light with a helium-neon (He-Ne) source. Certain difficulties are introduced by this choice, however, the benefits outweigh the operational complexity and equipment that would be required for focus in EUV. Moreover, a sounding rocket instrument requires robust, adjustment-free mounts to survive the launch environment. Such a de-

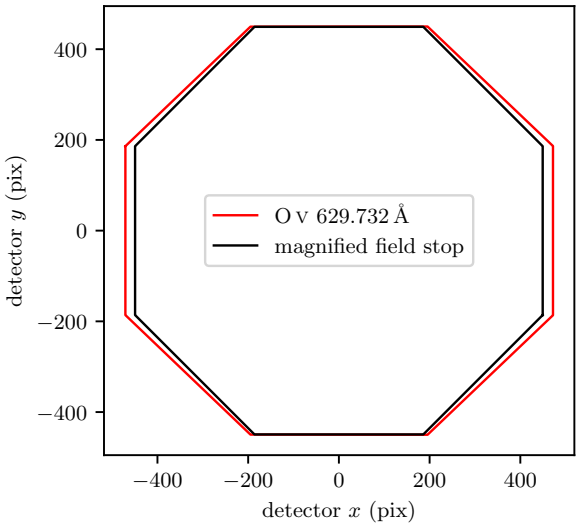


Figure 7. Plot of the magnified, undistorted field stop aperture vs. the distorted O v 629.732 Å image of the field stop aperture on the ESIS detector. The magnification factor used for the undistorted field stop aperture is the ratio of the grating exit arm to the grating entrance arm (4.01). The distorted image of the field stop aperture was calculated using the ESIS distortion model, described in Table 3.

Table 4. Estimated signal statistics per channel (in photon counts) for ESIS lines in CH, QS, and AR. Note that the SNR estimates are lower bounds since charge diffusion decreases the shot noise.

Source	V&R	V&R	V&R	CDS
Solar context	QS	CH	AR	AR
1 × 10 s exp.				
O v 629.732 Å	25	22	77	39
Mg x 624.941 Å	4	0	29	18
Total	29	22	106	57
Shot noise	5.4	4.7	10.3	7.5
Read noise	2.0	2.0	2.0	2.0
SNR	5.1	4.3	10.1	7.3
17 × 10 s exp.				
Total	493	369	1799	963
SNR	20.9	17.7	41.7	30.0

sign is not amenable to iterative adjustment in vacuum. The choice of alignment wavelength is arbitrary for most components; CCD response and multilayer coating reflectivity is sufficient across a wide band of visible wavelengths. The exceptions are the thin film filters (which were not installed until just before launch and have no effect on telescope alignment and focus) and the diffraction gratings. Visible light gratings were manufactured specifically for alignment and focus. These gratings are identical to the EUV flight version, but with a ruling pattern scaled to a 632.8 nm wavelength.

After alignment and focus had been obtained with the He-Ne source, the instrument was prepared for flight by replacing the visible gratings by the EUV flight versions. Each grating (EUV and alignment) is individually mounted to a backing plate using a bipod system similar to that of the primary mirror. The array of gratings on their backing plates are in turn mounted to a single part which we call the 'tuffet.' The backing plate is attached to the tuffet by bolts through spherical washers. With this mounting scheme, the gratings can be individually aligned in tip/tilt. The tuffet is attached to the secondary mirror mount structure (Figure 13). This enables the entire grating array to be replaced simply by removing/installing the tuffet, such as when switching between alignment and EUV gratings.

Properly positioning the gratings was expected to be the most difficult aspect of final telescope assembly. Table **??** shows that the telescope is very sensitive to defocus. The depth of field is an order of magnitude smaller in EUV than in visible light. Moreover, the sensitivity to defocus at the gratings is $M^2 + 1 = 17$ times greater than at the detectors, where $M = 4.01$ is the ratio of the grating exit arm to the grating entrance arm. Another sensitive aspect of the telescope is grating tip/tilt. A tolerance of $\sim \pm 0.34$ mrad is needed to ensure that the entire image lands on the active area of the CCD.

Once the visible gratings were aligned and focused, the challenge was to transfer this alignment to the EUV flight gratings. We performed this transfer using a 4D Phasecam 6000 interferometer. We aligned the interferometer to each visible grating such that the tip, tilt, and defocus aberrations were zero. Then, placing the corresponding flight grating in the visible grating's place, we shimmed the mounting screws of the flight grating to match the tip, tilt, and defocus of the alignment grating. Since the mounting interface for the tuffet is extremely repeatable by design we were able to swap tuffets and compare their alignment and focus to ensure everything was transferred correctly.

When transferring the focus and alignment two key details were considered. First, there is nothing about the tuffet that constrains the grating roll. Therefore, we needed to ensure each flight grating had the same roll as each visible grating. This was accomplished by using a He-Ne laser diverged through a cylindrical optic that illuminated each grating with a line perpendicular better make sure I know which direction is implied here to the

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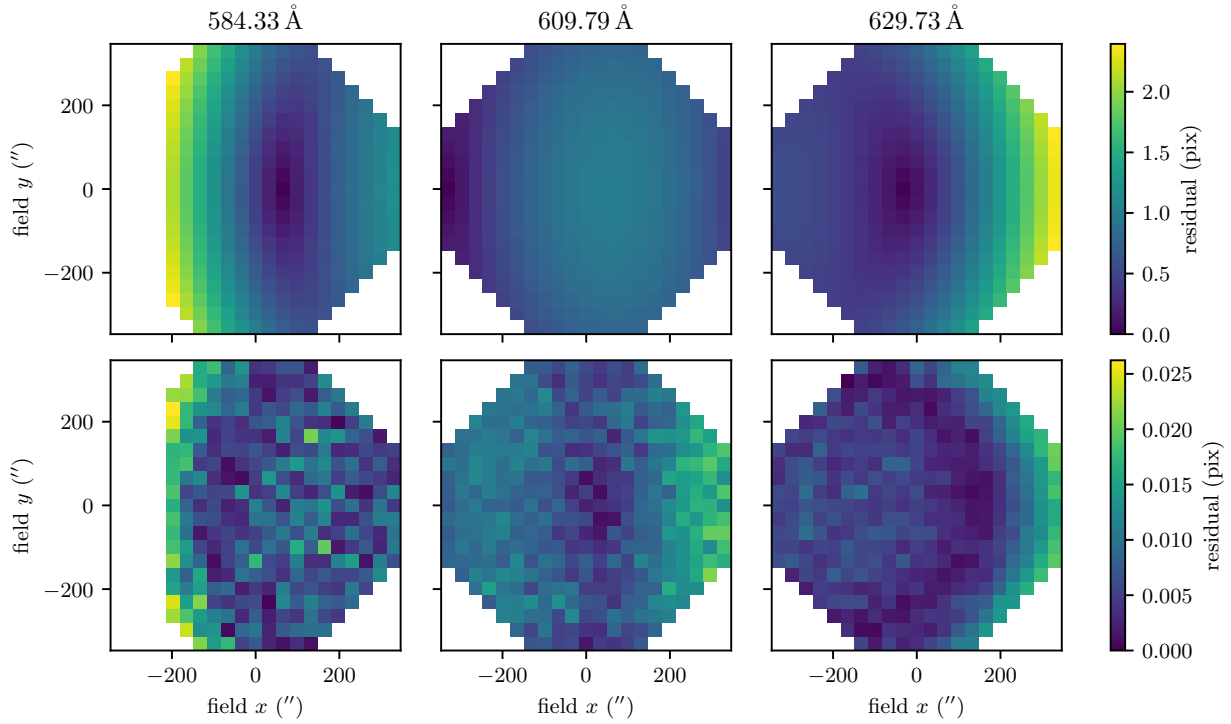


Figure 8. Magnitude of the residual between a linear distortion model and the raytrace model (top) and between a quadratic distortion model and the raytrace model (bottom). This figure demonstrates that a quadratic distortion model is sufficient to achieve sub-pixel accuracy.

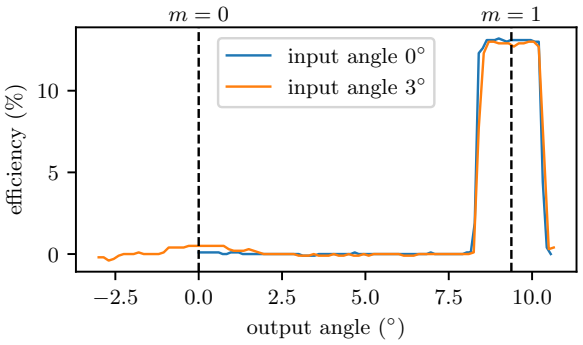


Figure 9. Measured efficiency at 63 nm of the Channel 2 grating as a function of reflection angle on 2018 January 21. Note flat response in first order over instrument FOV and suppression of zero order.

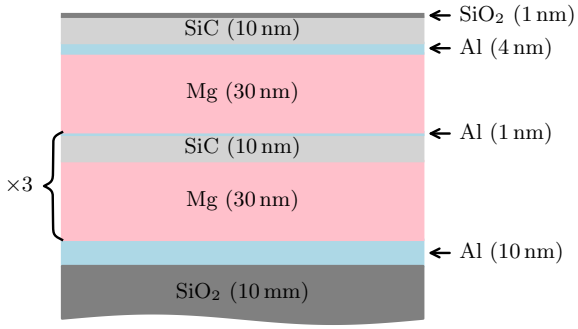


Figure 10. Schematic of the Al/SiC/Mg multilayer with $N = 4$ layers.

grating blaze direction. The line of light was reflected back onto a ruled target that could be compared between gratings. Since our alignment gratings were ruled to image light of approximately an order of magnitude longer wavelength the laser and cylindrical optic were placed at the position of Littrow for the 10th order image in the visible grating, and the 1st order for the EUV gratings. Second, during testing we measured slight differences in radius of curvature between each grating. Therefore each flight grating was prescribed a specific amount of defocus to account for the difference in radius of curvature between each optic when transferring alignment and focus.

4.7. *Apertures and Baffles*

Needs a big rework Each channel of ESIS was designed with two apertures: one at the surface of the grating and

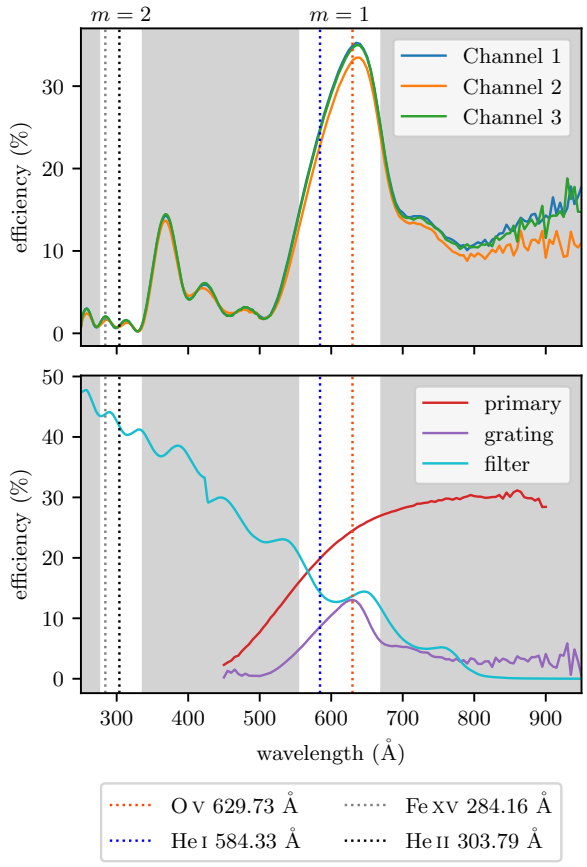


Figure 11. (Top) Measured reflectance for several multilayer coated witness samples at an incidence angle of 4° on 2018 January 21. The white regions indicate wavelengths that intercept the detector and the gray regions indicate wavelengths that miss the detector. Note the suppression of second order relative to the first order and the consistency of the coatings between each channel. The Channel 4 grating measurement is missing due to issues in the measurement apparatus. (Bottom) Comparison of the efficiency of the three main ESIS optical components: primary mirror, grating and filter. The primary mirror efficiency is based on measurements of a Si witness sample taken on 2018 May 10 at an angle of incidence of 4°. The grating efficiency is from a measurement of the Channel 2 grating taken on 2018 January 21 at an angle of incidence of 0°. The filter efficiency is a theoretical model that includes the filter mesh, 100 nm of Al and 4 nm of Al oxide.

a stop in front of the primary mirror. The purpose of the aperture at the grating is to mask the out-of-figure margins at the edges of these optics. This provides a well defined edge to the clear aperture of each grating while also keeping unwanted rays from being reflected from the grating margins and back onto the CCDs. The dimensions of the grating aperture match those of the grating clear aperture shown in Figure 4c.

The stop at the primary mirror was designed to be the stop of each channel. Its area was maximized under the constraint that no rays be vignetted anywhere else in the system. The gratings and their clear apertures were the most significant areas of concern for potential vignetting, so the shape of the stop was largely influenced by the shape of the grating clear aperture. The inner extent of the stop was defined by the occultation of the primary by the shadow cast from the gratings and their mounts. This presented an intricate geometry problem, as the occultation is a function of the incoming field angle, the radial extent of the grating mount, and the distance of the mount to the primary mirror along the optical axis. Hence, the inner extent of the stop was solved for iteratively with the optimization described in Section **??**, which affected the placement of the gratings relative to the primary mirror.

After final optimization, the stop geometry was analyzed with the optical model to check for vignetting at the grating. A footprint diagram was generated at the grating from multiple grids of rays. The incidence angle of each grid of rays corresponded to the extremes of the FOV defined by the positions of the eight points of the octagonal field stop. The footprint diagram showed that, with the stop completely filled, no ray landed outside of the grating clear aperture, and no ray was intercepted by the central obscuration.

This non-vignetting stop left considerable surface area of the primary mirror unused, which could only be put to use by relaxing the vignetting constraint. Before flight, the stop was removed to increase the sensitivity of the instrument at the expense of vignetting (Section 4.2), which made the grating aperture the stop of each channel. The footprint of the beam of each channel on the primary mirror in Figure 4b is that of the instrument as flown, without the stop.

The ESIS baffles are designed to block direct light paths between the front aperture plate and the CCDs for any ray <1.4° from the optical axis. This angle is purposefully larger than the angular diameter of the sun (∼0.5°) so that any direct paths are excluded from bright sources in the solar corona. All baffles are bead-blasted, anodized Al sheet metal oriented perpendicular to the optical axis. The size and shape of the cutouts were determined using a combination of the ray trace from Section **??** and 3D modeling. Between the primary mirror and the gratings, each cutout is the footprint of the rays of that trace where they cross the baffle. Between the gratings and the CCDs, the light path is defined as the volume that connects each vertex of the

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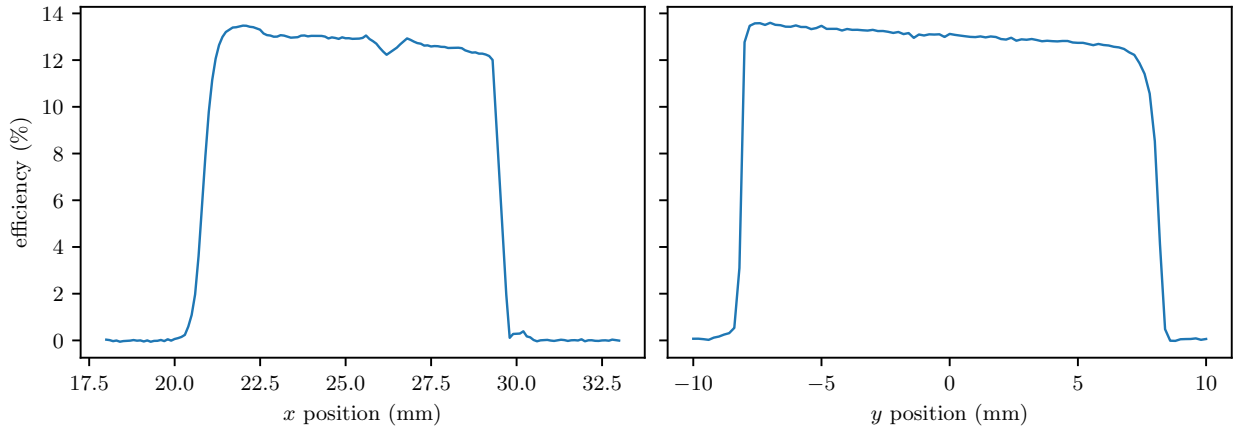


Figure 12. Channel 2 grating efficiency at 63 nm vs. position for two orthogonal slices across the optical surface on 2018 January 21.

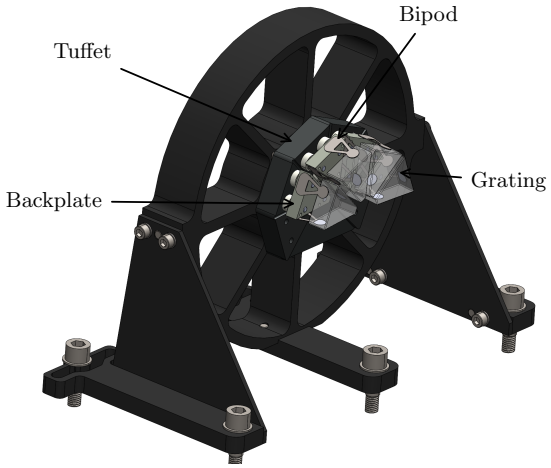


Figure 13. A solid model of the ESIS secondary mount. Each trapezoidal grating is supported and attached to its backplate by three bonded, titanium bipods. All mounted gratings sit on the "tuffet" and are mounted with three spherical washers. This allows independent tip/tilt adjustment of each grating, and for all of them to be removed/reinstalled without compromising alignment and focus.

grating clear aperture to every vertex of the image of the octagonal field stop on the CCD. This is a conservative definition that ensures no rays within the FOV are excluded, and therefore unintentionally vignetted by the baffles. The cutouts there are the projection of this volume onto the baffle. A conservative 1 mm margin is added to each cutout to prevent unintentional vignetting. A model of the six baffles, showing their cutouts and positions on the optical bench, is displayed in Figure 14.

4.8. *Cameras*

The ESIS CCD cameras were designed and constructed by MSFC and were the latest in a series of camera systems developed specifically for use on solar space flight instruments. The ESIS camera heritage includes those flown on both the Chromospheric Lyman-alpha Spectro-polarimeter (CLASP) (R. Kano et al. 2012; K. Kobayashi et al. 2012) and Hi-C (K. Kobayashi et al. 2014).

The ESIS detectors are CCD230-42 astro-process CCDs from Teledyne/e2v. For each camera, the CCD is operated in a split frame transfer mode with each of the four ports read out by a 16-bit A/D converter. The central 2048×1024 pixels of the $2k \times 2k$ device are used for imaging, while the outer two regions are used for storage. Two overscan columns on either side of the imaging area and eight extra rows in each storage region were intended to monitor read noise and dark current. When the camera receives the trigger signal, it transfers the image from the imaging region to the storage regions and starts image readout. The digitized data are sent to the data acquisition and control system (DACS) through a SpaceWire interface immediately, one line at a time. The frame transfer takes 50 ms, and readout takes 1.1 s. The cadence is adjustable from 2 s to 600 s in increments of 100 ms, to satisfy the requirement listed in Table 1. Because the imaging region is continuously illuminated, the action of frame transfer (transferring the image from the imaging region to the storage regions) also starts the next exposure without delay. Thus the exposure time is controlled by the time period between triggers. Camera 1 (Figure 15) generates the sync trigger, which is fed back into Camera 1's trigger input and provides independently buffered triggers to the remaining three cameras. The trigger signals are synchronized

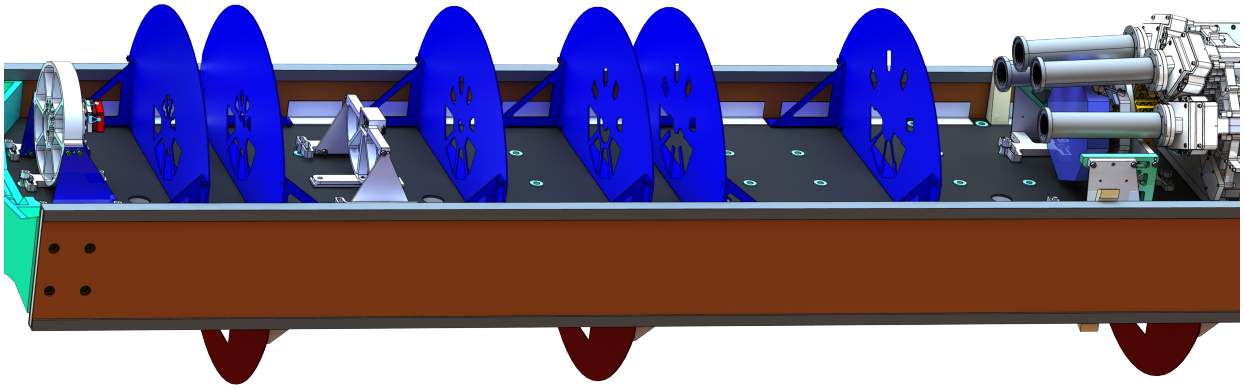


Figure 14. Model view of ESIS baffle placement and cutouts.

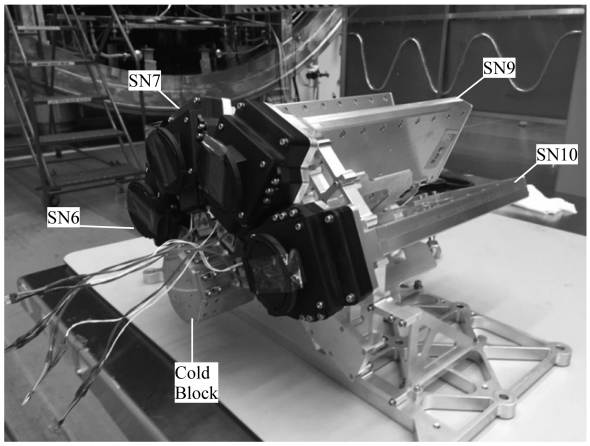


Figure 15. ESIS camera assembly as built by Marshall Space Flight Center (MSFC). Thin film filters and filter tubes are not installed in this image.

to better than ±1 ms. Shutterless operation allows ESIS to observe with a 100 % duty cycle. The cadence is limited only by the 1.1 s readout time.

MSFC custom designed the camera board, enclosure, and mounting structure for ESIS to fit the unique packaging requirements of this experiment (Figure 15). The front part of the camera is a metal block which equalizes the temperature across the CCD while fastening it in place. The carriers of all cameras are connected to a central two-piece copper (3 kg) and aluminum (1 kg) thermal reservoir (cold block) by flexible copper cold straps. The flexible cold straps allow individual cameras to be translated parallel to the optical axis (by means of shims) up to ∼13 mm to adjust focus in each channel prior to launch. The centrally located cold block was cooled by liquid nitrogen (LN_2) flow from outside the payload until just before launch. The LN_2 flow was controlled automatically by a ground support equipment (GSE) computer so that all cameras were maintained above survival temperature but below the target temperature of −55°C to ensure a negligible dark current level.

The gain, read noise, and dark current of the four cameras were measured at MSFC using an ^{55}Fe radioactive source. Cameras are labeled 1, 2, 3, and 4 with associated serial numbers SN6, SN7, SN9, and SN10 respectively in Figure 15. Gain ranges from 2.50 to 2.63 e^- DN^{-1} in each quadrant of all four cameras. Table 5 lists gain, read noise, and dark current by quadrant for each camera.

The QE of the ESIS CCDs was not measured before flight. This paper models it with the measurements of a back-illuminated e2v CCD97 by I. Moody et al. (2017) and J. Heymes et al. (2020), which give a QE of 39 % at He I 584.334 Å and 30 % at O V 629.732 Å.

4.9. *Avionics*

The ESIS DACS is based on the designs used for both CLASP (R. Kano et al. 2012; K. Kobayashi et al. 2012; N. Narukage et al. 2016) and Hi-C (K. Kobayashi et al. 2014; L. A. Rachmeler et al. 2019). The electronics are a combination of military off-the-shelf (MOTS) hardware and custom designed components. The DACS is a 6-slot, 3U, open VPX PCIe architecture conduction cooled system using an AiTech C873 single board computer. The data system also includes a MOTS PCIe switch card, MSFC parallel interface card, and two MOTS Spacewire cards. A slot for an additional Spacewire card is included to accommodate two more cameras for the next ESIS flight. The C873 has a 2.4 GHz Intel i7 processor with 16 GB of memory. The operating temperature range for the data system is −40 °C to 85 °C. The operating system for the flight data system is Linux Fedora 23.

The DACS is responsible for several functions; it controls the ESIS experiment, responds to timers and uplinks, acquires and stores image data from the cam-

Table 5. ESIS camera properties.

Camera	Quad.	Gain ($e^- DN^{-1}$)	Read noise (e^-)	Dark current ($e^- s^{-1}$)
1 (SN6)	1	2.57	10.3	0.061
	2	2.50	10.5	0.051
	3	2.52	10.7	0.049
	4	2.53	9.7	0.061
2 (SN7)	1	2.55	10.4	0.117
	2	2.58	10.8	0.102
	3	2.57	10.7	0.156
	4	2.63	10.8	0.175
3 (SN9)	1	2.57	10.6	0.059
	2	2.53	10.4	0.050
	3	2.52	10.4	0.054
	4	2.59	11.1	0.067
4 (SN10)	1	2.60	10.1	0.117
	2	2.60	10.1	0.100
	3	2.54	10.7	0.125
	4	2.58	10.5	0.153

eras, downlinks a subset of images through telemetry, and provides experiment health and status. The DACS is housed with the rest of the avionics (power supply, analog signal conditioning system) in a 0.56 m to 0.43 m transition section outside of the experiment section. This relaxes the thermal and cleanliness constraints placed on the avionics. Custom DC/DC converters are used for secondary voltages required by other electronic components. The use of custom designed converters allowed additional ripple filtering for low noise.

4.10. *Pointing System*

The imaging target was selected prior to launch, the morning of the day of flight. During flight, pointing was maintained by the Solar Pointing Attitude Rocket Control System (SPARCS) (Lockheed Missiles and Space Co. Report 1969). Images from Camera 1 were downlinked and displayed in real time on the SPARCS control system console at intervals of ∼16 s to verify pointing was maintained during flight.

4.11. *Mechanical*

ESIS and MOSES are mounted on opposite sides of a composite optical table structure originally developed for the Solar Plasma Diagnostics Experiment (SPDE) (M. E. Bruner 1995). The layered carbon fiber structure features a convenient, precisely coplanar array of threaded inserts with precision counterbores. The carbon fiber layup is designed to minimize the longitudinal coefficient of thermal expansion. The optical table is housed in two 0.56 m diameter skin sections, with a total length of 3 m. A ball joint and spindle assembly on one end and flexible metal aperture plate on the other hold the optical table in position inside the skin sections. The kinematic mounting system isolates the optical table from bending or twisting strain of the skins.

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FLOWS, MAGNETIC TOPOLOGY, AND TEMPERATURE OF A TRANSITION REGION JET OBSERVED BY THE EUV SNAPSHOT IMAGING SPECTROGRAPH

Contribution of Authors and Co-Authors

Manuscript in Chapter 5

Author: Roy T. Smart
Contributions: **XX**

Co-Author: Charles C. Kankelborg
Contributions: **XX**

Co-Author: Jacob D. Parker
Contributions: **XX**

Co-Author: Dana W. Longcope
Contributions: **XX**

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Flows, Magnetic Topology, and Temperature of a Transition Region Jet Observed by the EUV Snapshot Imaging Spectrograph

Roy T. Smart [1], Charles C. Kankelborg [1], Jacob D. Parker [2] and Dana W. Longcope [1]

[1] *Montana State University, Department of Physics, P.O. Box 173840, Bozeman, MT 59717, USA*
[2] *NASA Goddard Space Flight Center, Greenbelt, MD 20771, USA*

ABSTRACT

Jets and explosive events in the solar transition region are thought to be driven by magnetic reconnection, but the spatial structure of their flows is difficult to observe, since a slit spectrograph records it one position at a time. The EUV Snapshot Imaging Spectrograph (ESIS) is a slitless, computed tomography imaging spectrograph which recorded the quiet Sun near disk center in O v 630 Å and several other transition region and coronal lines across a wide field of view during a sounding rocket flight on 2019 September 30. The largest event it observed, event E, is a jet with an inverted-Y morphology which lasted 14 min and extended 64 Mm. We invert the ESIS images into spectral cubes using the multiplicative algebraic reconstruction technique (MART), and measure Doppler shifts of up to **XX** toward and **XX** away from the observer. We compute the magnetic topology beneath the event from SDO/HMI magnetograms using magnetic charge topology, and compare the separatrices of the coronal field with the locations of the flows. We combine intensities from SDO/AIA, Hinode/XRT, and ESIS to compute differential emission measures of the event, and find a peak temperature of 2 MK. One or two sentences of conclusions: whether the flows are organized by the separatrices, and what the temperature implies about the coronal counterpart of the jet.

Keywords: Solar transition region (1532), Solar magnetic reconnection (1504), Solar magnetic fields (1503), Solar extreme ultraviolet emission (1493), Spectroscopy (1558), Doppler shift (401)

1. INTRODUCTION

The solar transition region (TR) is host to a variety of small, transient phenomena whose spectral line profiles reveal flows of order $100\,\mathrm{km\,s^{-1}}$. Explosive events, first observed by the High Resolution Telescope and Spectrograph (HRTS) as high-velocity jets in the quiet Sun (G. E. Brueckner & J.-D. F. Bartoe 1983), are compact, short-lived brightenings whose line profiles are broadened into one or both wings by Doppler shifts of about this size (K. P. Dere et al. 1989). D. E. Innes et al. (1997) found that the blue- and red-shifted emission of these events is separated along the slit, as expected of the bidirectional outflows from a reconnection site, and explosive events have since been widely interpreted as the signature of magnetic reconnection in the TR (K. P. Dere et al. 1991; D. E. Innes et al. 2015). They are associated with the evolution of the photospheric magnetic field (K. Muglach 2008), and a compact TR brightening has been found to occur in a magnetic field with a fan-spine topology (L. P. Chitta et al. 2017).

Jets are related, collimated phenomena which are observed across a wide range of temperatures, from network jets in the TR (H. Tian et al. 2014) to coronal X-ray jets (K. Shibata et al. 1992; N. E. Raouafi et al. 2016). Coronal jets commonly have an inverted-Y morphology, which is interpreted as reconnection between a small closed bipole and the surrounding large-scale field (K. Shibata et al. 1992; N. E. Raouafi et al. 2016), at a null point whose fan surface divides the two flux systems (E. Pariat et al. 2009). They are divided into standard and blowout jets (R. L. Moore et al. 2010), and many have been found to be driven by the eruption of a minifilament, in coronal holes (A. C. Sterling et al. 2015), in active regions (A. C. Sterling et al. 2016), and in the quiet Sun, where the eruption is triggered by flux cancellation (N. K. Panesar et al. 2016). The breakout model unifies these observations as reconnection at a null point above an erupting filament channel (P. F. Wyper et al. 2017, 2018). Whether explosive events, network jets, and coronal jets are the same process at different

Corresponding author: Roy T. Smart
Email: roytsmart@gmail.com

scales remains open (Y. Chen et al. 2019), and one part of that question is whether explosive events have a coronal counterpart at all (L. Teriaca et al. 2002).

These models make specific predictions about where flows should appear relative to the magnetic structure, and testing them requires the spectrum at every point of an event at the same time. A slit spectrograph samples only one position at a time, so a raster across an evolving event conflates its spatial and temporal structure, while an imager records the morphology of the event but measures flows only in the plane of the sky. The EUV Snapshot Imaging Spectrograph (ESIS) was designed to resolve this ambiguity. It is a slitless spectrograph which records several dispersed images of the Sun at once, each dispersed in a different direction, from which a spectral cube of the whole field can be recovered in the manner of a computed tomography imaging spectrograph (CTIS) (T. Okamoto & I. Yamaguchi 1991; M. Descour & E. Dereniak 1995). It is the successor to the Multi-Order Solar EUV Spectrograph (MOSES) (C. C. Kankelborg & R. J. Thomas 2001), whose flights recorded explosive events in He II 304 Å (J. L. Fox et al. 2010; T. Rust & C. C. Kankelborg 2019) and showed how the spectral content of overlapping images can be recovered (H. T. Courrier & C. C. Kankelborg 2018; J. D. Parker & C. C. Kankelborg 2022).

ESIS flew for the first time on 2019 September 30. J. D. Parker et al. (2022) reported the first results of that flight, including the Doppler shifts of several explosive events in O V 630 Å, and identified among them event E, the largest and most complex velocity event the instrument observed. In the images of event E they found a strong redshift at its brightest point and faint, blue-shifted material rising from it along a thin structure which is also visible in the Atmospheric Imaging Assembly (AIA) images, which they suggested might be a small jet or minifilament eruption (A. C. Sterling et al. 2015). Its complexity defeated an interpretation from the images alone, and they deferred its analysis to a future publication. Here we recover its spectra by inverting the ESIS images with the multiplicative algebraic reconstruction technique (MART) (R. Gordon et al. 1970; T. Okamoto & I. Yamaguchi 1991), which reconstructs the spectral cube at every exposure.

If event E is driven by reconnection, its flows should originate where reconnection can occur. In magnetic charge topology (MCT) (D. W. Longcope & I. Klapper 2002; D. W. Longcope 2005), the photospheric field is represented by discrete magnetic sources, and the potential coronal field above them is divided into domains of distinct connectivity by separatrix surfaces, which intersect along separators where reconnection transfers flux between domains. MCT has been applied to active regions (G. Barnes et al. 2005), and the X-ray brightness of coronal bright points is consistent with the energy released by reconnection at the separator between a converging pair of sources (D. W. Longcope et al. 2001). Computed from the Helioseismic and Magnetic Imager (HMI) magnetograms (P. H. Scherrer et al. 2012) beneath event E, the separatrices predict where its bidirectional flows should be, and the Doppler maps from ESIS show where they are.

A second question is how hot the event becomes. Differential emission measures (DEMs) are routinely computed from the extreme ultraviolet (EUV) channels of AIA (J. R. Lemen et al. 2012) using a variety of methods (I. G. Hannah & E. P. Kontar 2012; J. Plowman et al. 2013; M. C. M. Cheung et al. 2015; J. Plowman & A. Caspi 2020), but those channels are dominated by coronal lines and constrain the TR poorly, while the X-Ray Telescope (XRT) (L. Golub et al. 2007) is sensitive to the hottest plasma. The ESIS passband contains lines of He I, O III, O IV, O V, and Mg X, which together span the TR and the low corona, so that combining all three instruments constrains the DEM of the event from the TR to the corona.

In this paper we analyze event E with all three of these tools. Section 2 describes the observations from ESIS, AIA, HMI, and XRT and gives an overview of the event. Section 3 describes the inversion of the ESIS images and the Doppler shifts of the event, Section 4 the magnetic topology computed from HMI, and Section 5 the DEMs. We discuss what they imply about the mechanism of the event in Section 6 and conclude in Section 7.

2. OBSERVATIONS

Outline.

- **The ESIS flight** (J. D. Parker et al. 2022). Launched from White Sands Missile Range at 18:04 UT on 2019 September 30, with 10 s exposures from 18:06:11 to 18:11:01 UT. The octagonal field of view (FOV), about 11.5′ across, was pointed near disk center during an exceptionally quiet period, in which GOES had detected no flare of B class or above since 2019 July 7. The passband and its lines: He I 584 Å, O III 600 Å, O IV 608 Å and 610 Å, Mg X 610 Å and 625 Å, and O V 630 Å, with the temperature at which each is formed.
- **Coordinated observations.** AIA EUV images at a 12 s cadence (J. R. Lemen et al. 2012); HMI LOS magnetograms at a 45 s cadence, about seven of which cover the flight (P. H. Scherrer et al. 2012); XRT images through its Al_poly filter, the only X-ray filter it used during the flight, of a field 6.6′ across near disk center at 1.03″ per pixel, at a cadence of about 18.5 s throughout the flight (L. Golub et al. 2007; T. Kosugi et al. 2007). The Interface Region Imaging Spectrograph (IRIS) ran coarse rasters near disk center during

the launch window, but its slit never crossed event E, so there are no IRIS spectra of it; its 1400 Å slit-jaw images show most of event E, which is cut off by the edge of their field.

- **Co-alignment.** ESIS to AIA 304 Å (which correlates best with He I), and HMI and XRT to AIA, with the residual misalignment that bounds the comparisons in Sections 4 and 5.
- **Overview of event E.** Its location, (47.8″, −87.8″), about 100″ from disk center, and its lifetime (14 min, from 18:02:54 to 18:16:30 UT), including the parts before and after the ESIS exposures from AIA, over which its base is brighter in AIA 304 Å, the channel in which it brightens first, than before the event by more than 10 % of its peak brightening. Its inverted-Y morphology in O V and in the AIA channels, an order of magnitude brighter than the surrounding network, and the northward drift of its blue-shifted side by about 11″ over four minutes. Its length (64 Mm), in the plane of the sky from its base to its northern footpoint, where the fainter arc which continues its spire ends. Its light curves in every EUV channel of AIA and in XRT (Figure 1), from which its lifetime is measured.
- **Phases of the event**, as seen in the preliminary inversions: (1) blue shifts of about 60 km s^{-1} directly beside strong red shifts near the right footpoint of the inverted Y; (2) the blue-shifted component migrates to the cusp while the red-shifted component stays at the footpoint; (3) a blue-shifted blob moves along the right leg at an apparent speed of 100 km s^{-1} to 200 km s^{-1} in the plane of the sky.
- **Figures:** the full ESIS field with event E boxed; the light curves of event E (Figure 1); a time series of event E in ESIS O V, AIA 304 Å, 171 Å and 193 Å, XRT, and HMI.

3. SPECTRAL INVERSION AND DOPPLER SHIFTS

Outline.

- **Forward model.** Each ESIS channel records a dispersed image of the field, so the images are a linear function of the spectral radiance $I(x, y, \lambda)$. Describe the model of that function built from the flight optics (the `esis` and `optika` packages), including the point-spread function and the vignetting.
- **The MART inversion** (R. Gordon et al. 1970; T. Okamoto & I. Yamaguchi 1991), implemented in the `ctis` package. The update rule, the initial guess, and the stopping criterion. The reconstructed cubes sample the sky every 0.75″ and the line of sight every 17.5 km s^{-1} over ±210 km s^{-1} about each line, in five spectral windows, and are reconstructed independently at each exposure. How the blend of O IV and Mg X near 610 Å is handled, and why the radiometry of the channels is relative, so that velocities are trusted further than intensities.

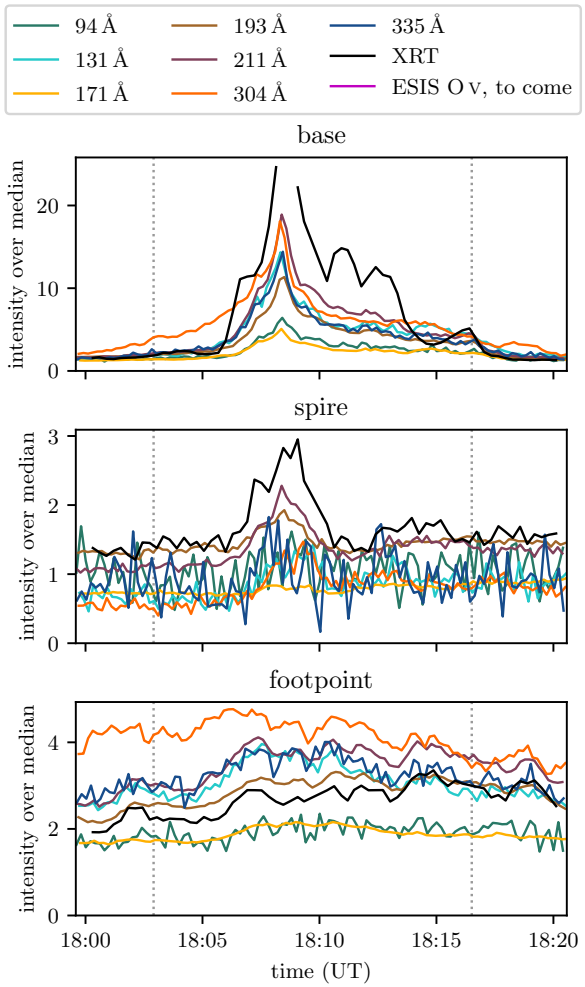


Figure 1. The light curves of the base of event E (top), the hottest part of its spire (middle), and its northern footpoint (bottom), the three points circled in Figure 4, in each EUV channel of AIA, labeled by its wavelength, and in XRT Al_poly, at the middle of every exposure from 17:59:35 to 18:20:34 UT. Each is the mean intensity of the pixels within 3″ of the point, or within 1.5″ of the spire, which is narrower, over the median intensity of its image, so that 1 is the median in every channel. The curve of XRT at the base has a gap where XRT saturated. The dotted lines mark the lifetime of event E, from 18:02:54 to 18:16:30 UT, over which the base is brighter in AIA 304 Å than in the first 2 min of the curve by more than 10 % of its peak brightening. The ESIS O V curves are left out until the ESIS data are ready.

- **Validation.** Invert synthetic ESIS images of a known scene containing a bidirectional jet, and report how well the recovered Doppler shifts and widths match the truth as a function of brightness and velocity. This sets the uncertainties quoted for event E, and the speed

beyond which a fast component is poorly constrained by the spectral window.

- **Doppler shifts of event E.** Line-of-sight velocity and width maps from the inverted cubes, from moments and from fits of two Gaussian components to each O V profile, a core near rest and a fast component. Peak speeds of the blue and red components (**XX** and **XX**). Preliminary fits find blue wings reaching about 150 km s^{-1} and a separate red component near 100 km s^{-1}, while the median Doppler shift, which moves only part of the way toward a fast component in one wing, peaks near 60 km s^{-1}; settle which of these the article quotes.
- **Other lines.** The red-shifted footpoint appears in O III and O IV, while the blue-shifted material at the cusp is seen only in O V and faintly in He I; whether this is a matter of brightness or of temperature.
- **Motion.** The plane-of-sky speed of the blue-shifted component, combined with its line-of-sight speed into a three-dimensional velocity.
- **Figures:** the inverted line profiles at a few points in the event; a time series of Doppler maps of event E; the plane-of-sky track of the fast components.

4. MAGNETIC TOPOLOGY

Outline.

- **Magnetograms.** The HMI LOS magnetograms (P. H. Scherrer et al. 2012) around the time of the flight, their co-alignment with ESIS, and the evolution of the photospheric flux beneath event E over the preceding hours: is there a parasitic polarity at the footpoint of the inverted Y, and is it emerging or cancelling (N. K. Panesar et al. 2016)?
- **MCT** (D. W. Longcope & I. Klapper 2002; D. W. Longcope 2005). Partition the magnetogram into unipolar sources, represent each as a point charge, and compute the potential field they produce (G. Barnes et al. 2005). Locate its null points, spines, separatrix surfaces, and separators, and in particular any fan-spine structure above the event and the height of its null.
- **Separatrices and flows.** Overlay the separatrix footprints and the location of any null point or separator on the Doppler maps of Section 3. Test whether the blue and red components lie on opposite sides of a separatrix, whether the cusp of the inverted Y lies beneath the null, and whether the migration of the blue-shifted component to the cusp follows a separatrix.
- **Energetics.** The flux transferred across the separator during the event, and the energy that its reconnection could release (D. W. Longcope et al. 2001).
- **Robustness.** How the topology changes with the partitioning threshold and from one magnetogram to the next, and whether the features that matter to event E persist in the weak fields of the quiet Sun.
- **Figures:** the magnetogram with its sources and separatrix footprints; the same footprints overlaid on the Doppler maps; a three-dimensional view of the separatrices and the null.

5. TEMPERATURE

Outline.

- **Data.** AIA EUV intensities (J. R. Lemen et al. 2012), XRT intensities (L. Golub et al. 2007), and the ESIS line intensities, co-aligned and averaged over the event and over a background region of network, at several times through the event.
- **Temperature responses.** The AIA and XRT responses and the contribution functions of the ESIS lines, all computed from CHIANTI (G. Del Zanna et al. 2021) with the same abundances and ionization equilibrium so that the instruments can be combined. The ESIS lines of O III, O IV, and O V extend the coverage of AIA and XRT down into the TR, and Mg X overlaps the coolest coronal channels of AIA. He I and AIA 304 Å are optically thick and are left out of the inversion. The AIA and XRT responses the DEMs are inverted from so far are plotted in Figure 2.
- **Radiometric calibration.** The ESIS intensities are needed in absolute units here, unlike in Section 3; how the channels are cross-calibrated against AIA and the uncertainty this contributes.
- **DEM inversion** (J. Plowman et al. 2013; J. Plowman & A. Caspi 2020), implemented in the `utu` package, cross-checked against an independent method (I. G. Hannah & E. P. Kontar 2012; M. C. M. Cheung et al. 2015). Report the DEMs with and without each instrument, to show what each contributes (Figure 3).
- **Results.** The DEM of event E compared with that of its surroundings, its peak temperature (2 MK, where the DEM of the base from AIA and XRT in Figure 3 peaks), and whether the event has a coronal component visible to XRT or is confined to the TR, the question raised by L. Teriaca et al. (2002). The DEM of the base also has a second, cooler peak in the TR, which AIA and XRT constrain poorly and the ESIS lines will pin down. Whether the temperature differs between the cusp and the footpoint of the inverted Y.
- **Evolution.** The DEM of event E from AIA and XRT for every XRT image from 18:00:12 to 18:19:58 UT (Figure 4), inverted as in the DEM tutorial of the `hinode` package, at the full resolution of XRT, with and without the background subtracted.
- **Figures:** the temperature responses of the three instruments (Figure 2); the DEMs of event E and of the background (Figure 3); the DEM of event E through its lifetime (Figure 4).

6. DISCUSSION

Outline.

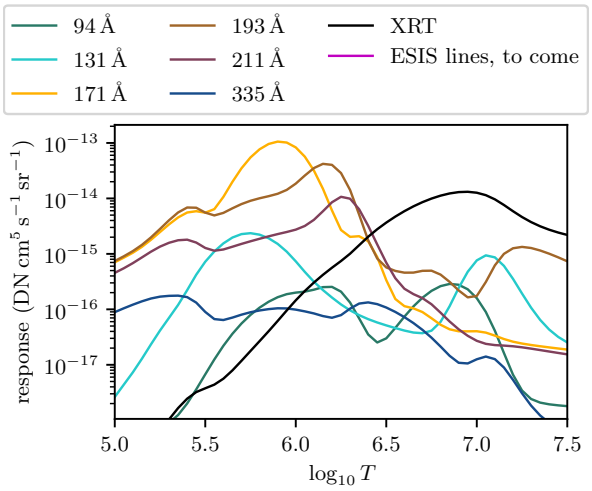


Figure 2. The temperature responses from which the DEM of event E is inverted, per unit emission measure and per steradian, from $\log_{10} T = 5.0$ to 7.5: each EUV channel of AIA, labeled by its wavelength, as computed by SolarSoft (P. Boerner et al. 2012), normalized to the EUV Variability Experiment (EVE) (T. N. Woods et al. 2012) and corrected for the emission missing from CHIANTI, and XRT Al_poly, as computed by xrtpy (N. Narukage et al. 2011), with the contamination on the date of the flight. AIA 304 Å is left out, since He II is optically thick. The ESIS lines of O III, O IV, O V, and Mg X are left out until the ESIS line list is ready. The AIA and XRT responses come from different versions of CHIANTI, where the outline calls for one.

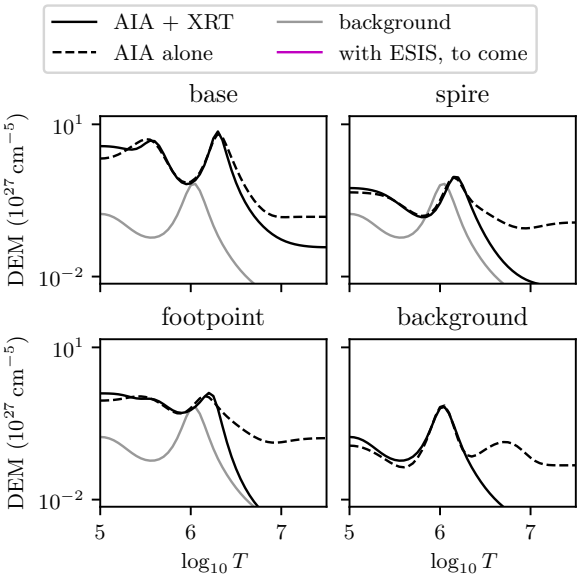


Figure 3. The DEMs of the base of event E, the hottest part of its spire, its northern footpoint, and a background region of network at 18:08:00 UT, the time of the still of Figure 4, each inverted from the mean intensities of the pixels within 3″ of the point, 1.5″ for the spire, or within 5″ of (95″, −10″) for the background, from the six EUV channels of AIA alone (dashed) and with XRT Al_poly (solid). The DEM of the background from AIA and XRT is repeated in each panel (gray). The DEMs from AIA alone predict 1.6 to 8.7 times the XRT intensity observed, since AIA alone leaves room for a tail of hot plasma which XRT rules out, and those with XRT predict it to within 8 %. The DEMs with the ESIS lines are left out until the ESIS line list is ready.

- **A reconnection jet?** Bring together the flows, the topology, and the temperature into a single picture of event E, and ask whether it is consistent with reconnection at a null point or separator driving bidirectional outflows (D. E. Innes et al. 1997; E. Pariat et al. 2009).
- **A minifilament eruption?** Compare the three phases of the event with the minifilament eruption picture (A. C. Sterling et al. 2015, 2016): adjacent blue and red jets as the rising minifilament reconnects internally, the blue component moving to the cusp as it reconnects with the surrounding field, and the blob of the third phase as a plasmoid. Compare the size of the dome, the height of the null, and the speed and duration of the jet with the breakout model (P. F. Wyper et al. 2017, 2018), and decide between standard and blowout (R. L. Moore et al. 2010).
- **Energetics.** The kinetic and thermal energy of the event compared with the magnetic energy that the topology suggests was released.
- **Context.** Where event E falls among TR explosive events (G. E. Brueckner & J.-D. F. Bartoe 1983; K. P. Dere et al. 1989), network jets (H. Tian et al. 2014; Y. Chen et al. 2019), and coronal jets (K. Shibata et al. 1992; N. E. Raouafi et al. 2016), and what it says about whether explosive events have a coronal counterpart (L. Teriaca et al. 2002).
- **Snapshot spectroscopy.** What a slit raster would have recorded of the same event, and which conclusions depended on observing the whole event at once.

7. CONCLUSIONS

Outline.

- Summarize the measured flows, topology, and temperature of event E.
- State the answer to the question posed in the introduction: whether the bidirectional flows of event E are organized by the separatrices of the coronal field.
- Implications for future snapshot spectrographs, including the second flight of ESIS.

ACKNOWLEDGMENTS

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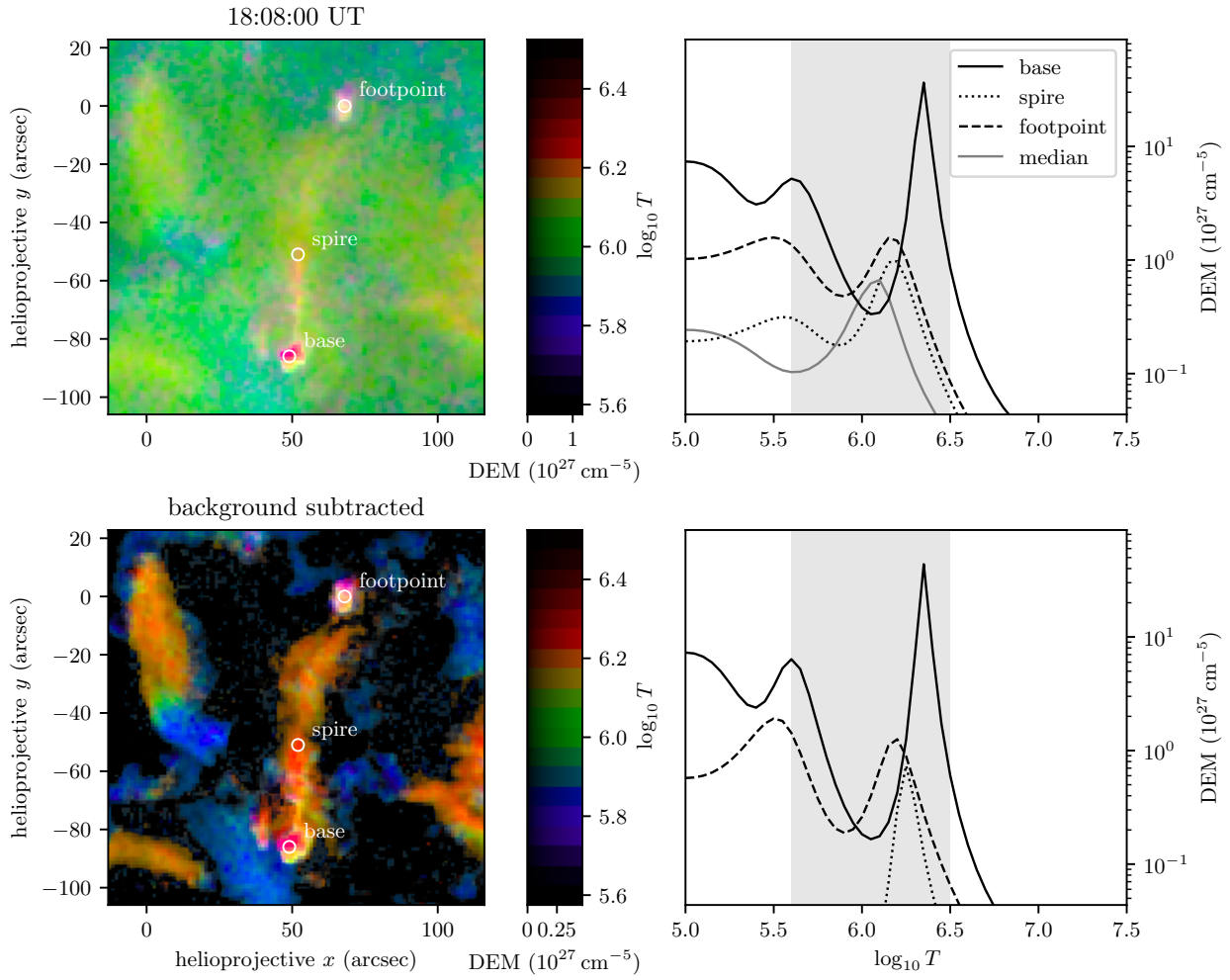


Figure 4. The DEM of event E from the six EUV channels of AIA and from XRT Al_poly at 18:08:00 UT, as a false-color image (left), inverted from the intensities as observed (top) and with the background of each pixel subtracted first (bottom): in each channel, the mean of the 61 by 61 pixels of XRT (63″) around it, without the brightest and the faintest 25 % of them, which leaves out event E. Each pixel is colored as if the DEM were the spectrum of the light it emits, with $\log_{10} T$ from 5.6 to 6.5 mapped onto the visible spectrum, from violet for the coolest plasma to red for the hottest (middle). Every temperature and every frame of a row shares one linear scale, up to the 99.5th percentile of the row over the whole movie, so the brightness of a pixel is how much plasma it has and its color is at what temperature. White pixels are those where XRT saturated, which have no DEM, and black pixels in the bottom row are those no brighter than their background in any channel. The DEM of the pixel nearest each of the three circled points, the base of event E, the hottest part of its spire, and its northern footpoint, is plotted over every temperature of the inversion (right), against the median DEM of the pixels in the image (gray line, top only, since the median pixel is no brighter than its background), with the temperatures colored in the image shaded gray. The animation shows the DEM for each of the 64 XRT images which began from 18:00:12 to 18:19:58 UT, at 4 frames per second, and runs for 16 s. The animation can be watched online.

the AIA and HMI data are courtesy of NASA/SDO and the AIA and HMI science teams. *Hinode* is a Japanese mission developed and launched by ISAS/JAXA, with NAOJ as domestic partner and NASA and STFC (UK) as international partners. It is operated by these agencies in co-operation with ESA and NSC (Norway). This article is an executable paper: every figure, table, and number in it is computed from the data when the article is built, and the code that builds it is available at https://github.com/esis-mission/esis-jet-paper.

APPENDIX

A. BACKGROUND SUBTRACTION

For our collaborators, to compare two sizes of the background; probably to be removed before submission. The bottom row of Figure 4 is inverted from the intensities of each pixel less their background, the trimmed mean of the 61 by 61 pixels of XRT (63″) around it. Figures 5 and 6 compare that background, at the time of the still, with the trimmed mean of the 31 by 31 pixels (32″) around each pixel. The wider background is smoother, and leaves more of the loops around event E behind when it is subtracted. Under the base of event E, either background is a small part of the intensity in every channel. Under the spire, either is most of it, and its DEM is close to that of the spire itself, so the DEM of the spire with the background subtracted is the one which depends the most on how the background is chosen.

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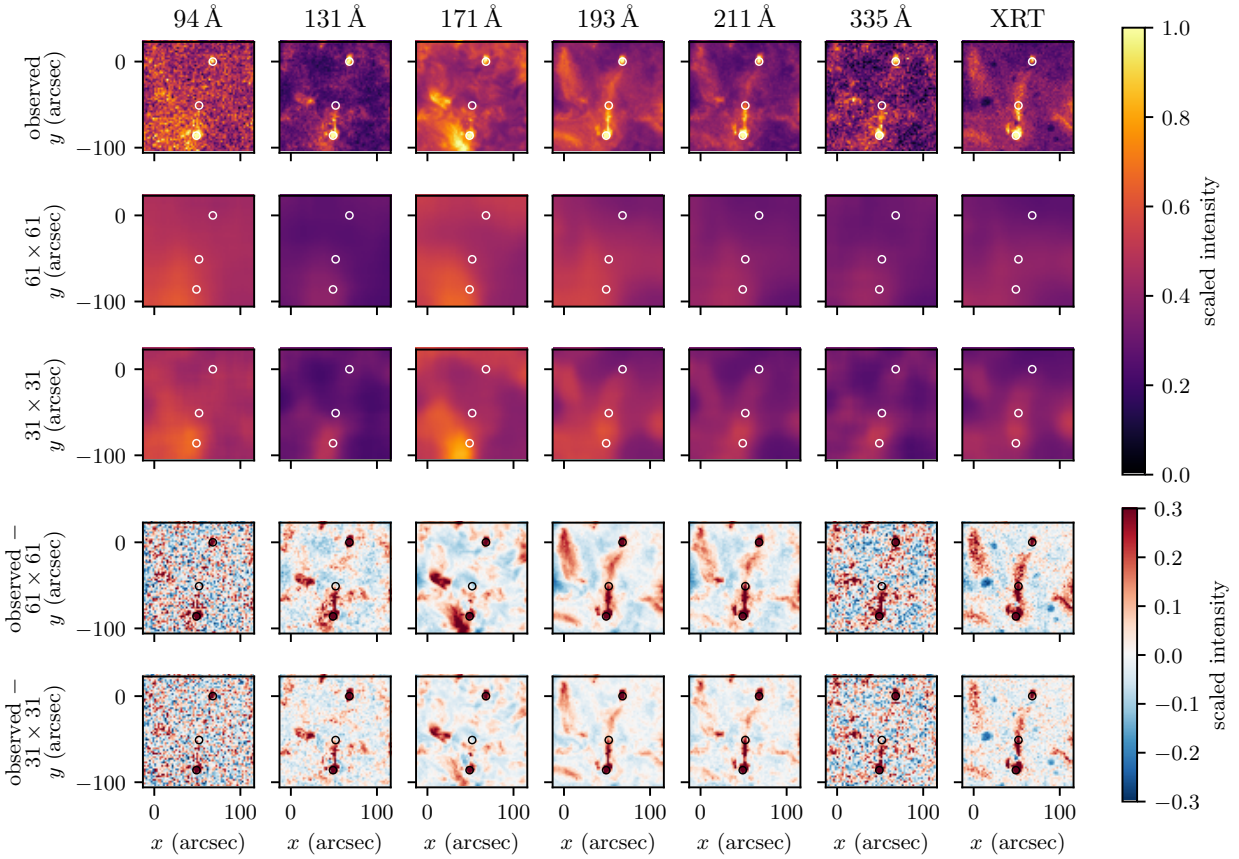


Figure 5. The intensities of event E at 18:08:00 UT, the time of the still of Figure 4, in each EUV channel of AIA, labeled by its wavelength, and in XRT Al_poly (top row), on the pixels of XRT from which the DEM is inverted. Beneath them are two backgrounds, the mean of the 61 by 61 pixels (63″) around each pixel, which the bottom row of Figure 4 subtracts, and that of the 31 by 31 pixels (32″) around it, each without the brightest and the faintest 25 % of them, and beneath those, what is left when each is subtracted (bottom two rows, red where the intensity is above the background and blue where it is below). Each channel is scaled to the 99.5th percentile of the intensity observed in it. The base of event E, the hottest part of its spire, and its northern footpoint are circled.

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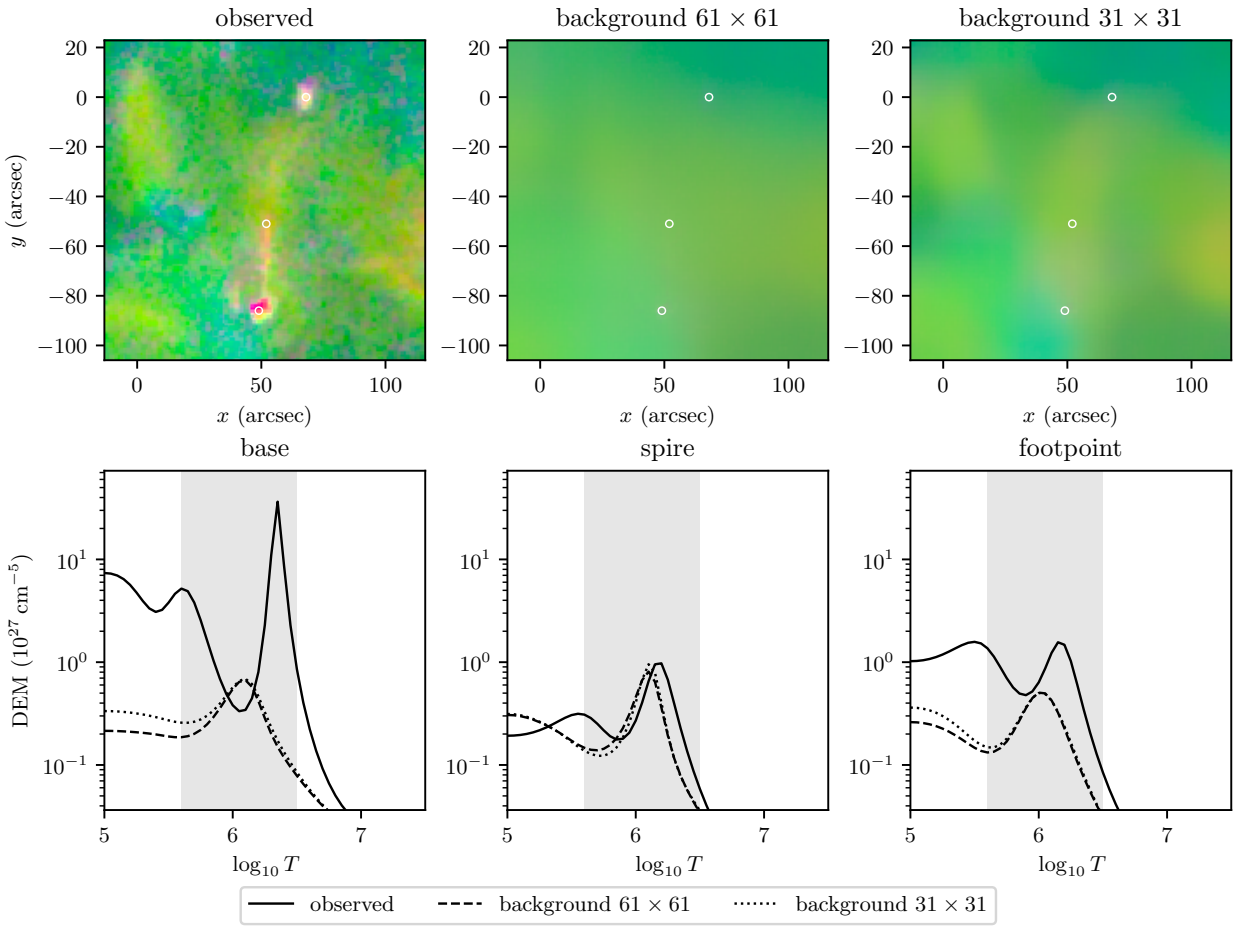


Figure 6. The DEM of event E at 18:08:00 UT, as in the top row of Figure 4 (left), and the DEMs inverted from its backgrounds of 61 by 61 (middle) and 31 by 31 pixels (right) in Figure 5, as false-color images colored as in Figure 4, on the brightness scale of its top row (top), and in the pixel nearest each of the three circled points (bottom), with the temperatures colored in the images shaded gray. White pixels are those where XRT saturated, which have no DEM.

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CONCLUSION

Outline.

- Link the manuscripts thematically.
- Future directions of research.

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